Beyond Average Effects: The Uneven Promise of Child Marriage Bans

Preliminary Version[*]

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Abstract

Child marriage remains widespread, although most countries have set a legal minimum marriage age of 18. Previous studies find that minimum-age reforms rarely delay marriage on average, but evidence on how their effects vary across contexts and how they affect women's life-cycle outcomes remains limited. I study child marriage bans (CMBs), which set 18 as the minimum age without parental-consent exceptions, using a regression discontinuity design based on birth-month eligibility. Combining WORLD policy data with Demographic and Health Surveys, I cover 20 countries that adopted a CMB in 1995–2023, the largest sample to date. While CMBs do not reduce child marriage on average, they can do so where prevalence was already low, with weaker evidence where informal unions were less common or the previous legal age was close to 18, and among girls aged 16 and 17. Independently of marriage timing, CMBs lead women to marry more educated husbands and reduce their justification of intimate partner violence. CMBs thus do not by themselves overturn established marriage practices but can reinforce social change already underway.

Keywords: Child marriage; child marriage bans; regression discontinuity design; heterogeneous treatment effects; women's life-cycle outcomes; intimate partner violence
JEL Classification: C21, I26, J12, J13, J16, O15

[*]This is a preliminary draft of my Job Market Paper. The most current version can be accessed *here*. A declaration on the use of AI tools can be found in Appendix Section A

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1 Introduction

Child marriage remains widespread despite the global commitment to eliminating it by 2030 (UN General Assembly, 2015). Approximately 12 million girls younger than 18 years marry globally each year, corresponding to one girl every three seconds (UNFPA, 2026).[1] Beyond constituting a human rights violation (UN General Assembly, 2019), child marriage is a disruptive economic and social institution, capable of fundamentally reshaping girls' life trajectories. Marrying before adulthood is associated with lower educational attainment (Sunder, 2019), earlier and higher fertility (Wodon et al., 2016; Onagoruwa and Wodon, 2018), an elevated risk of experiencing intimate partner violence (IPV) (Kidman, 2016; Omidakhsh and Heymann, 2020), and reduced labour-force participation (Wilson, 2022). These consequences make child marriage more than a demographic outcome: it is an important constraint on women's human-capital accumulation and economic opportunities, with implications extending to households and societies more broadly.

In an effort to protect minors from these adverse consequences, governments have increasingly adopted child marriage bans (hereafter CMBs), and as of 2023, two thirds of countries had implemented a CMB that establishes 18 years as the minimum legal age of marriage for women, irrespective of parental consent (WORLD Policy Analysis Center (WORLD) (2023)). In spite of this, as of 2025, still one in five women aged 20 to 24 worldwide was married before the age of 18 (UNICEF, 2025), and recent evidence shows that CMBs frequently fail to bind (Collin and Talbot, 2023; Batyra and Pesando, 2021). While the adverse effects of child marriage are well documented, existing evidence on CMBs remains limited, mostly estimating average effects and examining only a few potential sources of heterogeneity (Wilson, 2022; Collin and Talbot, 2023). Such averages can conceal substantial variation, and consequently, it remains largely unknown under which conditions CMBs induce behavioural change and whether they reshape women's lives even if marriage is not delayed.

In this paper, I show that while CMBs do not reduce child marriage on average, they can do so where it was already less common, and that they reshape marriage markets and attitudes towards IPV even where marriage is not delayed. I combine the most recent WORLD (2023) policy dataset, extended with enactment months, with all available waves of the Demographic and Health Surveys (DHS) for 20 countries that implemented a CMB between 1995 and 2023, and use a regression discontinuity design (RDD) around reform-specific birth-month cutoffs.[2] In countries with a low pre-reform child marriage rate, CMB eligibility reduces the probability of marrying as a minor by 6.8 percentage points, or about 39%, and even in the full pooled sample where the average effect is null, it increases husbands' educational attainment and reduces

[1] I define child marriage as a formal marriage or informal union resulting in cohabitation. I do not distinguish between the two due to data limitations (see Section 3). Although child marriage affects both boys and girls, this paper focuses on girls and women, who are six times more likely to marry before adulthood and disproportionately bear its adverse effects (UNFPA, 2026).

[2] I employ the robust bias-corrected local-polynomial framework of Calonico et al. (2014). Eligibility depends on a woman's birth month relative to the reform date, a running variable individuals are unlikely to manipulate, so that comparing women born just before and just after the cutoff isolates the policy effect from regional and cohort trends. I correct for multiple hypothesis testing across the many subsamples and outcomes I consider.

women's justification of IPV. This study contributes to the literature by assembling the largest cross-country dataset on CMBs to date, evaluating them with one consistent definition and design, documenting heterogeneity along previously unstudied margins, and analysing a broad set of life-cycle outcomes. Taken together, my results suggest that CMBs do not by themselves overturn established marriage practices but can reinforce social change that is already underway, which speaks to a central debate in development and gender economics on when legal reforms generate behavioural change.

Three main reasons explain why legal prohibitions often fail to prevent child marriage, and why their effectiveness reasonably depends on the local context. First, laws are often inconsistently enforced, with Collin and Talbot (2023) detecting enforcement in only around 10% of countries. I therefore consider both the effectiveness of the government as well as proxies of the social acceptance of child marriage as evidence indicates that laws are less likely to influence behaviour when they are in conflict with the prevailing norm (Acemoglu and Jackson, 2017).[3] Second, legal prohibitions can shift marriages involving minors from formal into informal or customary unions, as recorded Bellés-Obrero and Lombardi (2023) for Mexico and by Melnikas et al. (2021) for Malawi. I thus also consider the share of informal unions as a potential moderator of CMB effectiveness, as when they are more common, a substitution effect from formal to informal is more likely. Third, many legal frameworks contain exceptions, most prominently allowing marriage below 18 with parental consent (Arthur et al., 2018).[4] Because parents are commonly involved in arranging child marriages, such provisions significantly undermine the effectiveness of a legal minimum age. This is precisely why my definition of a CMB excludes reforms that retain a parental-consent exception. Importantly, weak enforcement does not imply that CMBs cannot change behaviour, as laws can act as signals of societal values that may shift customary practice (McAdams, 2000; Bénabou and Tirole, 2026).

My analysis reveals four main findings. *First*, CMBs do not reduce child marriage on average. This null result is robust to alternative bandwidths, functional forms and a coarser annual running variable, and is consistent with the existing evidence base (Collin and Talbot, 2023; Batyra and Pesando, 2021). *Second*, this average masks relevant heterogeneity in where CMBs bind. In countries with a low pre-reform child marriage rate, eligibility reduces child marriage by about two fifths relative to the control group, an effect that survives correction for multiple testing but is driven disproportionately by one country and depends on monthly identification. I find smaller, suggestive reductions in countries where informal unions are less common, where the previous legal minimum age was already close to 18, and in predominantly Muslim countries, while the effect is reversed where informal unions are more common. This pattern is consistent with CMBs being binding only when they do not conflict with prevailing marriage norms and where informal unions cannot easily substitute for formal ones. In contrast to Wilson (2022) and

[3] I consider the baseline prevalence of child marriage and the previous legal age.

[4] Other exceptions include the girl being pregnant or having given birth, religious or customary laws, and the approval of a court or similar entity (Arthur et al., 2018).

Collin and Talbot (2023), I find no robust heterogeneity by urban residence or education.

Third, CMBs may delay marriage among women just below the new legal age. While the effect at 18 is null on average, I find suggestive reductions of about 6 to 10% in the share of women married at ages 16 and 17, relative to the control group, in an extended 24-country sample.[5] This is in line with Wilson (2022), who finds a positive effect on continuous age at marriage despite a null effect on marrying before 18. *Fourth*, CMBs shape marriage markets and attitudes even when they do not delay marriage. Eligibility lowers the justification of IPV by about a tenth, and by about a fifth in countries with a low child marriage rate. In addition, CMBs shift marital sorting toward more educated husbands, increasing the proportion with secondary or higher education. These effects appear even in the full pooled sample and in subsamples where the effect on marital timing is null, which suggests that legal reform can operate through channels other than solely delayed marriage itself, for instance, by shifting social norms on IPV or by altering which unions form and are formalised. In contrast, I find no effects on women's own education, fertility, employment, bargaining power, or reported experience of IPV.

This paper contributes to the existing literature in four ways. *First*, I construct what is, to the best of my knowledge, the largest cross-country dataset assembled to study the consequences of CMB legislation without parental-consent exception. Sample sizes of previous papers range from single-country case studies (e.g., Bellés-Obrero and Lombardi, 2023, for Mexico; McGavock, 2021, for Ethiopia) to a maximum of 17 countries (Le et al., 2025; Wilson, 2022). My sample extends this to 20 countries by using the latest WORLD Policy Analysis Center (WORLD) (2023) release, which existing evaluations do not yet incorporate. This provides the statistical power and institutional variation needed for my heterogeneity analysis. To analyse the effects on girls aged 16 and 17, I additionally construct an extended sample of 24 countries by relaxing the restriction to age 18 used elsewhere in the paper.

Second, I provide a unified causal evaluation applying a consistent CMB definition and econometric approach throughout. Existing quasi-experimental studies vary in how they define a CMB, most prominently in whether they account for parental-consent exceptions. That this matters is highlighted by Kim et al. (2013) who find that CMBs without such exceptions significantly reduce adolescent fertility while those retaining them are indistinguishable from having no legal minimum age at all. Existing studies also alternate between difference-in-differences and RDDs with substantial variation in bandwidth, clustering, and polynomial specifications, limiting comparison and generalisability.[6] I apply a single CMB definition, excluding reforms with parental-consent exceptions, and identify causal effects by employing a robust bias-corrected RDD following Calonico et al. (2014) around reform-specific age cutoffs, pooling reform-eligibility variation across the full sample. Observing CMBs, births, and marriages at the monthly level

[5] The extended sample adds countries with sufficient data at these younger ages. These effects are only marginally significant in the main sample. Placebo cutoffs show no significant effects.

[6] Table A1 gives an overview of the most relevant studies for this paper.

further sharpens identification around the cutoff relative to standard annual measures.[7]

Third, I provide the most comprehensive evidence to date on treatment heterogeneity. Treating average effects as representative risks masking exactly the variation that determines when legislation binds, and while existing studies estimate small or null average effects across or within countries, evidence on treatment heterogeneity is recent and limited to a narrow set of margins. I both test whether existing evidence on treatment heterogeneity holds in this paper's larger sample, and provide novel evidence on previously not considered dimensions, including the baseline prevalence of child marriage, and informal unions, as well as regime type or rural population share.

Fourth, I test whether CMBs reshape women's life-cycle and marriage market outcomes, even where they fail to delay marriage itself. Existing evidence on downstream effects is fragmented across studies with a single or only a few outcomes, including early childbirth (Kidman et al., 2024; Kim et al., 2013; Maswikwa et al., 2015; Rokicki, 2021), attitudes towards IPV (Omidakhsh and Heymann, 2020), educational attainment and employment (Wilson, 2022), and infant mortality (Le et al., 2025). Instead, I consider a comprehensive set of outcomes, covering fertility, education, employment, IPV, household decision-making, and marriage-market outcomes within a unified framework. To do so, I rely primarily on ITT effects via reduced-form estimates, interpreting fuzzy local average treatment effects (LATE) only where the first stage is sufficiently strong. This approach allows me to identify meaningful downstream impacts even when CMB exposure does not directly reduce child marriage.

Beyond the child marriage literature, this paper relates to two broader literatures. First, it speaks to research on how laws shape behaviour and norms (Acemoglu and Jackson, 2017; McAdams, 2000; Bénabou and Tirole, 2026). Second, it relates to the economics of marriage markets, where marriage timing affects premarital investments, matching, and bargaining within the household (Bergstrom and Bagnoli, 1993; Chiappori et al., 2009, 2017), and where marriage payments and household constraints drive early marriage (Corno et al., 2020; Corno and Voena, 2023). By showing that CMB eligibility shifts women's husbands towards higher education and lowers the justification of IPV even where marital timing does not change, my results suggest that minimum-age laws can reshape partner selection and household norms through channels beyond delayed marriage.

The remainder of this paper is organised as follows. Section 2 provides an overview of the institutional setting and conceptual framework. Section 3 describes the data employed in this analysis and Section 4 outlines the empirical strategy. Section 5 presents and discusses the the main results and robustness checks. Section 6 concludes.

[7] Only Batyra and Pesando (2021) similarly employ monthly timing in an RDD context. Collin and Talbot (2023) rely on a coarser annual measure due to reporting heaping. I use monthly data in my baseline specification and test for digit preference by country. Birth months show heaping in nine countries, most plausibly due to recall and imputation, and excluding them does not strengthen the first stage. I also report results using an annual running variable (see Section 5.3).

2 Background

2.1 Institutional Setting

According to the *Convention on the Rights of the Child*, a child is any human being under the age of 18 years (United Nations, 1989). Hence, in this paper, child marriage refers to any marriage or similar union, such as informal or customary, entered into by a person under 18 years of age. In 1995, governments articulated a global commitment to eradicate child marriage in the *Beijing Declaration and Platform for Action* (UN Women, 1995). The declaration explicitly targets forced and underage marriage by calling on governments to take legislative action.[8] Since then, many countries have enacted legislation aimed at protecting minors from child marriage. By 2023, 92% of countries had set 18 as the legal minimum age of marriage, or 67% once parental-consent exceptions are taken into account, the latter representing an increase of 19 percentage points compared to 2013 (WORLD, 2013). Globally, between 1999 and 2024, the share of women aged 20 to 24 who married before 18 declined by 6 percentage points, from 25% to 19%, while the share who married before age 15 fell by half, from 8% to 4% (UNICEF, 2025). Despite this progress, child marriage remains prevalent in many regions globally, for instance, 31% in Sub-Saharan Africa and 6% in Central Asia (UNICEF, 2025).

The potential consequences of not abiding by CMBs vary by country. While some countries solely set a minimum age without explicitly criminalising it, others enforce clear measures for perpetrators, including annulment of the marriage, warnings, fines, and imprisonment (UNFPA and UNICEF, 2020). According to international standards, no minor should face any form of punishment, but parents, guardians, officials, or other adults involved in the child's marriage (UNFPA and UNICEF, 2020). For instance, in Benin any responsible adult relative may face both prison sentences of several years as well as a fine, and between 2019 and 2023, of 3,310 registered child marriages 3,063 received some form of intervention (CGRA, 2025).[9] Nonetheless, even though the minor bride does not face legal consequences, both refusing the marriage as well as reporting it may have socio-economic consequences for her, due to family rejection, social exclusion, stigma, or the mental distress of having caused legal action against family members.

Similarly to potential consequences, the social norms around child marriage also differ across contexts. Although studies show that child marriage is more common when parents and communities support it, internalised norms do not necessarily equal social pressure to marry before adulthood (Psaki et al., 2021). For example, Steinhaus et al. (2019) find that while in a region with a high rate of child marriage in Malawi the empirical expectation for women in their community to marry before age 18 is very high (89%), the normative expectation to do so is significantly lower (53%). Taken together, the continued prevalence of child marriage paired with the variation in enforcement and socio-economic contexts underscores the need to

[8] Paragraph 274e: "Enact and strictly enforce laws to ensure that marriage is only entered into with the free and full consent of the intending spouses; in addition, enact and strictly enforce laws concerning [. . .] the minimum age for marriage and raise the minimum age for marriage where necessary" (UN Women, 1995)

[9] The form of intervention was not defined. Since the passing of the CMB in 2004, different laws have defined different fine amounts and imprisonment lengths.

understand when CMBs effectively delay marriage to adulthood, a global goal re-emphasised by Sustainable Development Goal (SDG) target 5.3, which seeks to eliminate child marriage by 2030 (UN General Assembly, 2015).

2.2 Conceptual framework

The diffusion of CMBs has been driven largely by international norms and commitments such as the United Nations Committee on the Rights of the Child (CRC) and the Committee on the Elimination of Discrimination against Women (CEDAW), so that CMBs have also been adopted where marriage below 18 years of age remained a common practice (Kim et al., 2013; Ebetürk, 2021). Economically, CMBs regulate a decision that is typically taken by parents but whose consequences fall largely on daughters. In particular in low-income households and regions where a bride price is customary, an early marriage can provide parents with a marriage payment, reduce the number of household members to support, and serve as informal insurance against income shocks (Corno et al., 2020; Corno and Voena, 2023). At the same time, early marriage frequently ends girls' schooling and thus reduces their human capital (Becker, 1964; Field and Ambrus, 2008), a cost borne mainly by the daughter herself. Since a minor cannot credibly commit to compensating her parents later for the costs of postponing her marriage, even altruistic parents might choose a marriage timing that is earlier than would be optimal for the family as a whole (Bergstrom, 1989; Baland and Robinson, 2000).

Child marriage can also persist as a coordination failure. In contexts where age at marriage signals a bride's quality, communities may remain in an early-marriage equilibrium even if all families would prefer and benefit from a delay, since no single family can deviate on its own without incurring a cost (Wahhaj, 2018; Buchmann et al., 2023). As a CMB applies to all families at once, it can in principle shift such an equilibrium. Crucially, weak formal enforcement therefore does not imply that no effect is to be expected. Laws can act as focal points and signals of societal values (McAdams, 2000; Bénabou and Tirole, 2026) through which they can potentially shift customary practice even if the CMB is implemented inconsistently (Aldashev et al., 2012). Beyond the forces of law and order, enforcement itself also relies on private reporting and is thus shaped by local norms (Acemoglu and Jackson, 2017).

The conditions under which CMBs bind, described in Section 1, can be tested in several dimensions. CMBs should be less binding where enforcement capacity is weak, where child marriage is common, and where informal and customary unions are widespread, as families can substitute formal marriage by informal ones (Melnikas et al., 2021; Bellés-Obrero and Lombardi, 2023). The effects may also differ with the rural and religious composition of a country, as location has been shown to shape enforcement (Wilson, 2022) and religion can influence attitudes towards family planning and consequently shape family behaviour (Heaton, 2011; Westoff and Bietsch, 2015). By raising the expected age at marriage, CMBs may, moreover, affect premarital investments and matching, and thus whom women marry (Bergstrom and Bagnoli, 1993; Chiappori et al., 2009).

However, the welfare effects of CMBs are ambiguous. As early marriage can be a rational response to missing credit and insurance markets and limited alternatives (Corno and Voena, 2023), a ban that removes this option without an adequate alternative may leave households worse off, as documented for the case of child labour bans (Bharadwaj et al., 2020), while it is more likely to raise women's welfare where viable alternatives exist, such as schooling and employment (Jensen, 2012; Baird et al., 2011; Bandiera et al., 2020). In addition, governments have also justified exceptions to a minimum marriage age with cases of elopement and premarital pregnancy, an argument that has been contested (Asadullah and Wahhaj, 2016). Importantly, this paper does not estimate welfare effects, which depend on individual circumstances the data do not capture, but evaluates whether and where CMBs change marriage behaviour and a broad set of welfare-relevant outcomes.

3 Data

3.1 Data Sources & Sample

To identify countries with a CMB, I use data at the country-level of WORLD (2023). As exceptions for parental consent effectively legalise marriage below the stated minimum age (see Section 1, I define a CMB as a legislative change establishing a minimum legal age of marriage for women 18 years or older, irrespective of parental consent. WORLD (2023) tracks whether marriage at a given age is legally permitted across all UN member states on an annual basis since 1995 based on primary legal sources. Furthermore, and crucial for this analysis, it distinguishes between legal marriage age with and without parental consent. Because identification is based on the age in months relative to CMB implementation, I extend the dataset by constructing a new variable that captures the month of enactment of each CMB within this sample. These enactment months are extracted from the original legislative sources.

As of 2023, 129 countries set the minimum legal age of marriage at 18 years or older (WORLD, 2023). To identify the year of CMB implementation, longitudinal data is required at the year level, as otherwise I can only assess whether there was a change between the three available datasets dating from 2023, 2019, and 2013, without the possibility of assessing changes prior to 2013. This data is available for 114 countries, predominantly located in the Global South.[10] Within this subset, 53 countries implemented a CMB between 1995 and 2023, of which 45 are featured in the DHS.[11]

The source of the individual and household-level data used in this empirical analysis stems from the women's questionnaire of the DHS. The DHS provides nationally representative survey data for 91 countries on a wide array of socio-demographic variables, including education, family

[10]This subset of countries excludes almost universally high-income countries, small states and microstates, similar to the DHS, with 95.6% of DHS countries included in the WORLD longitudinal subset.

[11]Following verification of CMB years using WORLD (2023) and supplementary legislation, Peru and Nigeria were excluded. Peru's cited CMB year could not be confirmed. Ley Nº 31945 instead set a CMB in November 2023, too recent for this analysis. Nigeria's 2003 national CMB is not adopted by nine states and coexists with parallel Sharia legal systems (Fang et al., 2024), preventing clear identification of eligible individuals.

planning, IPV, and domestic violence attitudes (ICF, 2025).[12] I include all available surveys for countries where a survey was conducted after the implementation of the CMB, ensuring the sample contains both eligible and non-eligible respondents for each country. As the DHS data includes responses from women aged between 15 and 49, I restrict the sample to those aged 18 and above at the time of interview, since for younger respondents it cannot yet be known whether they will marry before turning 18.[13] Subsequently, I restrict the sample to countries with at least 36 months of birth-cohort coverage on each side of the cutoff, defined as the birth month of women who turned 18 in the month the CMB was implemented.[14] This yields a final sample of 20 countries.

Table 1 provides an overview of the countries included in the analysis with their respective CMB implementation date, the cutoff birth cohort, the DHS rounds employed, and their share of the full sample. In addition to the 20 countries included in the main sample, it also includes four additional countries that meet the data coverage requirements when considering the effects on women aged 17 and younger. The cutoff cohort corresponds to individuals who turned 18 in the month of CMB enactment, thereby generating a discontinuity in eligibility for legal protection under the reform.[15]

The majority of countries in the main sample are located in Sub-Saharan Africa (14), with the remainder in West Asia and North Africa (2), South Asia (2), Central Asia (1) and Europe (1). Because Sub-Saharan Africa is also the region where child marriage remains most common (UNICEF, 2025), the prevalence of child marriage in this sample is higher than the global average. Figure 1 depicts, by birth year, the share of women in the sample married before age 18 and before age 15, both declining steadily across cohorts, from around 50% to 30% and from 12% to 7% respectively. Figure 2 complements this by comparing marriage timing directly between women eligible and ineligible for protection under a CMB, restricting the sample to those born within three years of their country's cutoff so the two groups are drawn from comparably-aged cohorts. Eligible women are consistently less likely to be married at any given age than ineligible women, with the gap opening early and persisting through age 35.

3.2 Child Marriage & Life-Cycle Outcomes

Marital Timing. To measure marital timing, I rely on the self-reported age at first marriage or cohabitation. As DHS data only distinguishes between formal and informal for the current union, not for the first one, this variable captures union formation more broadly, including civil,

[12] Where available, I use consistently coded datasets from the Integrated Public Use Microdata Series (IPUMS) (Boyle et al., 2024). The remaining datasets were downloaded and integrated manually.

[13] Without this restriction, respondents aged 15 to 17 could only be coded zero (if already married) or missing (if not, since it is not yet known whether they will marry before 18), never one, mechanically inflating the measured child-marriage rate and biasing cohort comparisons, since eligibility is itself defined by birth cohort.

[14] The threshold corresponds to the median of the country-level MSE-optimal bandwidths for the effect on marital timing, so that each included country has enough data to support its own optimal estimation window, rather than merely a nominal presence on both sides of the cutoff. This affects four countries.

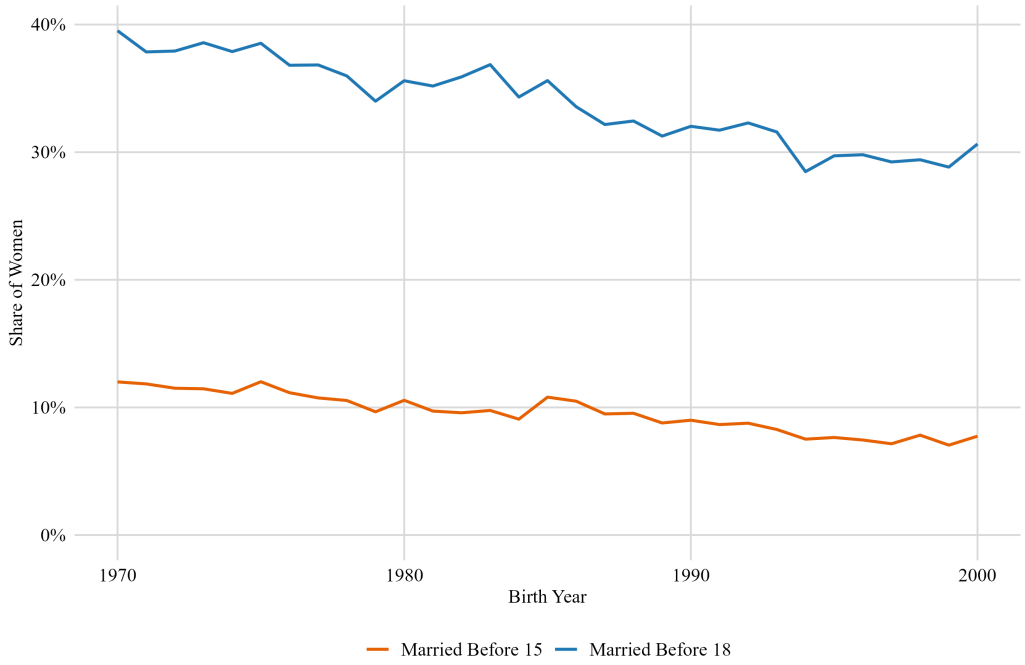
[15] I define eligibility using age 18 rather than the pre-reform legal minimum, since several countries lacked a clearly defined minimum age prior to reform, and separately examine heterogeneity by the pre-reform minimum age.

Table 1: Sample Countries and CMB Implementation Dates

	Child Marriage Ban		DHS Survey Rounds		
Country	CMB Date	Cutoff Cohort	Oldest	Newest	% of Sample
Albania	May 2003	May 1985	2008	2017	2.32
Benin	Sep 2004	Sep 1986	1996	2017	6.83
Chad†	Mar 2015	Mar 1997	1997	2015	
Comoros	Jan 2002	Jan 1984	1996	2012	0.88
DR Congo	Jan 2009	Jan 1991	2007	2024	6.47
Egypt	Jun 2008	Jun 1990	1988	2014	7.78
Ethiopia	Jul 2000	Jul 1982	2000	2019	8.13
Gabon†	Oct 2021	Oct 2003	2000	2021	
Gambia	Jul 2005	Jul 1987	2013	2019	2.72
Ghana	Jan 1999	Jan 1981	1988	2022	5.13
Guinea	Sep 2008	Sep 1990	1999	2021	4.01
Jordan	Apr 2001	Apr 1983	1990	2023	9.66
Kazakhstan†	Jan 1999	Jan 1981	1995	1999	
Kenya	May 2014	May 1996	1998	2022	9.53
Liberia	Feb 2012	Feb 1994	2007	2019	2.39
Madagascar	Aug 2007	Aug 1989	1992	2021	8.41
Maldives	Jan 2001	Jan 1983	2009	2016	2.00
Mozambique	Dec 2019	Dec 2001	2003	2022	3.13
Namibia	Jul 1996	Jul 1978	1992	2013	3.64
Nepal	Sep 2002	Sep 1984	1996	2022	7.64
Sierra Leone	Sep 2007	Sep 1989	2008	2016	3.71
Tajikistan	Jan 2012	Jan 1994	2012	2023	3.39
Togo	Jul 2007	Jul 1989	1988	2017	2.23
Zambia†	Aug 2022	Aug 2004	1992	2024	

Notes: † featured in the extended sample only. CMB date refers to the date on which the minimum legal age of marriage for women irrespective of parental consent was increased to a minimum of 18 years. The cutoff cohort refers to the birth month of women who turned 18 in the month the CMB was implemented, the first cohort eligible for protection under the CMB. % of Sample is calculated over the 20 countries in the main sample only, and is therefore left blank for the four countries featured in the extended sample only.

Figure 1: Share of Women Married Before Age 18 and 15, by Birth Year



Notes: Share of women married before age 18 (blue) and before age 15 (orange), by birth year from 1970 to 2000, in the full sample (20 countries, 615,771 women). Solid lines are weighted, dashed unweighted.

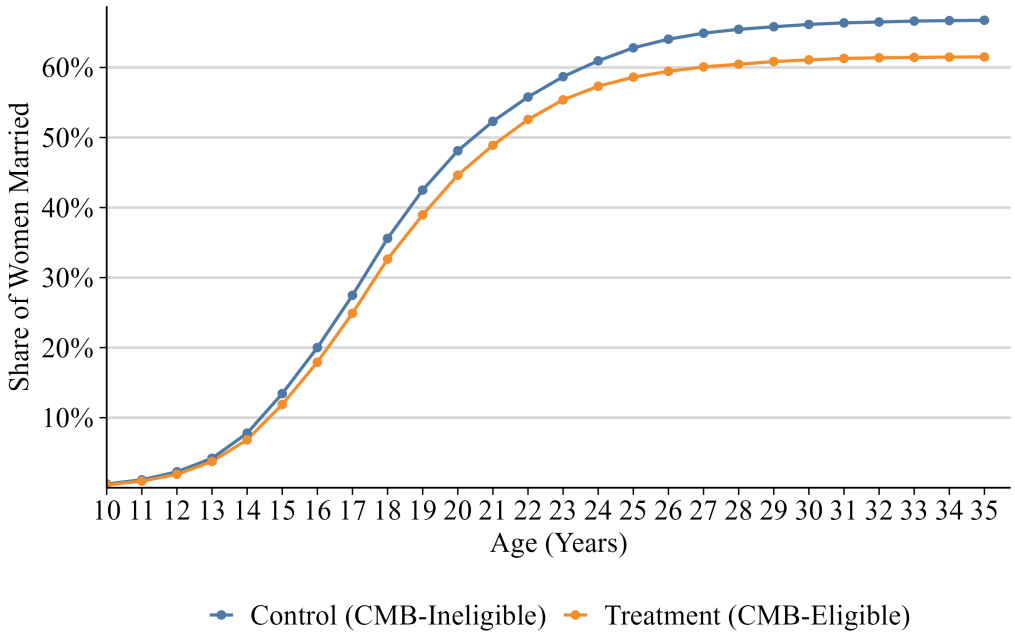
religious, and customary unions, as well as informal cohabitation. Although this more accurately captures the reality of child marriage, it does potentially mask a substitution effect from formal to informal marriage.[16] Consequently, my main outcome is a binary indicator equal to one if a woman first married or cohabited at or after age 18, or has never married, and zero otherwise, so that a positive estimate indicates a reduction in child marriage. I use this binary indicator rather than continuous age at first marriage, since the latter is only defined for women who are currently or were formerly married and would exclude women not yet married at the time of interview. I test robustness using age at first marriage as the outcome instead (see Section 5.3). To further assess differences by marital status, I restrict the sample to ever-married women and the exclude marriages potentially driven by pregnancy. As both are themselves potentially affected by the reform, I do not treat them as a heterogeneity in the main results, but instead report them as robustness checks in Section 5.3.

Life-Cycle Outcomes. To assess the effect of CMBs on women's life-cycle outcomes, I consider the following variables, categorised into four panels.[17] Each life-cycle outcome is assessed as the ITT effect of CMB eligibility and, only in subsamples with a first-stage F-statistic of at least 10, additionally as the LATE for compliers, women whose marriage timing responds to eligibility (see Section 4). While existing evidence points to adverse effects of child marriage on each of these outcomes, less is known about the effects of CMBs themselves, as discussed

[16] Among women currently in a union, 88.7% are married and 11.3% are cohabiting.

[17] See Table A2 for the respective DHS variable code, and descriptive statistics for the full sample and by treatment status.

Figure 2: Cumulative Share of Women Married, by Age and Eligibility



Notes: Weighted cumulative share of women married by each age, separately for the CMB-eligible (treatment) and CMB-ineligible (control) groups, restricted to a three-year bandwidth around each country's eligibility cutoff so the two groups are drawn from comparably-aged birth cohorts.

below, and how they differ by the considered heterogeneity dimensions.

Panel A: Fertility. Child marriage is associated with earlier childbearing (Wodon et al., 2016) and higher completed fertility (Onagoruwa and Wodon, 2018). As CMBs seek to reduce child marriage, they should mechanically delay the onset of fertility. Hence, I first consider a binary indicator, *Birth After 18*, equal to one if the respondent's first birth occurred at or after age 18, so that a positive coefficient corresponds to a lower incidence of adolescent childbearing. For completed fertility, this mechanism is less direct, as even if the first birth occurs later, women could compensate by having children in closer succession thereafter. For this, I additionally consider *Total Fertility*, measured as the total number of children the respondent has given birth to as of the time of the survey, including children who have since died.

Panel B: Education and Employment. Research shows that child marriage is also associated with lower educational attainment (Sunder, 2019) and reduced labour-market participation (Wilson, 2022). Early marriage may negatively affect schooling by imposing domestic responsibilities, resulting in them dropping out of school before completing their education. Consequently, CMBs should, by delaying marriage, and the associated responsabilities, raise educational attainment. However, this mechanism presumes that marriage timing itself drives the schooling decision, while it might also be the case that girls who never attended school or dropped out at an early age are more likely to marry early regardless of the legal minimum age, in which case CMBs would have no effect on educational attainment. Because women are already above primary-school age at the age-18 cutoff, I only assess effects on completing secondary education (*Secondary Education*), where remaining in school rather than marrying is the plausible margin.

Similarly, child marriage may constrain a woman's ability to engage in paid work, both directly through domestic responsibilities and indirectly through the education channel above, as reduced human-capital accumulation lowers the returns to, and likelihood of, labour-market participation (Becker, 1964). Conversely, Dhamija and Roychowdhury (2020) find that while delayed marriage raises education, in the case of India it does not significantly affect labour market outcomes. The authors attribute this to a backlash effect, whereby older brides face greater resistance from their husbands that offsets the labour-market gains otherwise expected from more schooling. To assess whether CMBs affect employment, I use a binary indicator, *Worked Last Year*, equal to one if the respondent engaged in any paid work outside the household in the past year.

Panel C: Decision-Making & IPV. Evidence suggests that child marriage reduces women's intra-household bargaining power and increases their risk of experiencing IPV. Tauseef and Sufian (2024) find that women who marry later have higher empowerment scores, while Kidman (2016) show that women who marry as children face an increased risk of experiencing IPV, with significant variation across countries. If a later age at marriage increases a woman's relative bargaining power within the household, CMBs should, by delaying marriage, reduce exposure to and acceptance of IPV. *Decision-Making* captures intra-household decision making and bargaining power by assessing who usually makes household decisions, such as on purchases or the respondent's own health care, across seven questions. Each response is assigned one point if the respondent decides alone, 0.5 if jointly, and zero if she is not involved, and the index is the average across all available answers. IPV attitudes are captured via *IPV Justification*, an index on whether the respondent considers it justified for a man to beat his wife or partner in specific circumstances, such as arguing or burning food. The indicator captures the fraction of yes answers, ranging from 0 to 1. A higher value means greater justification of IPV. The variables that capture the personal experience of IPV stem from an optional module that is not available for all countries or survey rounds, resulting in a smaller sample size for these outcomes of approximately 22% of the full sample. The module contains eleven questions asking whether the respondent has experienced different forms of IPV, including emotional, physical, and sexual abuse. *IPV Experience* is a binary indicator equal to one if the respondent reports any of the eleven forms and zero otherwise. To assess differential effects on different dimensions of IPV, I further include binary indicators equal to one if she reports any emotional, physical or sexual IPV (*IPV Experience: Emotional*, *IPV Experience: Physical* and *IPV Experience: Sexual*).

Panel D: Marriage Market Outcomes. Evidence on how child marriage and CMBs shape partner selection is more limited than that for the other panels. Standard marriage-market reasoning suggests that women who marry later have greater agency in partner selection (Becker, 1973), so CMBs may affect not only the timing of marriage but also whom a woman marries through positive assortative matching (Chiappori et al., 2017), predicting better-educated husbands and a narrower spousal age gap.[18] However, using a reform-based design in Bangladesh,

[18]Husband characteristics refer to different-sex couples only, reflecting the DHS survey design, which collects spousal information exclusively within the context of heterosexual marriage.

Tauseef and Sufian (2024) find that delayed marriage raises bargaining power without producing observably higher-quality spouses, suggesting the two need not move together, making this panel a complement to the previous panel. Husband characteristics are only observed for currently married women and reflect their current rather than necessarily their first husband. If CMBs mainly reduce the probability of marrying rather than change partner selection, this comparison is drawn on an endogenously selected subsample, so estimates here reflect sorting among women who remain married, not unconditional effects of the reform. Reassuringly, CMB eligibility does not predict the likelihood of being married by age 30 (coefficient ≈ 0, $p = 0.998$), suggesting no differential selection into the currently-married sample used for this panel. I thus assess whether eligible women marry more educated husbands using a continuous variable ranging from 0 (no education) to 3 (higher education), *Husband Education*. In the appendix, I additionally report a dummy for each education level to show where the shift occurs. Whether CMB eligibility's effect on husbands' education is a favourable outcome for the wife is itself unclear, and existing evidence is divided on this. From a bargaining perspective, relative spousal education, a wife's education relative to her husband's, rather than either spouse's absolute level, is commonly used as a proxy for her intra-household bargaining power, so a widening gap in the husband's favour could plausibly signal reduced rather than improved standing for the wife (Frempong and Stadelmann, 2024). Conversely, more educated husbands are found to be less likely to justify or perpetrate IPV, consistent with a protective, more gender-equitable effect of male education, though this relationship is inconsistent across countries and can even reverse (Tsala Dimbuene et al., 2024). Given this ambiguity, I do not take a stance on which mechanism dominate. Similarly, as I cannot observe the respondents actual welfare, I do not interpret an effect on husband's education as neither positive nor negative. Lastly, evidence suggests that women who marry before adulthood tend to marry significantly older men (Dommaraju, 2024; Pru et al., 2024). I therefore construct a binary indicator for an age gap of at least five years between the respondent and her current husband (*Age gap* $\geq$ *5 years*).[19]

3.3 Heterogeneity Analysis

This subsection describes the individual- and country-level characteristics used to examine heterogeneity in CMB effects.[20] These subsamples test whether CMB effectiveness and downstream life-cycle outcomes vary systematically with individual characteristics and the institutional environment. To avoid conditioning on outcomes of the reform, I restrict individual-level heterogeneity to characteristics that are fixed, or very likely to be fixed, before age 18. Country-level variables refer to the year of CMB implementation or to the weighted average of the five birth cohorts preceding the reform, unless otherwise indicated. To account for the large number of subsamples, 26 in total, Benjamini-Hochberg (BH) adjusted p-values are reported for

[19] Because the DHS reports information only on the current husband, this may bias the age-gap results downward, since women who married significantly older men may since have been widowed or remarried a younger husband.

[20] See Table A2 and Table A3 for descriptive statistics of the individual- and country-level variables for the full sample and by treatment status.

all samples (see Section 4).

Individual-Level Heterogeneity. The first two considered statifications, location and religion, are considered for both hetergeneity and as control variables (see Section 4). Wilson (2022) shows that location matters for CMB enforcement. Using this paper's larger sample and unified estimation strategy, I test whether this finding holds by distinguishing between women living in rural and urban areas (*Rural vs. Urban*). Further, I split the sample by whether the respondent identifies as *Christian* or *Muslim*, the two prominent religions in the sample as research shows that religion can shape fertility and family-planning behaviour (Heaton, 2011; Westoff and Bietsch, 2015). Both variables are recorded at the time of survey, and even though religious affiliation is typically fixed in childhood, I cannot exclude that women moved residence or converted to a different religion since. I therefore complement both splits with a country-level measure on the respective populations share below.

To assess whether CMB impacts vary by age, I stratify the sample into three roughly equally sized groups by age at time of the survey: *Age 18–24*, *Age 25–34* and *Age 35+*. This allows me to assess separately the short-, medium- and long-term consequences of CMBs. A potential concern is that this stratification reflects differences in which countries implemented their CMB when, rather than genuine variation by age. This is not the case, however, since I pool all available DHS survey rounds, and each age subsample includes all 20 sample countries. Osili and Long (2008) shows that higher education is associated with later marriage, while Ahonsi et al. (2019) find that girls who never attended school are more likely to marry as children than those who ever attended. Unfortunately, I do not observe the education level at the time of first marriage. Given the reverse causality underlying the relationship, I only consider primary education, as it is credibly pre-determined. Only 1.04% of women in the sample married before age 12, a common benchmark for primary school completion.[21] Accordingly, I distinguish between women without and with completed primary education (*No Primary Education* / *Primary Education*).

Country-Level Heterogeneity. Legislation does not immediately change the underlying socio-economic context, so a CMB may bind less where child marriage is more common (see Section 2). As the baseline prevalence of child marriage captures the extent to which a CMB departs from the prevailing marriage norm, understanding how it affects policy effectiveness is crucial to anticipate when a CMB can have the intended protective effect. I thus hypothesise that CMBs have a smaller effect in countries with a higher pre-reform child marriage rate, as the perceived risk of non-compliance may be lower where the practice is relatively widespread. Accordingly, I classify countries into three groups based on the World Bank estimate of the share of women aged 20 to 24 who were married before 18, for the year closest to the reform: 0-20% (*Low Child Marriage Rate*), 20-40% (*Medium Child Marriage Rate*), and above 40% (*High Child Marriage Rate*). I use round percentage thresholds rather than sample-specific terciles, since several countries' rates cluster too tightly for a data-driven cutoff to be robust. Further, fixed

[21] By contrast, secondary school is typically completed around age 17 or 18, by which point 23.6% and 32.1% of women in the sample, respectively, had already married.

thresholds are more easily interpretable and comparable across studies.

Besides the prevalence of child marriage at the time of CMB implementation, I use two other proxies to investigate differences by existing marital practice: The prevalence of informal unions and the previous legal age. To account for the potential substitution into informal unions discussed above, I classify countries by the prevalence of informal unions (*Low / High Informal Unions*). Besides facilitating the circumvention of the protective effect of CMBs, a substitution towards informal unions also results in women losing the legal rights associated with marriage. As the DHS does not record the type of a woman's first union, I use the share of women currently in a union who report cohabiting rather than being married, measured among the same cohorts and split at the cross-country median. I further consider the extent of legislative change, as behaviour may differ when a narrow gap is closed as opposed to when a CMB is introduced where previously no legal age existed. Following Wilson (2022), I split the sample into three subsamples based on the previous legal age: *Previous Age 16/17*, *Previous Age 14/15* and *No Previous Age*. This allows me to assess whether the pattern observed by Wilson (2022) holds across my larger sample and unified estimation strategy.

To complement the individual-level distinction between women residing in urban or rural areas, which is not credibly fully pre-determined, I classify countries by the share of women living in rural areas (*Low / High Rural Share*), measured among the five birth cohorts preceding the reform, and split countries at the cross-country median of this share. Further, I classify countries as predominantly Christian or predominantly Muslim based on whether half of women in the relevant cohorts identify with the respective religion (*Predominantly Christian / Predominantly Muslim*).[22]

Besides socio-economic conditions, the effectiveness of policy implementation also depends on state capacity, institutional quality, and governance (Besley and Persson, 2009, 2010). To capture this, I consider the following stratifications. First, I categorise countries into *Autocracies*, *Anocracies*, and *Democracies* based on their Polity Project Score (Marshall and Gurr, 2020). This score ranks regimes on a 21-point scale from hereditary monarchy to consolidated democracy. Second, as a proxy for a state's administrative and judicial capacity to enforce the CMB, I classify countries by the World Bank Government Effectiveness indicator (Kaufmann et al., 2011), which captures the quality of public service delivery, policy formulation, and implementation (*Low / High Government Effectiveness*).[23]

Strong First Stage Countries. Finally, I estimate the effect on marital timing separately for each country to construct a subsample of countries where the first stage is statistically significant at the 10% level, with controls, and positive, or in other words, countries in which CMBs successfully delayed marriage. As this selects countries on their own estimate, p-values in this category are conditional on the selection and it is not part of the BH family.

[22]This classifies 10 countries as predominantly Christian and 9 as predominantly Muslim. Nepal, where most women are Hindu, belongs to neither group.

[23]This stratification uses each country's value from the year closest to its own reform year, split at the cross-country median of that single-year measure.

4 Empirical Strategy

To estimate the causal effect of CMBs on marital timing and on women's life-cycle outcomes, I exploit the discontinuity in eligibility for protection under a CMB, comparing women who were just young enough to be protected by the reform with those who were just too old (Lee and Lemieux, 2010). Although governments can prohibit child marriage, they cannot perfectly ensure compliance, as informal and customary unions may still take place despite the prohibition of legal unions involving minors and the establishment of potential penalties (Collin and Talbot, 2023; Bellés-Obrero and Lombardi, 2023). Accordingly, I primarily estimate the effect of CMB eligibility on marital timing (1) and on life-cycle outcomes (2) as ITT effects, which correspond to the policy-relevant quantity. Where eligibility significantly affects marital timing, I additionally use it as an instrument for marrying at 18 or later in a fuzzy RDD (3). This scales the ITT effect on life-cycle outcomes by the effect on marital timing and identifies the effect of delayed marriage for compliers, women near the cutoff who marry at 18 or later if eligible but before 18 otherwise. Although compliers are the women the CMB targets, they are a specific subpopulation, so the fuzzy estimate speaks only to them. I report fuzzy estimates only for subsamples with a strong first stage, defined as an F-statistic of at least 10, and never for the full sample.

$$\text{Married}_{ijt}^{18+} = \alpha_0 + \alpha_1 Z_i + \alpha_2 (B_{ij} - c_j) + \alpha_3 (B_{ij} - c_j) Z_i + \boldsymbol{\alpha}_4' \mathbf{X}_{ij} + \mu_j + \lambda_t + e_{ijt} \qquad (1)$$

Equation (1) estimates the effect of CMB eligibility on marital timing, where, following Batyra and Pesando (2021), $\text{Married}_{ijt}^{18+}$ is a binary indicator for the absence of child marriage. It equals one if individual i in country j, observed in DHS survey year t, reports that her first marriage or cohabitation occurred at or after age 18, or if she has never married, and zero if it occurred before age 18. The coefficient α_1 is the ITT effect of eligibility on this probability and is also the first stage of the fuzzy design introduced below. A positive coefficient thus indicates that eligibility delays marriage.

Eligibility is assigned by comparing the respondent's birth month to a country-specific cutoff, set so that eligible respondents had not yet turned 18 when the CMB took effect. Formally, $Z_i = \mathbb{1}[B_{ij} \geq c_j]$, where B_{ij} is the individual's month of birth and c_j is the country-specific cutoff birth month. The running variable is the month of birth centred relative to the country-specific cutoff, $B_{ij} - c_j$, so that the coefficient of interest, α_1, can be interpreted as the size of the discontinuity in the probability of marrying at 18 or later at the cutoff.[24]

The specification includes an interaction between the centred running variable and the eligibility indicator Z_i to allow for separate linear trends on either side of the threshold, as recommended by Cattaneo and Titiunik (2022). All specifications include country fixed effects, μ_j, and survey-year fixed effects, λ_t, which account for time-invariant institutional differences across countries and global survey-year shocks, respectively, and the main specification additionally

[24] As a robustness check, I additionally construct the running variable using year of birth rather than month, since birth month is more prone to recall error and heaping in DHS data. See Section 5.3.

includes the controls $\mathbf{X}_{ij}$ described below.

$$Y_{ijt} = \gamma_0 + \gamma_1 Z_i + \gamma_2(B_{ij} - c_j) + \gamma_3(B_{ij} - c_j)Z_i + \boldsymbol{\gamma}_4'\mathbf{X}_{ij} + \mu_j + \lambda_t + v_{ijt} \tag{2}$$

Equation (2) estimates the corresponding ITT effect on each life-cycle outcome Y_{ijt}, regressing it directly on the eligibility indicator Z_i and using the same running variable, interaction term and fixed effects as (1). The coefficient γ_1 captures the effect of being eligible for protection under a CMB regardless of whether the individual's own marriage timing responded to the reform. Because γ_1 requires neither monotonicity nor the exclusion restriction to be interpreted causally, it remains valid even where the effect on marital timing is weak, unlike the fuzzy estimate in (3).

$$Y_{ijt} = \beta_0 + \beta_1 \widehat{\text{Married}}_{ijt}^{18+} + \beta_2(B_{ij} - c_j) + \beta_3(B_{ij} - c_j)Z_i + \boldsymbol{\beta}_4'\mathbf{X}_{ij} + \mu_j + \lambda_t + u_{ijt} \tag{3}$$

Equation (3) is the fuzzy second stage, in which Y_{ijt} is regressed on the predicted probability of marrying at 18 or later, $\widehat{\text{Married}}_{ijt}^{18+}$, obtained from (1). β_1 identifies the LATE for compliers and equals the Wald ratio γ_1/α_1, as both are estimated jointly on a single bandwidth for the effect on marital timing and on the outcome.

Estimation. Equations (1) to (3) are estimated using the robust bias-correction procedure of Calonico et al. (2014), implemented through local linear regression within a data-driven mean squared error (MSE) optimal bandwidth on both sides of the cutoff. This procedure selects the bandwidth that minimises the MSE of the local-linear estimate by balancing squared bias against variance. It then subtracts an estimate of the remaining curvature bias, obtained from a local quadratic regression on a separate pilot bandwidth, and computes standard errors that account for the additional uncertainty from estimating this bias, so that confidence intervals remain valid at the MSE-optimal bandwidth. Robustness to a CER-optimal bandwidth and a local quadratic specification is reported in Section 5.3. I estimate each equation in two specifications. First, I only include country and survey-year fixed effects. Second, I add controls $\mathbf{X}_{ij}$ for rural residence and religion. As the DHS records current rather than childhood residence, it is only partly pre-determined, but it shapes CMB enforcement (Wilson, 2022). Religion enters through separate indicators for Christian and Muslim, as it can shape gender norms and attitudes towards family planning (Makate and Makate, 2018; Turner and Götmark, 2023), and it is usually determined in childhood.[25] Both variables are observed for 96.4% of respondents.[26] The specification with controls is my main specification and uses only these women. I report the specification without controls, which uses all women and sets the terms $\boldsymbol{\alpha}_4'\mathbf{X}_{ij}$, $\boldsymbol{\gamma}_4'\mathbf{X}_{ij}$ and $\boldsymbol{\beta}_4'\mathbf{X}_{ij}$ to zero, alongside it throughout and in the appendix. Residence and religion are also used separately as stratifications in the heterogeneity analysis.

[25] The baseline category comprises women with no religion and those affiliated with other religions, such as Judaism, Hinduism or traditional and spiritual beliefs.

[26] Religion is missing for 3.6% of respondents, mostly in entire surveys from Egypt, Liberia, Mozambique, Kenya and Comoros.

Standard errors are clustered at the primary sampling unit level, the finest geographic unit available in the DHS. Survey weights are normalised within each country-survey cell so that every survey contributes equally to the pooled sample, regardless of its underlying size. For each fuzzy estimate, I report the first-stage F-statistic at the fuzzy bandwidth. In the few cases where an estimate exceeds half a standard deviation of the outcome, I re-estimate it excluding country-survey cells observed on one side of the cutoff only and mark it in the tables.[27]

Although the marital timing outcome and most life-cycle outcomes are binary, I estimate them using local linear regression rather than a logit or probit model. Since the conditional probability is left unrestricted outside the local window, the linear approximation within the bandwidth imposes no global functional form (Lee and Lemieux, 2010). Given the large number of subsamples and outcomes considered throughout the results section, I additionally report BH-adjusted p-values, which control for the expected share of false positives among the rejected hypotheses. The correction is computed separately for each specification and outcome across the full sample and the 26 subsamples. The countries with a strong first stage and the sample restrictions used as robustness checks are not part of this family.

Identification Assumptions. Identification of the ITT effects in (1) and (2) rests on the following two assumptions.[28] First, individuals cannot manipulate their recorded birth month to change eligibility, so that the density of the running variable is smooth through the cutoff. This fails if individuals are able to manipulate their recorded birth month and subsequently their eligibility status. While actual birth and marriage months are not manipulable, imputation of missing information as well as misreporting or recall bias in the self-reported DHS data remains possible. I test for this directly using the manipulation test of Cattaneo et al. (2020), run both on the pooled sample and separately for each country. The pooled test rejects continuity of the density at the cutoff and I detect heaping in nine countries after correcting for multiple testing, most plausibly driven by recall or imputation, but I cannot rule out more systematic misreporting. In Section 5.3, I show that excluding these countries does not strengthen the first stage and that the main results are largely robust to excluding observations close to the cutoff.

Second, since identification requires conditional expectations to be continuous at each country's own reform date, a confound coinciding with any single country's cutoff would bias that country's local estimate, and I assess how far any single country drives the pooled estimate with a leave-one-country-out analysis (Section 5.3). A confound biasing the pooled estimate systematically, rather than through one country, would additionally require coinciding independently with several countries' own idiosyncratic reform dates and affecting precisely the age cohort that was just eligible at each of them. As the identification relies on staggered, monthly data that is unrelated to one another, this is unlikely but I cannot exclude it entirely. As an indirect test, I check covariate balance at the cutoff (Section 5.3).

[27] I report the re-estimated value where it is plausible. Estimates that remain implausible are marked as unstable and excluded from the multiple-testing correction.

[28] Detailed results for the respective tests can be found in Section 5.3.

Identification of the LATE in (3) additionally rests on the following three assumptions. Under these assumptions, the fuzzy RDD estimate identifies the LATE for compliers, the subset of women near the cutoff whose marriage timing responds to the reform, and not the average treatment effect for the full population. First, the instrument must be relevant, resulting in a genuine discontinuous change in the probability of marrying at 18 or later, so that α_1 in (1) is not zero. I therefore report fuzzy estimates only where the effect on marital timing is significant and rely on the reduced form (2) elsewhere, including in the full sample.

Second, monotonicity requires that crossing the eligibility cutoff weakly increases the probability of marrying at 18 or later for every woman, so that no woman is induced to marry earlier by becoming legally eligible for protection (Hahn et al., 2001; Imbens and Lemieux, 2008). This is plausible given the direction of the legal change, since a binding minimum-age law raises the cost of marrying before 18 and cannot lower it.

Third, the exclusion restriction requires that eligibility affects outcomes only through marrying at 18 or later. This is a strong assumption, since eligibility may also operate through other channels, such as partner selection or attitudes towards IPV, and the ITT effects I find where marital timing is unaffected suggest it may not hold. I therefore interpret fuzzy estimates cautiously and treat the ITT as my primary estimates.

5 Results

5.1 Child Marriage Bans and Early Marriage

This section examines the effect of CMB eligibility, defined by birth month relative to the reform date, on marital timing, measured by the probability of not marrying or cohabiting before age 18, so that a positive estimate indicates a reduction in child marriage. This section presents and discussed the main results. Detailed results can be found in Appendix Section D.1 and robustness checks in Section 5.3.

Full Sample. Table 2 reports the effect in the full pooled sample across five specifications. Column (1) includes neither fixed effects nor controls. Column (2) adds country and survey-year fixed effects, columns (3) and (4) additionally control for religion and for rural residence respectively, and column (5) includes both and is my main specification. Religion is not observed for 3.6% of women (24,991 of 687,055), who almost exclusively reside in Egypt, Liberia, Mozambique, Kenya and Comoros. To reflect differences by specification rather than by sample, all five columns use the same sample of women with rural residence and religion observed. In every specification, CMB eligibility has no perceptible effect on child marriage, with coefficients ranging from 0.4 to 1.0 percentage points, all statistically indistinguishable from zero. This null effect also holds under alternative specifications, including an annual running variable, CER-optimal bandwidth, a local quadratic polynomial, and fixed bandwidths (see Section 5.3). In all cases coefficients are at most 1.5 percentage points, and none is significant at the 5% level.

Table 3 reports the full sample estimates without and with controls, including fixed effects for country and survey-year. While the first specification uses all observations, once controls

Table 2: Effect of CMB Eligibility on Marital Timing: Full Sample by Specification

	(1)	(2)	(3)	(4)	(5)
Estimate	0.010	0.006	0.004	0.006	0.004
Standard Error	(0.009)	(0.008)	(0.008)	(0.008)	(0.008)
p-value	[0.230]	[0.501]	[0.652]	[0.455]	[0.574]
Country and survey-year FE	No	Yes	Yes	Yes	Yes
Religion	No	No	Yes	No	Yes
Rural	No	No	No	Yes	Yes
Effective observations	165,060	147,214	162,034	165,060	174,994
Bandwidth (months)	50.4	44.6	49.3	51.0	54.0
Control mean	0.666	0.665	0.666	0.666	0.665

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Each column is a separate local linear regression discontinuity estimate of the effect of eligibility on not being married before age 18. Robust bias-corrected standard errors clustered at the primary sampling unit level in parentheses, p-values in square brackets. Effective observations within the MSE-optimal bandwidth. All columns use the same sample, women with rural residence and religion observed, so that differences between columns reflect the specification rather than the sample. Column (2) therefore differs from the fixed-effects estimate on all women reported in Table 3.

are considered, the sample is restricted to those respondents where residence and religion are observed. Without controls, the estimated effect is 1.9 percentage points and significant at the 5% level. However, this effect does not survive correction for multiple testing (BH p = 0.117) and, as seen in column 2 of Table 2, drops to 0.6 percentage points once women with missing religion are excluded. The estimate without controls is therefore sensitive to the sample and I do not interpret it as evidence of an average effect. With controls, the estimate is 0.4 percentage points and is identical to column (5) of Table 2.

This average null result matches the findings in the existing literature. The difference-in-differences design by Wilson (2022) in 17 countries using a continuous measure of the intensity of the local ban does not find a significant average effect on the probability of marrying before 18. Similarly, Collin and Talbot (2023) find detectable enforcement in only around 10% of countries and Batyra and Pesando (2021) find no decline in child marriage following the reform in four of the six countries studied. However, my results show that this null average effect masks substantial country-level heterogeneity and that CMBs alter life-cycle outcomes even where they do not delay marriage.

Figure 3 presents the effect on marital timing graphically for the full sample and across all heterogeneity dimensions including controls and depicting their 95% confidence intervals. With controls, nine of the 26 subsamples show a significant effect at the 10% level. However, only one of them survives the correction for multiple testing (Table 3). The complete results for all subsamples and both specifications are reported in Table A5.

Table 3: Effect of CMBs on Marital Timing: Full Sample and Significant Subsamples

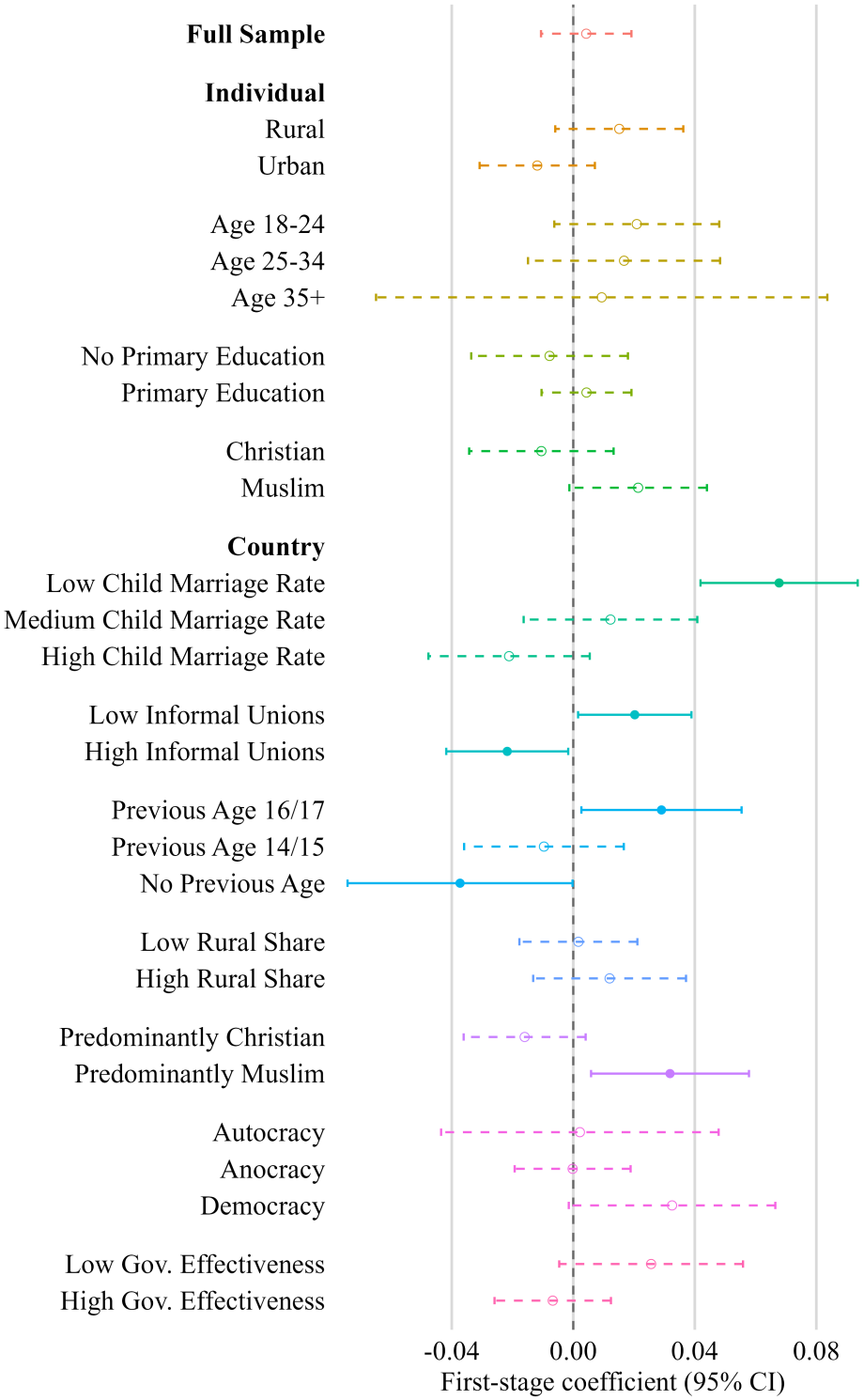
	Without controls			With controls		
Sample	Estimate	p-value	BH p-value	Estimate	p-value	BH p-value
Full Sample	0.019*	0.017	0.117	0.004	0.574	0.674
	(0.008)			(0.008)		
Rural	0.018+	0.097	0.374	0.015	0.159	0.330
	(0.011)			(0.011)		
Muslim	0.016	0.144	0.485	0.021+	0.065	0.218
	(0.011)			(0.012)		
Low Child Marriage Rate	0.048***	<0.001	0.006	0.068***	<0.001	<0.001
	(0.014)			(0.013)		
Low Informal Unions	0.008	0.413	0.695	0.020*	0.033	0.182
	(0.010)			(0.010)		
High Informal Unions	-0.008	0.489	0.695	-0.022*	0.034	0.182
	(0.012)			(0.010)		
Previous Age 16/17	0.014	0.210	0.596	0.029*	0.031	0.182
	(0.011)			(0.013)		
No Previous Age	-0.161***	<0.001	<0.001	-0.037*	0.049	0.218
	(0.024)			(0.019)		
Predominantly Muslim	0.029+	0.063	0.287	0.032*	0.016	0.182
	(0.015)			(0.013)		
Democracy	-0.090***	<0.001	<0.001	0.033+	0.061	0.218
	(0.021)			(0.017)		
Low Government Effectiveness	0.025+	0.064	0.287	0.026+	0.097	0.291
	(0.014)			(0.015)		

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Robust bias-corrected RDD estimates of the effect of eligibility on not being married before age 18. Each column is a separate specification. Robust standard error in parentheses below are clustered at the primary sampling unit level. P-value and the BH-adjusted p-value reported alongside. Both specifications include country and survey-year fixed effects. The specification without controls uses all women, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. Subsamples are shown if the effect is significant at the 10% level in either specification. The BH correction is computed within each specification across the full sample and all 26 subsamples.

Low Child Marriage Rate. In countries with a pre-reform child marriage rate below 20%,[29] CMB eligibility raises the probability of not marrying before age 18 by 6.8 percentage points. Figure 4 illustrates this graphically by comparing the RDD plot for this subsample to that for the full pooled sample. In the full sample, the probability of not marrying before 18 shows no visible discontinuity at the cutoff, whereas in countries with a low child marriage rate, a discontinuity at the cutoff is visible, increasing the probability marrying in adulthood for women born after the reform date. This corresponds to a reduction in child marriage of 39%, given that 17.5% of women in the control group marry before 18. This effect is significant at the 0.1% level, both

[29] Albania, Egypt, Jordan, Maldives, Namibia and Tajikistan.

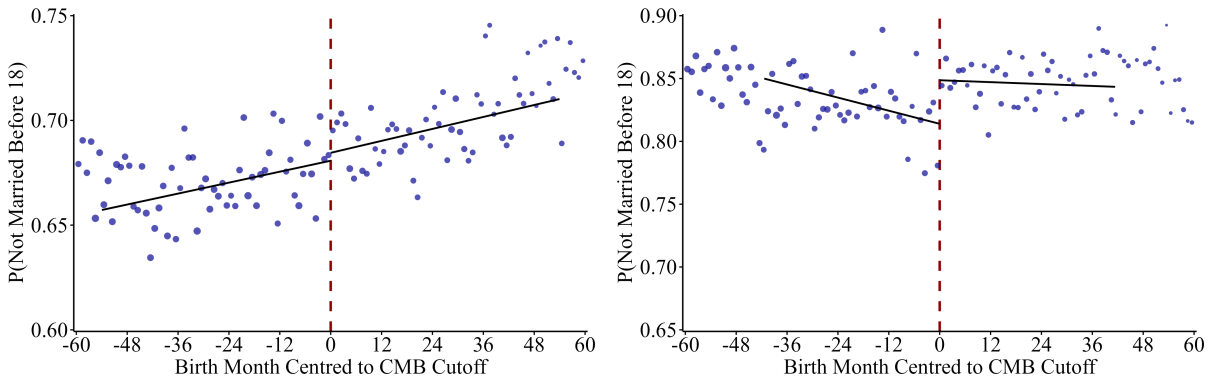
Figure 3: Effect of CMBs on Marital Timing: Heterogeneity



Notes: Each point is the robust bias-corrected local-linear RDD coefficient (Calonico et al., 2014) for the probability of not marrying before age 18 at the reform cutoff, estimated separately within each subsample. Horizontal bars are 95% confidence intervals. All regressions include country and survey-year fixed effects and control for rural residence and religion, and standard errors are clustered at the primary sampling unit level. Solid lines and filled markers denote significance at the 5% level, dashed lines and hollow markers denote non-significance. The upper block shows the full sample and individual-level subsamples, the lower block shows country-level subsamples.

with and without controls, and is the only one that survives the correction for multiple testing with controls (BH $p < 0.001$). While the result is robust to alternative bandwidths and functional forms with controls, leave-one-out checks show that it is driven disproportionately by Tajikistan. It is also not detectable when using a coarser, annual running variable, meaning that the result depends on this finer timing. This suggests that monthly variation matters for identification, as an annual variable might blur a concentrated discontinuity by pooling eligible and ineligible women born in the same year. It is therefore the most credible effect in the sample, although it rests heavily on a single country and on monthly variation.

Figure 4: Marital Timing at the CMB Cutoff



(a) Full Sample **(b) Low Child Marriage Rate**

Notes: Each dot is the mean of the outcome in a birth-month bin, adjusted for country and survey-year fixed effects, rural residence and religion, and the line is a local linear fit on either side of the cutoff within the MSE-optimal bandwidth. The dashed vertical line marks the CMB cutoff. Women born to the right of the cutoff are eligible.

Suggestive Effects. Several further subsamples show significant effects at conventional levels but these do not survive the correction for multiple testing (see Table 3), and I read them as suggestive rather than confirmed.

The starkest contrast can be observed when distinguishing by the prevalence of informal unions, which the age at first union in the DHS does not separate from formal ones. In countries with a low share of informal unions at the time of CMB implementation, the effect is 2.0 percentage points, a relative reduction in child marriage of 5.7%. Conversely, in countries with a high share, the effect is negative (-2.2 percentage points), an increase in child marriage of 6.8%. However, both effects depend on country composition. A possible interpretation of this is that CMBs can only pose a binding constraint when informal unions cannot easily substitute formal ones, supporting the findings of Bellés-Obrero and Lombardi (2023).

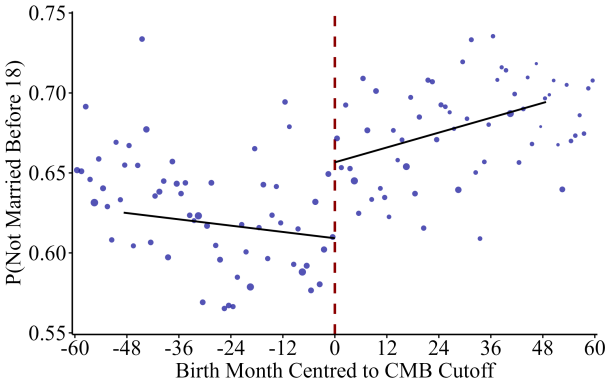
In countries where the previous legal minimum age was 16 or 17, CMB eligibility raises the probability of not marrying before 18 by 2.9 percentage points, a relative reduction in child marriage of 8.4%. This estimate is sensitive to the exclusion of Albania. In predominantly Muslim countries, the estimated effect is 3.2 percentage points and depends on the inclusion of Tajikistan, while among Muslim women across all countries it is smaller (2.1 percentage points) and only significant at the 10% level. In countries with low government effectiveness, the effect

is 2.6 percentage points, but only significant at the 10% level. Finally, democracies and countries with no previous minimum age are significant but reverse sign between the specifications with and without controls. As both rest on few countries and the No Previous Age estimate changes sign when a single country is excluded, I do not interpret them as evidence of a backfire effect.

Taken together, my results indicate that CMBs have the clearest potential to alter marital timing where child marriage was already comparatively rare before the reform, or in other words, where the newly passed reform is close to the prevailing norm. The suggestive patterns for countries where informal unions are relatively less common and with a previous legal age of 16 or 17 point in the same direction but do not meet the same robustness. Thus, CMBs seem most likely to succeed when they accompany social change already underway rather than attempting to shift a well-established social institution through legislation alone.

Countries with a Strong First Stage. To study the life-cycle consequences of a delay in marriage, I additionally define a group of countries in which eligibility has a significantly positive effect on marital timing at the 10% level with controls. This includes three countries, namely Benin, Nepal, and Tajikistan.[30] In this group, eligibility increases the probability of not marrying before 18 by 4.8 percentage points with controls, a reduction in child marriage of 12.9%, and by 5.7 percentage points without. Figure 5 shows the respective discontinuity at the cutoff graphically. As the group is selected on its own first stage, its estimate is upward biased by construction, and it is therefore not part of the BH family. It serves only to condition the analysis in Section 5.2.

Figure 5: Marital Timing at the CMB Cutoff: Strong First Stage Countries



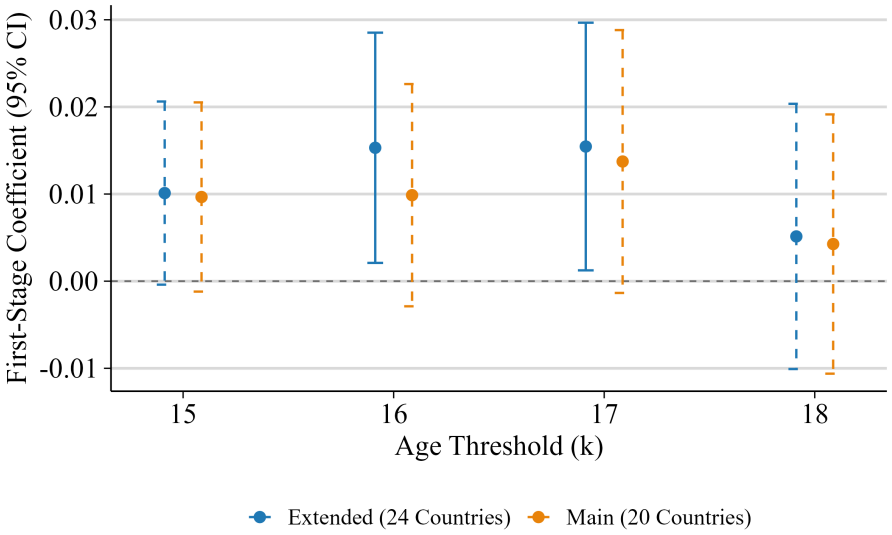
Notes: Figure includes countries with a significant positive first stage, namely Benin, Nepal and Tajikistan. Each dot is the mean of the outcome in a birth-month bin, adjusted for country and survey-year fixed effects, rural residence and religion, and the line is a local linear fit on either side of the cutoff within the MSE-optimal bandwidth. The dashed vertical line marks the CMB cutoff. Women born to the right of the cutoff are eligible.

Effect by Age Cohort. The above results investigate whether CMB eligibility shifts marriage probability by age 18, the new legal minimum age of marriage. A separate question is whether the same eligibility delays marriage at younger ages even when it does not, on average, delay marriage by age 18. An effect at a younger age could, for instance, reflect that marriages are

[30] Full results by country in Table A4.

delayed to an age closer to the new legal one, out of fear of consequences if the gap is larger. For this purpose, I include, for each age threshold, only respondents who have already reached that age, which allows the inclusion of four additional countries that now have sufficient observations on each side of the cutoff, and I additionally provide results for the main sample of 20 countries for comparability. Figure 6 presents the effect of CMB eligibility on the probability of not being married at ages 15 to 18.[31]

Figure 6: Effect of CMBs on Not Marrying Before Ages 15 to 18



Notes: Robust bias-corrected coefficients with 95% confidence intervals. Treatment is fixed at the true eligibility cutoff. Blue is the extended 24-country sample, orange the main 20-country sample. Solid lines and filled markers denote significance at the 5% level, dashed lines and hollow markers denote non-significance.

While the effect at age 18 itself, and at ages 20 and beyond, remains indistinguishable from zero throughout, I observe a significant positive effect at both ages 16 and 17 of 1.5 percentage points in the extended sample, corresponding to reductions of 9.8% and 6.4% in the share of women married by those ages, as well as a marginally significant effect of 1.0 percentage point at age 15 (11.9%). Within the main sample, the coefficients remain similar in size, but the effect at age 16 loses statistical significance, and age 17 remains only marginally significant. This indicates that while CMBs do not delay marriage at the new legal age of 18 itself, they may have the potential to delay marriage among cohorts just below this age, although this effect is sensitive to the four additional countries that enter the sample once the age floor is relaxed below 18.

Given these findings, I re-estimate the effect of CMB eligibility on not being married by ages 16 and 17 for each subsample.[32] The results show a pattern distinct from that at age 18. Rural residence is significant at both ages in both samples, with effects of 2.3 to 3.0 percentage points, while the effect at age 18 is concentrated in countries with a low child marriage rate, which show no significant effect at ages 16 or 17. The most notable instability concerns countries with no previous minimum marriage age. At age 17, this subsample is highly significant in both

[31] Table A6 provides complete results at these ages as well as at ages 20, 25 and 30.
[32] Full results in Table A7 and Table A8.

samples but with opposite signs, a 7.5 percentage-point decline in the extended sample and an 11.0 percentage-point increase in the main sample. As this reversal follows from a change in country composition alone, heterogeneity at these ages should be interpreted with caution.

This pattern is worth reading alongside Wilson (2022), who finds a positive average effect on continuous age at marriage despite, like this paper, no significant average effect on the probability of marrying before 18. A continuous outcome, or a lower age threshold as used here, can pick up marginal delays that a single binary cutoff at 18 misses. Read together, the evidence across both papers suggests that CMB eligibility may delay marriage timing at the margin for some cohorts and countries, but does not prevent child marriage entirely.

5.2 Child Marriage Bans and Life-Cycle Outcomes

This section investigates the effects of CMB eligibility on women's life-cycle outcomes across the four panels discussed in Section 3.2. I report reduced-form ITT estimates as the primary specification.[33] As the previous section established that the effect of eligibility on marital timing is on average and across most subsamples statistically insignificant and weak (F<10), scaling the reduced form by it would yield an unreliable, and potentially misleading, LATE, since under a weak instrument two-stage least squares estimates are biased towards OLS and have poor finite-sample properties (Bound et al., 1995). Every magnitude and significance level discussed below therefore refers to the ITT effect. Table 4 presents reduced-form estimates for the full sample, and Figure 7 depicts them graphically. Figure 8 shows the same estimates for the two subsamples with the most credible effect identified in the previous section: countries with a low child marriage rate and those with a significant and positive first stage. Full results for each outcome and subsample are reported in Appendix Section D.2, including BH-adjusted p-values computed within each outcome across the full sample and all 26 subsamples. The following paragraphs provide an overview of the results by panel.

Panel A: Fertility. On average, CMB eligibility has no significant effect on adolescent or on total fertility. While the effect is null on average, I find positive effects on the first birth occurring after age 18 in seven of the 26 subsamples (after correcting for multiple testing), most of them subsamples with a null first stage. This includes, for instance, women residing in rural locations (2.6 percentage points), Christian women (4.0 percentage points), as well as women residing in both predominantly Christian (4.2 percentage points) and predominantly Muslim countries (4.8 percentage points). All these effects survive the correction for multiple testing, however they should be read cautiously, as the estimate for predominantly Muslim countries changes sign in most leave-one-out exclusions. No subsample shows a robust effect on total fertility. This suggests that under certain conditions, CMBs can delay the onset of fertility, reducing adolescent childbearing. However, total fertility remains unaffected, indicating a catch-up effect in adulthood.

[33] Fuzzy RDD estimates are reported in the appendix only for the two groups with a strong first stage and only for outcomes with a first-stage F-statistic of at least 10 (Table A22 and Table A23).

Table 4: Reduced-Form Effects on Life-Cycle Outcomes: Full Sample

Outcome	Estimate (SE)	p-value	BH p	Mean	BW	N
Panel A: Fertility						
Total Children	0.031 (0.023)	0.181	0.466	2.097	49.4	181,145
Births After Age 18	0.014 (0.009)	0.109	0.239	0.535	42.6	145,215
Panel B: Education and Employment						
Completed Secondary Education	0.005 (0.006)	0.434	0.836	0.212	59.5	210,171
Worked Last Year	0.005 (0.008)	0.564	0.801	0.584	53.5	160,957
Panel C: Decision-Making & IPV						
Decision-Making	0.002 (0.006)	0.708	0.894	0.434	43.2	103,015
IPV Justification	-0.021*** (0.006)	<0.001	0.004	0.196	27.6	101,330
IPV Experience	-0.003 (0.014)	0.843	0.980	0.330	66.9	54,001
IPV Experience: Emotional	-0.006 (0.013)	0.629	0.986	0.225	57.0	47,407
IPV Experience: Physical	0.001 (0.012)	0.920	0.926	0.236	68.2	55,545
IPV Experience: Sexual	0.009 (0.008)	0.254	0.649	0.077	50.5	41,672
Panel D: Marriage Market Outcomes						
Husband Education Level	0.051** (0.017)	0.003	0.013	1.403	49.0	117,729
Husband Education: None	-0.008 (0.007)	0.233	0.676	0.255	56.4	134,255
Husband Education: Primary	-0.021** (0.007)	0.004	0.014	0.213	52.9	124,658
Husband Education: Secondary	0.020* (0.010)	0.043	0.179	0.410	43.6	104,783
Husband Education: Higher	0.014* (0.006)	0.027	0.094	0.119	65.2	154,216
Spousal Age Gap ≥ 5 yrs	-0.011 (0.010)	0.273	0.717	0.599	54.4	127,787

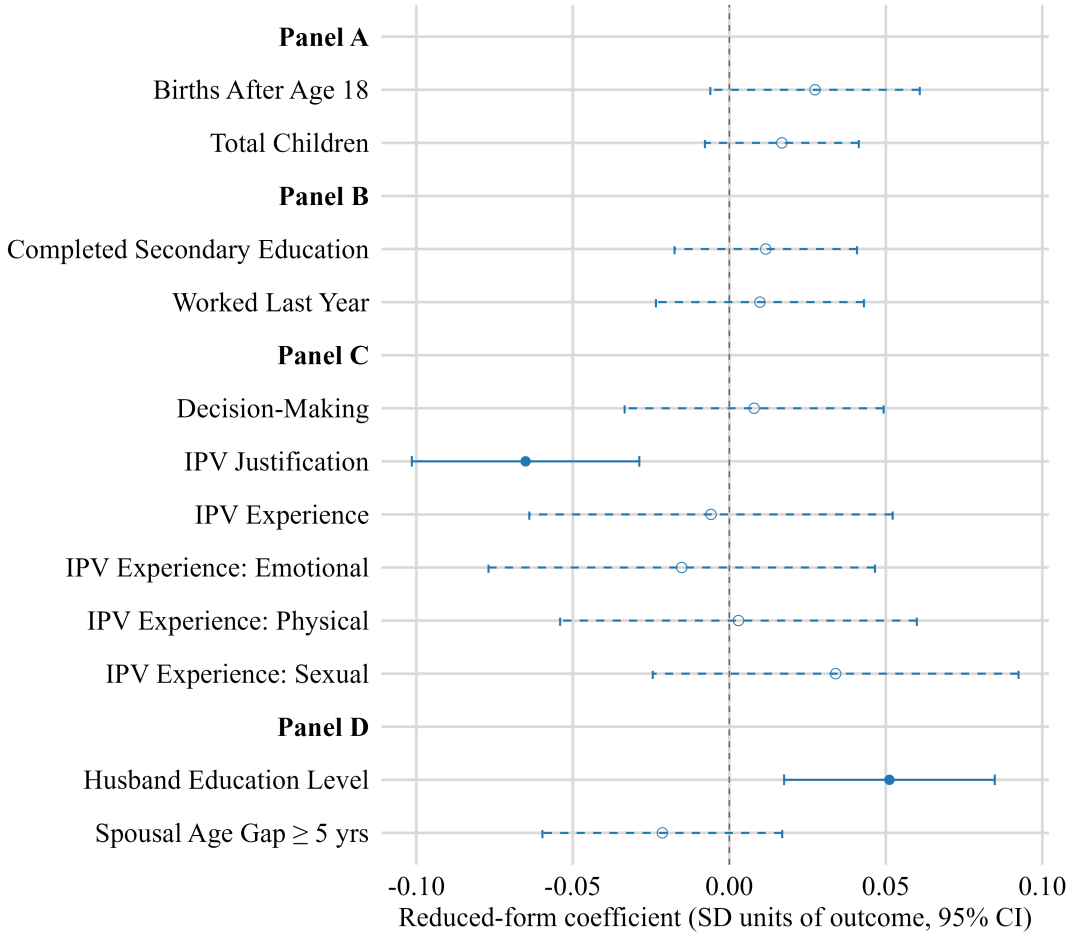
Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected reduced-form RDD coefficients (SE in parentheses) of CMB eligibility on each outcome, using Calonico et al. (2014), for the full sample. The specification includes country and survey-year fixed effects and controls for rural residence and religion, and uses the women for whom both are observed. BH p is the BH-adjusted p-value within the family of the full sample and all 26 subsamples. Mean is the weighted mean of the outcome among untreated-side observations within the estimation bandwidth, BW the bandwidth in months. N is the total number of effective observations. Standard errors are clustered at the primary sampling unit level.

Panel B: Education and Employment. I find no significant effects on completed secondary education or on employment in the past year, in the full sample or credibly in any subsample. The only estimates that survive the correction for multiple testing with controls are in the Democracy subsample, where secondary education falls by 7.8 and employment by 9.4 percentage points. However, neither estimate remains significant when any one of the subsample's countries is excluded.

Panel C: Decision-Making and IPV. This panel yields the clearest full-sample result. CMB eligibility reduces the justification of IPV by 2.1 percentage points, relative to a control mean of 0.196, a decline of 10.7%. This effect is significant at the 0.1% level, survives the correction for multiple testing, is similar without controls (2.2 percentage points) and holds under the exclusion of most single countries.[34] This reduction in IPV justification is also detectable across several subsamples. Among the 25 subsamples with a stable estimate, it is significant after correction

[34]The effect is most sensitive to Guinea.

Figure 7: Standardised Reduced-Form Life-Cycle Effects: Full Sample



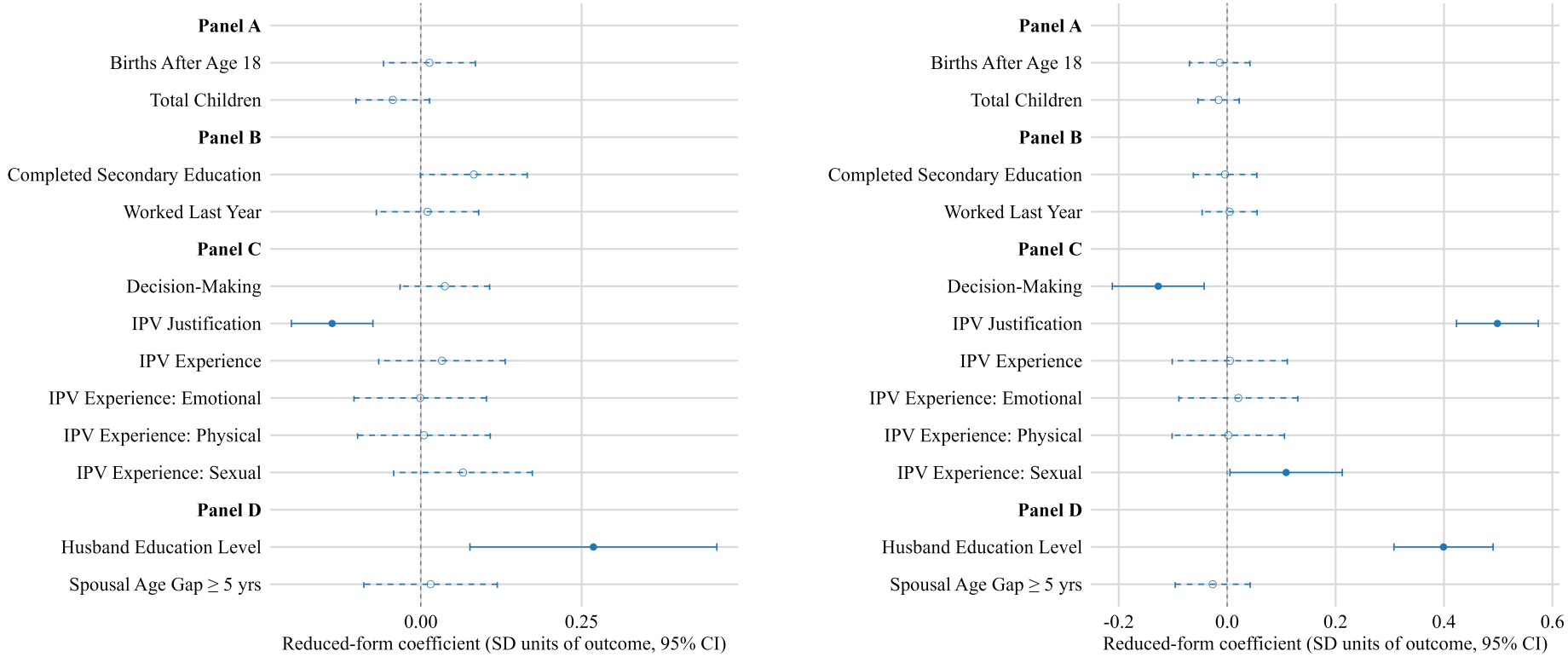
Notes: Each point is the robust bias-corrected reduced-form (ITT) RDD coefficient (Calonico et al., 2014), standardised by the outcome's own standard deviation. Solid lines and filled markers denote significance at the 5% level, dashed lines and hollow markers denote non-significance. All regressions control for rural residence, religion, and country and survey-year fixed effects, and cluster standard errors at the primary sampling unit level.

in nine, eight of them negative, with declines of 10 to 13% relative to the control mean. This includes rural location, anocracies, predominantly Muslim countries and among women with primary education.[35] Intra-household decision-making power and the experience of IPV, whether as combined measure or by type, being emotional, physical or sexual, are unaffected on average. A few subsample estimates for IPV experience are large but reverse sign across specifications and do not hold up to leave-one-out checks, which is why I do not interpret them. Importantly, as IPV is known to suffer from underreporting, with more educated and empowered women more likely to disclose experience of IPV (Cullen, 2023), any effect on stated IPV experience cannot be read directly as a change in incidence and technically, an increase in reporting could mask a reduction in IPV itself.

Panel D: Marriage Market Outcomes. On average, eligible women in the full pooled sample marry more educated husbands, with the husband's education level rising by 0.051 points on

[35]The one positive estimate is for women aged 25 to 34.

Figure 8: Standardised Reduced-Form Life-Cycle Effects: Countries with a Significant First Stage



(a) Low Child Marriage Rate **(b) Strong First Stage Countries (Benin, Nepal, Tajikistan)**

Notes: Each point is the robust bias-corrected reduced-form (ITT) RDD coefficient (Calonico et al., 2014), standardised by the outcome's own standard deviation, in (a) Low Child Marriage Rate countries and (b) the countries with a significantly positive first stage. Solid lines and filled markers denote significance at the 5% level, dashed lines and hollow markers denote non-significance. The strong first stage group is selected on its own first stage and consists of three countries. Its estimates for IPV justification and husband education level reverse sign without controls and are not interpreted. All regressions control for rural residence, religion, and country and survey-year fixed effects, and cluster standard errors at the primary sampling unit level.

the 0–3 scale, 3.6% of the control mean of 1.40. This effect survives the correction for multiple testing. As Table A20 shows, this is driven by a lower probability of marrying a husband with only primary education (2.1 percentage points) and, more tentatively, a higher probability of marrying a husband with secondary (2.0 percentage points) or higher education (1.4 percentage points). The probability of marrying a husband with no education and the probability of a spousal age gap of five years or more are unaffected. The education level is significantly higher in 15 of the 26 subsamples, all surviving BH correction, with increases of 4 to 10% relative to the control mean among Muslim women, women aged 18 to 24, urban and rural women, and in anocracies. The exceptions are Democracy and high government effectiveness, which reverse sign without controls and which I do not interpret.

Given the inclusion of country and survey-year fixed effects, as well as the absence of an effect on women's own education or fertility, the effect on husband's education is unlikely to reflect a trend, but rather a genuine effect of partner selection. This is further supported by the fact that the likelihood of getting married by age 30 is the same for eligible and non-eligible women (Table A6), which suggests no differential selection into the sample in which husband characteristics are observed. Importantly, throughout this analysis I cannot observe whether the shift in husband's education is favourable for the wife or not. Relative spousal education is sometimes used as a proxy for a wife's bargaining power, so a widening gap in the husband's favour could plausibly signal reduced rather than improved standing for the wife (Frempong and Stadelmann, 2024), while other evidence links more educated husbands to lower IPV perpetration (Tsala Dimbuene et al., 2024). Given this ambiguity, I report the effect as a marriage-market outcome without characterising it as an improvement or a setback.

Life-Cycle Outcomes in Countries with a Significant First Stage. The results above do not condition on a delay in marriage. I therefore turn to the two groups in which eligibility delays marriage: countries with a low child marriage rate and a strong first stage (Benin, Nepal and Tajikistan). In Low Child Marriage Rate countries, IPV justification falls by 4.0 percentage points relative to a control mean of 0.180, a decline of 22.2%. The effect survives the correction for multiple testing and is not driven by Tajikistan, although it loses significance when Jordan is excluded. The husband's education level rises by 0.20 points (10.5% of the control mean), driven by a higher probability of a husband with higher education (4.5 percentage points, 21.7%), but this estimate is unstable to the exclusion of single countries. Fertility, education, decision-making, and IPV experience are unaffected. The fuzzy estimate for IPV justification is negative (-0.69, Table A22), but as the first stage is only 6.8 percentage points, the fuzzy coefficient scales the reduced form by a small number and is inflated, so I treat it as indicative. In the group of countries with a first stage effect, the estimates that are consistent in sign across specifications are a lower probability of a husband with no education (9.0 percentage points), a higher probability of a husband with higher education (2.7 percentage points), lower decision-making power (3.8 percentage points) and a small increase in sexual IPV experience (2.7 percentage points). As the group is selected on its first stage, consists of three countries, and is not part of the BH family, I

read these as descriptive. The corresponding fuzzy estimates for IPV justification are negative and significant (-0.57, Table A23) but equally inflated.

Taken together, these life-cycle results point to a channel distinct from delayed marriage itself. CMB eligibility reshapes marriage markets and attitudes towards IPV broadly, including in subsamples where the first stage is null. Among the 17 subsamples without a significant first stage with controls, IPV justification, births after age 18 and husband's education are each significant at the 5% level in at least six, far more than the roughly one expected by chance. Fertility, education and employment are generally unaffected.

5.3 Robustness

This section examines whether the results above are driven by sample composition, by invalid identification, or by the specification. I focus on the first stage and on the two groups the paper conditions on, Low Child Marriage Rate and the strong first stage countries. I first vary the estimation sample, restricting it to ever-married women and to marriages that are not pregnancy-driven. I then test the validity of the design, using a manipulation test of the running variable, a covariate balance test, and placebo tests. Finally, I vary the specification, using age at first marriage as a continuous outcome, an annual running variable, alternative bandwidths, a local quadratic polynomial, and re-estimate the main results dropping one country at a time. The BH correction for multiple testing is applied throughout the main results and described in Section 5. All tables and figures referenced here are in Appendix Section D.3.

Ever-Married Subsample. I first restrict the sample to ever-married women, including those who are separated, divorced or widowed, to isolate the population for whom the treatment margin is actively binding. This rules out that pooling women who for unknown reasons never married biases the effect of the CMB on the probability of marrying at age 18 or later. Table A24 reports the full results and Figure A2a displays the significant subsamples. The overall first stage remains weak (0.012, p=0.14), but seven subsamples are significant at the 10% level, none of which survives the BH correction. These reproduce Low Child Marriage Rate (0.030), Previous Age 16/17 (0.027) and low informal unions (0.026) from the main estimation and add women aged 18–24 (0.045), rural residence (0.023), women with primary education (0.021) and anocracies (0.019). Restricting to ever-married women therefore does not itself recover a strong instrument, but it reproduces the main heterogeneity pattern.

Excluding Pregnancy-Driven Marriage. Since a pregnancy-driven marriage plausibly does not respond to a CMB the way other marriages do, I re-estimate the first stage excluding women whose first birth occurred within the 12 months preceding marriage or the first 8 months following it, which drops 10.3% of the full sample. Some child marriage legislation exempts marriages linked to pregnancy or childbirth, so these marriages may occur regardless of the CMB. Further, the negative consequences of child marriage on women's human capital accumulation and life-cycle outcomes are here contrasted with evidence showing that marriage can benefit children through greater parental resources (Kearney and Levine, 2017; Yan, 2025). The measure

cannot perfectly identify pregnancy-driven marriages, but it substantially reduces the likelihood that observed marriages were directly linked to a pregnancy. Women with a pregnancy-driven marriage marry on average about four months earlier than the others (18.4 against 18.7 years in the five-year bandwidth), but only slightly more often before 18 (43.6% against 40.7%), so excluding them removes a group that is only marginally more exposed to the reform margin. Table A25 reports the full results and Figure A2b displays the significant subsamples. The overall first stage remains null (0.007, p=0.39), but ten subsamples are significant at the 10% level and six survive the BH correction. The suggestive effects of the main estimation strengthen, in particular for predominantly Muslim countries (0.059) and in the opposite directions for low and high informal unions (0.021 and −0.034), while Low Child Marriage Rate (0.034) and Previous Age 16/17 (0.024) remain significant. Two subsamples that are not significant in the main estimation become so, Medium Child Marriage Rate (0.071) and Previous Age 14/15 (0.103). As a direct complement, the first stage among women whose marriage was potentially caused by pregnancy is null (-0.016, p=0.48), consistent with these marriages being driven by the pregnancy itself rather than by the legal minimum age, and indicating that the reform does not detectably shift timing for this group either.

Manipulation Test. I run Cattaneo et al. (2020)'s density test on the running variable, pooled and by country.[36] The pooled test rejects continuity at the 1% level, a validity concern I do not want to understate, and nine countries are flagged at the 10% level after a BH adjustment.[37] However, excluding them does not strengthen the first stage with controls (0.007 against 0.004 at baseline, p=0.45), but it removes the full-sample estimate without controls (0.005 against 0.019), which supports not interpreting that estimate. A separate check shows only a modest excess of respondents reporting the same month for their birth and marriage (10.3% against 8.3% expected), concentrated in the same countries. Besides misreporting, this could reflect known imputation mechanisms when the respondent cannot recall either date. The Low Child Marriage Rate estimate is robust to a donut hole of 1 and 3 months around the cutoff, but not of 2 or 6 months. The strong first stage group is robust to donuts of up to 6 months. Both checks are consistent with a recall or imputation mechanism rather than intentional misreporting.

Covariate Balance. As I cannot rule out that another policy coincides with the reform cutoff, I test for continuity at the cutoff in the individual-level covariates I include as controls, namely rural residence and the two predominant religions, Christian and Muslim (Table A28). Rural residence and Christian women are balanced, while the share of Muslim women is imbalanced at the 10% level (0.019, p=0.09).

Continuous Age at First Marriage. I re-estimate the effect on age at first marriage in months, addressing the concern that the binary indicator collapses within-category variation in marital timing (Table A29). Because age at first marriage is only observed for ever-married women, this restricts the estimation sample. With controls, the full-sample effect is indistinguishable

[36] See Table A26 for the density table, Figure A3 and Table A27 for the country-level heaping table.
[37] Benin, Ghana, Comoros, Guinea, Tajikistan, Sierra Leone, Madagascar, Mozambique, Liberia.

from zero (1.1 months, p=0.22), consistent with the main specification. CMB eligibility delays age at first marriage by 5.5 months in Low Child Marriage Rate countries (p=0.002) and by 3.7 months among women aged 18–24 (p=0.005), the latter not evident in the binary specification. The estimate for Previous Age 16/17 is positive with controls (3.3 months) but negative without (−3.9 months). When interpreting these results it should be kept in mind that this check uses a different outcome and sample. It does show, however, that the direction of the main effects is not an artefact of the binary threshold.

Annual Running Variable. I collapse the running variable to years since the reform, in line with Collin and Talbot (2023) (Table A30). The full-sample first stage is unchanged (0.002). Low Child Marriage Rate, the most robust effect among my results, loses significance (0.004, p=0.81), and so does the strong first stage group (0.020), whereas Previous Age 16/17 strengthens (0.052, p<0.001). This is important, as it means that the result for the countries with a low child marriage rate depends on the finer monthly timing.

Bandwidth Selection and Functional Form. I re-estimate using Calonico et al. (2014)'s CER-optimal bandwidth and a local quadratic polynomial in place of the baseline MSE-optimal bandwidth and local linear specification (Table A32 and Table A33). The full-sample estimate stays insignificant under both (0.012 and 0.006), so neither under-fitting nor the choice of bandwidth explains the null average first stage. Low Child Marriage Rate (0.041 and 0.038) and the strong first stage group (0.046 and 0.037) remain significant under both. Previous Age 16/17 and predominantly Muslim countries remain significant under the CER-optimal bandwidth (0.032 and 0.034) but only at the 10% level under the quadratic specification (0.023 and 0.027). The CER-optimal bandwidth also yields a significantly negative estimate for medium child marriage rate countries (−0.072) and an implausible one for Democracy, which I do not interpret.

Fixed Bandwidth. As a check independent of any bandwidth selector, I re-estimate the first stage at fixed bandwidths of 2 to 10 years (Table A31 and Figure A4). Low Child Marriage Rate and the strong first stage group are significant at the 5% level at every bandwidth, the most uniformly robust results. Previous Age 16/17 requires at least five years, while predominantly Muslim countries are significant between three and six years and fade at ten. The full sample remains null throughout.

Placebo Tests. I estimate the first stage at fake cutoffs 12 and 24 months before and after the true cutoff, on one side of the true cutoff only. None of the twelve estimates for the full sample, Low Child Marriage Rate and the strong first stage group is significant at the 5% level, while the estimates at the true cutoff stand out for the two groups (Figure A5). Moving the legal age threshold between 15 and 20 shows the same for Low Child Marriage Rate, which is only significant at 18, whereas the strong first stage group is significant at 18 and at 20 (0.062), and the full sample is only significant at 16 (−0.044) (Figure A1).

Leave-One-Country-Out. I re-estimate the effect on marital timing and on every outcome significant with controls, dropping one country at a time, for the full sample, Low Child Marriage Rate and the strong first stage group (Figure A6, Figure A7 and Figure A8). In the full sample,

IPV justification is significant under exclusion of 17 of the 20 countries and most sensitive to Guinea, while the husband's education level is significant under all 20. For Low Child Marriage Rate, the first stage remains significant when any country other than Tajikistan is excluded, but falls to zero without Tajikistan, which is also part of the strong first stage group. The effects on IPV justification and husband's education in this subsample remain significant when five of the six countries are excluded in turn, but the husband's education estimate is unstable in size, as excluding Jordan inflates it beyond half of the scale's range. In the strong first stage group, the first stage is significant under every exclusion, but its life-cycle estimates collapse when any single country is excluded and I read them as descriptive.

5.4 Limitations and Future Research

Limitations. Several limitations should be noted. First, and most fundamentally, the first stage is weak in the full sample and in most subsamples, so the paper's causal claims rest on reduced-form estimates rather than fuzzy RDD LATE estimates throughout. Of the nine subsamples with a first stage significant at the 10% level with controls, only one, Low Child Marriage Rate, survives correction for multiple testing across the 27 estimates. The others are reported as suggestive rather than confirmed for this reason alone, independent of the country-level fragility discussed next. Even this result is itself fragile, as it is disproportionately driven by a single country and loses significance when the running variable is collapsed to years.

Second, the data themselves carry the usual limitations of retrospective, self-reported survey data. Birth and marriage dates are recalled rather than drawn from administrative registers, and the manipulation test detects heaping in the running variable in nine of twenty countries, most plausibly driven by enumerator or respondent date imputation rather than strategic manipulation, but not fully distinguishable from it in survey data alone. The IPV experience module is available only for a subset of DHS countries and rounds, so those specific outcomes are estimated on a smaller and potentially non-random subsample of countries and years. Husband characteristics are only observed for the current husband, which could understate effects on marriage-market outcomes if women who married substantially older men are disproportionately widowed or remarried by the time of survey. Individual characteristics such as education and religion, are measured at the time of survey rather than at the time of marriage, and while the narrow estimation bandwidth (typically three to five years around the cutoff) makes it plausible that these are close to predetermined, I cannot rule out that some respondents' circumstances changed between the CMB reform and the survey in ways correlated with treatment.

Third, while the cross-country design improves external validity relative to a single-country study, it necessarily abstracts from country-specific mechanisms that plausibly shape reform effectiveness: local enforcement capacity, administrative implementation, and existing social norms around marriage age all vary across the sample, both across and within countries. While I attempt to capture these via country-level subsamples, this could mask regional heterogeneity. For instance, one dimension I do not currently exploit is ethnic and customary variation in

marriage practices, in particular bride-price and dowry systems, which the economics literature on marriage markets (e.g. Corno et al. (2020) on bride price and drought-driven child marriage) links directly to the economic incentives underlying early marriage. Linking DHS clusters to ethnic-group-level marriage-custom data would allow a direct test of whether CMB effectiveness varies with the prevailing bride-price regime.

Fourth, this paper only considers effects on women. While girls and women are disproportionately affected by child marriage and its consequences, insights on the effects on boys and men would complement the findings of this paper. Extending the analysis to men is not straightforward, however, as the legal minimum age of marriage, and hence the timing of reform, often differs by sex, requiring CMB eligibility to be recoded separately for each country. The DHS Men's Questionnaires are also considerably less comprehensive than the Women's Questionnaires used throughout this paper, limiting the range of outcomes available for a male sample.

Finally, the outcomes studied here are downstream life-cycle outcomes, not direct measures of enforcement or compliance behaviour. I can document where the first stage is strong or weak, but I do not directly observe why. In other words, I cannot distinguish whether effects in the respective subsamples are due to administrative capacity, awareness campaigns, community-level enforcement, or shifting social norms. IPV outcomes are self-reported attitudes and experiences rather than objective incidence, a well-documented measurement concern in the domestic violence literature more broadly.

Future research. These limitations point to several avenues for future research. Administrative or registry-based birth and marriage data, where available, would sidestep the recall and heaping concerns inherent to retrospective survey data and allow a sharper manipulation test. Marriage histories that record more than the current union would also help establish whether the husband-characteristics results are understated by remarriage. Linking the existing sample to ethnic-level marriage-custom data would allow future work to test bride-price and dowry regimes as moderators of CMB effectiveness directly. Sub-national or district-level measures of state capacity and enforcement, where available, would allow a more precise test of the enforcement-gap mechanism than the rural versus urban or country-level institutional indices used here. Recoding CMB eligibility separately by sex and drawing on the DHS Men's Questionnaires, despite their more limited scope, would allow a direct test of whether CMBs affect marriage timing and downstream outcomes for men in the same way as for women. Longer panel follow-up would let future work test whether the paper's most robust reduced-form findings, on husband's education and attitudes towards IPV, persist into second-generation outcomes such as children's schooling and health. Finally, as additional DHS rounds become available, several of the underpowered subsamples identified here could be re-estimated on a larger and more stable sample.

6 Conclusion

Child marriage remains a major policy concern, yet evidence on when CMBs actually delay marriage, and on whether they change women's broader life trajectories, has remained limited.

The present study set out to determine, first, whether CMBs, reforms setting the minimum legal age of marriage at 18 without parental-consent exceptions, have the potential to alter marital timing under specific socio-economic conditions, and, second, whether they can alter women's life-cycle outcomes even when marital timing remains unchanged. In this paper I provide, to my knowledge, the most comprehensive cross-country evaluation of CMBs to date, covering reforms enacted between 1995 and 2023 by combining harmonised legal reform data from the WORLD Policy Analysis Center with nationally representative DHS data from 20 countries, and exploiting variation in reform eligibility across birth cohorts through a robust bias-corrected RDD.

Excluding reforms that retain parental-consent exceptions matters, as when minimum age legislation allows for an earlier age with consent of the parents, child marriage practically remains legal. This study contributes to the ongoing discussion in development and gender economics, among other disciplines, of when legal reforms translate into actual behavioural change by focusing on reforms that impose a binding constraint on the age of marriage. My results confirm the existing finding that CMBs do not delay marriage on average, but establish two novel findings the literature has not yet documented.

First, this average null effect masks meaningful heterogeneity. Where child marriage was already comparatively rare before the reform, CMB eligibility measurably reduces it further, by 39% relative to the prevalence observed in the control group. This effect survives correction for multiple testing, but it is driven disproportionately by one country and depends on the monthly timing of birth cohorts. I document similar patterns in countries where informal unions are less common, where the previous legal minimum age was already close to 18, and in predominantly Muslim countries, but treat them as suggestive rather than confirmed, since none of these subsamples survives correction for multiple testing. Further, while I find no average effect on marriage at 18, the new legal age, I find suggestive evidence of an effect at ages 16 and 17 in an extended 24-country sample, though only the effect at age 17 remains marginally significant in the paper's main sample. I find no robust evidence that individual characteristics, including rural location, and education, moderate CMB effectiveness.

Second, and independent of whether marriage is delayed, CMB eligibility reshapes marriage markets and attitudes towards IPV. Even in the complete pooled sample, where the reform has no detectable effect on marriage timing, CMB eligibility lowers the justification of IPV by 11% and raises the education of the husband a woman goes on to marry. Both effects survive correction for multiple testing, and appear even in subsamples where the first stage is entirely null. In countries with a low child marriage rate, where marriage is delayed, the decline in IPV justification is even larger, at 22%, suggesting that positive effects are enhanced when the CMB binds. Importantly, whether the shift in husband's education is welfare-improving cannot be assessed, but it shows that CMBs can alter women's life-cycle outcomes beyond marital timing. At the same time, I find no generalisable effect on women's own education, employment, or fertility, or on the experience of IPV itself.

A number of limitations should be noted. The first stage is weak throughout this paper, so the

causal claims rest on reduced-form ITT estimates rather than fuzzy RDD LATE estimates, which would be imprecisely estimated given this weakness, and the one confirmed first-stage result rests heavily on one country. Administrative data in place of self-reported data, a longer panel, and considering the smaller male DHS sample would each strengthen this evidence, particularly for the results on husband's education and IPV attitudes that do not depend on the fragile first stage.

Notwithstanding these limitations, this study points to the specific margin at which policy operates. While CMB legislation cannot overturn a marriage practice sustained by broader social and economic conditions, it can reinforce a shift already underway, and it seems to operate on the marriage market through channels other than its stated goal of delayed marriage. Policy design that treats a CMB as uniformly binding, or writes it off wherever it fails to move the headline marriage-timing statistic, misses both. Legal reform is thus best understood in this context as a complement to social change already in motion, not a substitute for it. If the goal is to alter marriage age itself, a minimum-age law on its own is unlikely to be sufficient. Its effectiveness depends on the conditions it meets, not on the law alone. Even where those conditions are not yet in place, however, this paper shows the law is not without consequence. It still shifts the marriage market and the attitudes surrounding it, an effect distinct from, and independent of, its stated goal.

References

Acemoglu, Daron and Matthew O. Jackson, "Social Norms and the Enforcement of Laws," *Journal of the European Economic Association*, April 2017.

Ahonsi, Babatunde, Kamil Fuseini, Dela Nai, Erika Goldson, Selina Owusu, Ismail Ndifuna, Icilda Humes, and Placide L. Tapsoba, "Child marriage in Ghana: evidence from a multi-method study," *BMC Women's Health*, December 2019, *19* (1), 126.

Aldashev, Gani, Imane Chaara, Jean-Philippe Platteau, and Zaki Wahhaj, "Using the law to change the custom," *Journal of Development Economics*, March 2012, *97* (2), 182–200.

Arthur, Megan, Alison Earle, Amy Raub, Ilona Vincent, Efe Atabay, Isabel Latz, Gabriella Kranz, Arijit Nandi, and Jody Heymann, "Child Marriage Laws around the World: Minimum Marriage Age, Legal Exceptions, and Gender Disparities," *Journal of Women, Politics & Policy*, January 2018, *39* (1), 51–74.

Asadullah, M Niaz and Zaki Wahhaj, "Child Marriage Law and Freedom of Choice in Bangladesh," *Economic and Political Weekly*, 2016, *51* (3), 24–26.

Baird, S., C. McIntosh, and B. Ozler, "Cash or Condition? Evidence from a Cash Transfer Experiment," *The Quarterly Journal of Economics*, November 2011, *126* (4), 1709–1753.

Baland, Jean-Marie and James A. Robinson, "Is Child Labor Inefficient?," *Journal of Political Economy*, August 2000, *108* (4), 663–679.

Bandiera, Oriana, Niklas Buehren, Robin Burgess, Markus Goldstein, Selim Gulesci, Imran Rasul, and Munshi Sulaiman, "Women's Empowerment in Action: Evidence from a Randomized Control Trial in Africa," *American Economic Journal: Applied Economics*, January 2020, *12* (1), 210–259.

Batyra, Ewa and Luca Maria Pesando, "Trends in child marriage and new evidence on the selective impact of changes in age-at-marriage laws on early marriage," *SSM - Population Health*, June 2021, *14*, 100811.

Becker, Gary S., *Human Capital: A Theoretical and Empirical Analysis with Special Reference to Education, First Edition*, National Bureau of Economic Research, 1964.

_ , "A Theory of Marriage: Part I," *Journal of Political Economy*, July 1973, *81* (4), 813–846.

Bellés-Obrero, Cristina and María Lombardi, "Will You Marry Me, Later?: Age-of-Marriage Laws and Child Marriage in Mexico," *Journal of Human Resources*, January 2023, *58* (1), 221–259.

Bergstrom, Theodore C., "A Fresh Look at the Rotten Kid Theorem–and Other Household Mysteries," *Journal of Political Economy*, October 1989, *97* (5), 1138–1159.

_ **and Mark Bagnoli**, "Courtship as a Waiting Game," *Journal of Political Economy*, February 1993, *101* (1), 185–202.

Besley, Timothy and Torsten Persson, "The Origins of State Capacity: Property Rights, Taxation, and Politics," *American Economic Review*, August 2009, *99* (4), 1218–1244.

_ **and** _ , "State Capacity, Conflict, and Development," *Econometrica*, 2010, *78* (1), 1–34.

Bharadwaj, Prashant, Leah K Lakdawala, and Nicholas Li, "Perverse Consequences of Well Intentioned Regulation: Evidence from India's Child Labor Ban," *Journal of the European Economic Association*, June 2020, *18* (3), 1158–1195.

Bound, John, David A. Jaeger, and Regina M. Baker, "Problems with Instrumental Variables Estimation when the Correlation between the Instruments and the Endogenous Explanatory Variable is Weak," *Journal of the American Statistical Association*, June 1995, *90* (430), 443–450.

Boyle, Elizabeth Heger, Miriam King, and Matthew Sobek, "IPUMS Demographic and Health Surveys: Version 11," 2024.

Buchmann, Nina, Erica Field, Rachel Glennerster, Shahana Nazneen, and Xiao Yu Wang,

"A Signal to End Child Marriage: Theory and Experimental Evidence from Bangladesh," *American Economic Review*, October 2023, *113* (10), 2645–2688.

Bénabou, Roland and Jean Tirole, "Laws and Norms," *Journal of Political Economy*, February 2026, *134* (2), 731–772.

Calonico, Sebastian, Matias D. Cattaneo, and Rocio Titiunik, "Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs: Robust Nonparametric Confidence Intervals," *Econometrica*, November 2014, *82* (6), 2295–2326.

Cattaneo, Matias D. and Rocío Titiunik, "Regression Discontinuity Designs," *Annual Review of Economics*, August 2022, *14* (1), 821–851.

_ , Michael Jansson, and Xinwei Ma, "Simple Local Polynomial Density Estimators," *Journal of the American Statistical Association*, July 2020, *115* (531), 1449–1455.

CGRA, "COI Focus Benin: Le mariage forcé," Technical Report, Office of the Commissioner General for Refugees and Stateless Persons (CGRA), Brussels 2025.

Chiappori, Pierre-André, Bernard Salanié, and Yoram Weiss, "Partner Choice, Investment in Children, and the Marital College Premium," *American Economic Review*, August 2017, *107* (8), 2109–2167.

_ , Murat Iyigun, and Yoram Weiss, "Investment in Schooling and the Marriage Market," *American Economic Review*, December 2009, *99* (5), 1689–1713.

Collin, Matthew and Theodore Talbot, "Are age-of-marriage laws enforced? Evidence from developing countries," *Journal of Development Economics*, January 2023, *160*, 102950.

Corno, Lucia and Alessandra Voena, "Child marriage as informal insurance: Empirical evidence and policy simulations," *Journal of Development Economics*, May 2023, *162*, 103047.

_ , Nicole Hildebrandt, and Alessandra Voena, "Age of Marriage, Weather Shocks, and the Direction of Marriage Payments," *Econometrica*, 2020, *88* (3), 879–915.

Cullen, Claire, "Method Matters: The Underreporting of Intimate Partner Violence," *The World Bank Economic Review*, January 2023, *37* (1), 49–73.

Dhamija, Gaurav and Punarjit Roychowdhury, "Age at Marriage and Women's Labour Market Outcomes in India," *Journal of International Development*, April 2020, *32* (3), 342–374.

Dimbuene, Zacharie Tsala, Bright Opoku Ahinkorah, and Dickson Abanimi Amugsi, "Men's education and intimate partner violence—Beyond the victim-oriented perspective: Evidence from demographic and health surveys in Central Africa," *PLOS ONE*, April 2024, *19* (4), e0302627.

Dommaraju, Premchand, "Age Gap Between Spouses in South and Southeast Asia," *Journal of Family Issues*, May 2024, *45* (5), 1242–1260.

Ebetürk, Irem, "Global diffusion of laws: The case of minimum age of marriage legislation, 1965–2015," *European Journal of Cultural and Political Sociology*, September 2021, *8* (3), 294–328.

Fang, Xiangming, Deborah Fry, Jingru Ren, Wuwenhao Jin, Yuchen Zhu, Ibrahim Sesay, Hadiza Abba, Amandine Bollinger, and Christine Wekerle, "The economic burden of child marriage in Nigeria," *Child Abuse & Neglect*, December 2024, *158*, 107135.

Field, Erica and Attila Ambrus, "Early Marriage, Age of Menarche, and Female Schooling Attainment in Bangladesh," *Journal of Political Economy*, October 2008, *116* (5), 881–930.

Frempong, Raymond Boadi and David Stadelmann, "Exploring Education-Induced Bargaining Power of Women on Household Welfare in Sub-Saharan Africa," *Economies*, October 2024, *12* (11), 293.

Hahn, Jinyong, Petra Todd, and Wilbert Klaauw, "Identification and Estimation of Treatment Effects with a Regression-Discontinuity Design," *Econometrica*, January 2001, *69* (1), 201–

209.

Heaton, Tim B., "Does Religion Influence Fertility in Developing Countries," *Population Research and Policy Review*, June 2011, *30* (3), 449–465.

ICF, "The DHS Program," 2025.

Imbens, Guido W. and Thomas Lemieux, "Regression discontinuity designs: A guide to practice," *Journal of Econometrics*, February 2008, *142* (2), 615–635.

Jensen, Robert, "Do Labor Market Opportunities Affect Young Women's Work and Family Decisions? Experimental Evidence from India *," *The Quarterly Journal of Economics*, May 2012, *127* (2), 753–792.

Kaufmann, Daniel, Aart Kraay, and Massimo Mastruzzi, "The Worldwide Governance Indicators: Methodology and Analytical Issues," *Hague Journal on the Rule of Law*, September 2011, *3* (02), 220–246.

Kearney, Melissa and Phillip Levine, "The Economics of Non-Marital Childbearing and The "Marriage Premium for Children"," Technical Report w23230, National Bureau of Economic Research, Cambridge, MA March 2017.

Kidman, Rachel, "Child marriage and intimate partner violence: a comparative study of 34 countries," *International Journal of Epidemiology*, October 2016, p. dyw225.

_ , Amy Raub, Alfredo Martin, Pragya Bhuwania, Bijetri Bose, and Jody Heymann, "Reducing child marriage in sub-Saharan Africa: Evaluating the joint potential of protective marriage and education policies," *Children and Youth Services Review*, September 2024, *164*, 107877.

Kim, Minzee, Wesley Longhofer, Elizabeth Heger Boyle, and Hollie Nyseth Brehm, "When Do Laws Matter? National Minimum-Age-of-Marriage Laws, Child Rights, and Adolescent Fertility, 1989–2007," *Law & Society Review*, September 2013, *47* (3), 589–619.

Le, Dung D., Teresa Molina, Yoko Ibuka, and Rei Goto, "The intergenerational health effects of child marriage bans," *Journal of Health Economics*, December 2025, *104*, 103075.

Lee, David S and Thomas Lemieux, "Regression Discontinuity Designs in Economics," *Journal of Economic Literature*, June 2010, *48* (2), 281–355.

Makate, Marshall and Clifton Makate, "Education and teenage childbirth in Uganda: Understanding the links from the Uganda Demographic and Health Survey," *International Journal of Social Economics*, 2018, *45* (5), 746–764.

Marshall, Monty G and Ted Robert Gurr, "Polity 5: Political Regime Characteristics and Transitions, 1800-2018," 2020.

Maswikwa, Belinda, Linda Richter, Jay Kaufman, and Arijit Nandi, "Minimum Marriage Age Laws and the Prevalence Of Child Marriage and Adolescent Birth: Evidence from Sub-Saharan Africa," *International Perspectives on Sexual and Reproductive Health*, 2015, *41* (2), 58.

McAdams, Richard H., "A Focal Point Theory of Expressive Law," *Virginia Law Review*, November 2000, *86* (8), 1649.

McGavock, Tamara, "Here waits the bride? The effect of Ethiopia's child marriage law," *Journal of Development Economics*, March 2021, *149*, 102580.

Melnikas, Andrea J., Nancy Mulauzi, James Mkandawire, and Sajeda Amin, "Perceptions of minimum age at marriage laws and their enforcement: qualitative evidence from Malawi," *BMC Public Health*, December 2021, *21* (1), 1350.

Omidakhsh, Negar and Jody Heymann, "Improved child marriage laws and its association with changing attitudes and experiences of intimate partner violence: a comparative multi-national study," *Journal of Global Health*, June 2020, *10* (1), 010707.

Onagoruwa, Adenike and Quentin Wodon, "Measuring the Impact of Child Marriage on Total

Fertility: A Study for Fifteen Countries," *Journal of Biosocial Science*, September 2018, *50* (5), 626–639.

Osili, Una Okonkwo and Bridget Terry Long, "Does female schooling reduce fertility? Evidence from Nigeria," *Journal of Development Economics*, August 2008, *87* (1), 57–75.

Pru, Mindy, C Michelle Brown, and Rama S Singh, "Cost of mate choice: Changing patterns of global age disparity in marriage and their consequences to women's health including maternal mortality and menopause," *Women's Health*, January 2024, *20*, 17455057241264687.

Psaki, Stephanie R., Andrea J. Melnikas, Eashita Haque, Grace Saul, Christina Misunas, Sangram K. Patel, Thoai Ngo, and Sajeda Amin, "What Are the Drivers of Child Marriage? A Conceptual Framework to Guide Policies and Programs," *Journal of Adolescent Health*, December 2021, *69* (6), S13–S22.

Rokicki, Slawa, "Impact of family law reform on adolescent reproductive health in Ethiopia: A quasi-experimental study," *World Development*, August 2021, *144*, 105484.

Steinhaus, Mara, Laura Hinson, A. Theodore Rizzo, and Amy Gregowski, "Measuring Social Norms Related to Child Marriage Among Adult Decision-Makers of Young Girls in Phalombe and Thyolo, Malawi," *Journal of Adolescent Health*, April 2019, *64* (4), S37–S44.

Sunder, Naveen, "Marriage Age, Social Status, and Intergenerational Effects in Uganda," *Demography*, December 2019, *56* (6), 2123–2146.

Tauseef, Salauddin and Farha Deba Sufian, "The Causal Effect of Early Marriage on Women's Bargaining Power: Evidence from Bangladesh," *The World Bank Economic Review*, August 2024, *38* (3), 598–624.

Turner, Nicola and Frank Götmark, "Human fertility and religions in sub-Saharan Africa: A comprehensive review of publications and data, 2010-2020," *African Journal of Reproductive Health*, 2023, *27* (1).

UN General Assembly, "Transforming our world: the 2030 Agenda for Sustainable Development," 2015.

UN Women, "Beijing Declaration and Platform for Action," in "Fourth World Conference on Women" United Nations Beijing, China September 1995.

UNFPA and UNICEF, "Technical Note on Child Marriage and The Law," Technical Report November 2020.

UNICEF, "Child marriage data (2015-2024)," April 2025.

United Nations, "Convention on the Rights of the Child," November 1989.

United Nations General Assembly, "Rights of the child," December 2019.

United Nations Population Fund (UNFPA), "Child marriage," January 2026.

Wahhaj, Zaki, "An economic model of early marriage," *Journal of Economic Behavior & Organization*, August 2018, *152*, 147–176.

Westoff, Charles F. and Kristin Bietsch, "Religion and Reproductive Behavior in Sub-Saharan Africa," DHS Analytical Studies No. 48, ICF International, Rockville, Maryland, USA 2015.

Wilson, Nicholas, "Child Marriage Bans and Female Schooling and Labor Market Outcomes: Evidence from Natural Experiments in 17 Low- and Middle-Income Countries," *American Economic Journal: Economic Policy*, August 2022, *14* (3), 449–477.

Wodon, Quentin, Minh Cong Nguyen, and Clarence Tsimpo, "Child Marriage, Education, and Agency in Uganda," *Feminist Economics*, January 2016, *22* (1), 54–79.

WORLD, "Child Marriage Laws," 2013.

_ , "Child Marriage Laws," May 2023.

Yan, Ji, "Does Marriage Still Make a Difference in Infant Health?," *Eastern Economic Journal*, April 2025, *51* (2), 246–267.

A Declaration on the use of AI tools

In the preparation of this paper "Beyond Average Effects: The Uneven Promise of Child Marriage Bans", I used Claude (Anthropic) in the following ways:

1. Language editing: I used the tool to edit my writing, including proofreading and reformulation. The aim was to reduce typos and grammar mistakes, as English is not my first language, and to improve readability, concision and organisation. After every use, I reviewed and edited the content as needed.

2. Coding assistance: I used the tool to help me write and debug R scripts, particularly for the comprehensive robustness tests I conduct. The empirical strategy, main estimation pipeline, and all reported results are my own work.

3. Literature research: I used the tool to conduct literature research using specific search terms.

All underlying analysis, findings, and interpretation are my own. I take full responsibility for the content, analysis, and conclusions of this paper, including all text and code produced with AI assistance, which I have reviewed and verified.

B Literature

B.1 Literature Overview

Appendix Table A1 provides an overview of the most relevant peer-reviewed articles for this analysis, considering the effects of child marriage reforms on women's socioeconomic outcomes, including health, education, and agency. Column *CMB* details whether the authors exclusively analyse child marriage reforms that set the legal minimum age at marriage for women at 18 years regardless of parental consent. Column *N* presents the number of countries included, with those above 17 including countries both with and without CMBs.

Table A1: Summary of Literature on Child Marriage Reforms and Socioeconomic Outcomes

Article	CMB	N	Data	Model	Outcomes
Batyra and Pesando (2021)	No	6	DHS, MICS, PROSPERED	Fuzzy RDD	Marriage
Bellés-Obrero and Lombardi (2023)	Yes	1	Ministry	Two-way FE DiD	Marriage, fertility, school enrolment
Collin and Talbot (2023)	Mixed	121	DHS, MICS, WORLD	RDD	Marriage
Kidman et al. (2024)	Yes	16	DHS, WORLD	Two-way FE	Marriage, fertility
Kim et al. (2013)	Mixed	115	Country level	FE time series	Fertility
Le et al. (2025)	Yes	17	DHS, WORLD	Generalized DiD	Infant mortality
Maswikwa et al. (2015)	Yes	12	DHS, GRID, WORLD	Multivariate log binomial, Log Poisson	Marriage, fertility
McGavock (2021)	No	1	DHS	DiD, event study	Marriage
Omidakhsh and Heymann (2020)	Yes	5	DHS, WORLD	DiD	IPV
Rokicki (2021)	No	1	DHS, WORLD	DiD, Synthetic control methods	SRH
Wilson (2022)	Yes	17	DHS	DiD	Education, employment, marriage, fertility

Notes: This table outlines literature directly relevant to the study, specifically focusing on CMBs. The overview is limited by the search parameters employed and will be expanded in ongoing work. "CMB" indicates whether the study explicitly analyses reforms setting the minimum marriage age at 18 regardless of parental consent. For studies defining distinct treatment and control groups, the country count N refers to treatment countries only. Abbreviations: Difference-in-Differences (DiD), Fixed-Effects (FE), Multiple Indicator Cluster Surveys (MICS).

C Descriptive Statistics

Table A2: Descriptive Statistics: Individual Characteristics and Outcomes (5 Year Bandwidth)

Variable	DHS ID	Statistic	Full	Control	Treatment
Individual Characteristics					
Age	v012	Mean (SD)	26.34 (5.84)	27.50 (6.12)	24.98 (5.16)
		Median	26.00	27.00	24.00
		[Min, Max]	[18, 47]	[18, 47]	[18, 42]
		n	223,741	121,298	102,443
Age at First Marriage	v511	Mean (SD)	18.72 (3.95)	18.82 (4.09)	18.59 (3.76)
		Median	18.00	18.00	18.00
		[Min, Max]	[4, 44]	[4, 44]	[6, 40]
		n	160,800	91,743	69,057
Married Before 18	v511†	no	134,592 (67%)	72,043 (66%)	62,549 (69%)
		yes	65,558 (33%)	37,301 (34%)	28,257 (31%)
		n	200,150	109,344	90,806
Years of Education	v133	Mean (SD)	6.39 (5.17)	6.12 (5.20)	6.72 (5.11)
		Median	7.00	6.00	7.00
		[Min, Max]	[0, 24]	[0, 24]	[0, 24]
		n	223,633	121,240	102,393
Education Level	v149†	No Education	62,516 (28%)	36,893 (30%)	25,623 (25%)
		Primary	52,618 (24%)	29,343 (24%)	23,275 (23%)
		Secondary	84,180 (38%)	42,691 (35%)	41,489 (41%)
		Higher	24,377 (11%)	12,347 (10%)	12,030 (12%)
		n	223,691	121,274	102,417
Rural	v025	no	95,439 (43%)	51,201 (42%)	44,238 (43%)
		yes	128,302 (57%)	70,097 (58%)	58,205 (57%)
		n	223,741	121,298	102,443
Christian	v130†	no	123,457 (56%)	66,542 (56%)	56,915 (57%)
		yes	95,509 (44%)	51,881 (44%)	43,628 (43%)
		n	218,966	118,423	100,543
Muslim	v130†	no	124,090 (57%)	67,133 (57%)	56,957 (57%)
		yes	94,876 (43%)	51,290 (43%)	43,586 (43%)
		n	218,966	118,423	100,543
Ever Married	v502†	no	39,350 (19%)	17,601 (16%)	21,749 (24%)
		yes	163,558 (81%)	92,896 (84%)	70,662 (76%)
		n	202,908	110,497	92,411
Pregnancy-Driven Marriage	v211,v212, v507,v508†	no	203,053 (91%)	109,269 (90%)	93,784 (92%)
		yes	20,688 (9%)	12,029 (10%)	8,659 (8%)
		n	223,741	121,298	102,443

(Continued on next page)

Appendix Table A2 *(continued)*

Variable	DHS ID	Statistic	Full	Control	Treatment
Panel A: Fertility					
Total Children	v201	Mean (SD)	2.11 (1.88)	2.37 (1.97)	1.80 (1.71)
		Median	2.00	2.00	1.00
		[Min, Max]	[0, 17]	[0, 17]	[0, 13]
		n	223,741	121,298	102,443
Birth After 18	v212†	no	95,054 (46%)	47,819 (43%)	47,235 (50%)
		yes	112,050 (54%)	64,324 (57%)	47,726 (50%)
		n	207,104	112,143	94,961
Panel B: Education and Employment					
Completed Secondary Education	v149†	no	167,195 (77%)	91,642 (77%)	75,553 (76%)
		yes	50,577 (23%)	26,698 (23%)	23,879 (24%)
		n	217,772	118,340	99,432
Worked Last Year	v731	no	79,725 (43%)	41,149 (42%)	38,576 (45%)
		yes	104,728 (57%)	57,901 (58%)	46,827 (55%)
		n	184,453	99,050	85,403
Panel C: Decision-Making & IPV					
Decision-Making	v739, v743a,b,d,f†	Mean (SD)	0.43 (0.27)	0.44 (0.28)	0.41 (0.27)
		Median	0.50	0.50	0.50
		[Min, Max]	[0, 1]	[0, 1]	[0, 1]
		n	142,588	80,928	61,660
IPV Justification	v744a–e†	Mean (SD)	0.21 (0.33)	0.21 (0.33)	0.20 (0.32)
		Median	0.00	0.00	0.00
		[Min, Max]	[0, 1]	[0, 1]	[0, 1]
		n	223,741	121,298	102,443
IPV Experience	d103, d104, d106-108, d110†	no	33,408 (66%)	17,328 (65%)	16,080 (67%)
		yes	17,453 (34%)	9,454 (35%)	7,999 (33%)
		n	50,861	26,782	24,079
IPV Emotional	d104†	no	38,649 (76%)	20,215 (75%)	18,434 (77%)
		yes	12,210 (24%)	6,565 (25%)	5,645 (23%)
		n	50,859	26,780	24,079
IPV Physical	d106, d107, d110†	no	38,525 (76%)	20,023 (75%)	18,502 (77%)
		yes	12,336 (24%)	6,759 (25%)	5,577 (23%)
		n	50,861	26,782	24,079
IPV Sexual	d105h, d105i, d108†	no	46,621 (92%)	24,495 (91%)	22,126 (92%)
		yes	4,235 (8%)	2,284 (9%)	1,951 (8%)
		n	50,856	26,779	24,077

(Continued on next page)

Appendix Table A2 *(continued)*

Variable	DHS ID	Statistic	Full	Control	Treatment
Panel D: Marriage Market Outcomes					
Spousal Age	v730, v012†	Mean (SD)	6.89 (6.69)	6.91 (6.77)	6.85 (6.59)
Gap		Median	5.00	5.00	5.00
		[Min, Max]	[-22, 77]	[-21, 77]	[-22, 77]
		n	143,145	80,929	62,216
Spousal Age	v730, v012†	no	57,884 (40%)	32,722 (40%)	25,162 (40%)
Gap ≥ 5 yrs		yes	85,261 (60%)	48,207 (60%)	37,054 (60%)
		n	143,145	80,929	62,216
Husband	v701	No Education	35,906 (25%)	21,174 (26%)	14,732 (23%)
Education		Primary	31,862 (22%)	18,369 (22%)	13,493 (21%)
Level		Secondary	59,464 (41%)	32,827 (40%)	26,637 (42%)
		Higher	17,662 (12%)	9,566 (12%)	8,096 (13%)
		n	144,894	81,936	62,958

Descriptive statistics for individual characteristics and outcomes based on DHS data for the full sample and by reform eligibility for a bandwidth of 5 years on either side of the cutoff. † Author-constructed variable: a recode or composite index derived from DHS survey items, rather than a direct DHS survey item. IPV experience variables are binary indicators. *Abbreviations:* Demographic and Health Surveys (DHS), Standard Deviation (SD), Intimate Partner Violence (IPV).

Table A3: Descriptive Statistics: Country-Level Characteristics

Variable	Source	Statistic	Full	Control	Treatment
Child Marriage Rate (%)	WB WDI	Mean (SD)	32.91 (15.59)	32.51 (15.47)	33.38 (15.71)
		Median	34.91	34.91	34.91
		[Min, Max]	[5.6, 57.34]	[5.6, 57.34]	[5.6, 57.34]
		n	223,741	121,298	102,443
Informal Unions (% of Unions)	DHS	Mean (SD)	13.27 (15.69)	13.48 (16.07)	13.01 (15.24)
		Median	9.88	9.88	9.88
		[Min, Max]	[0, 68.78]	[0, 68.78]	[0, 68.78]
		n	223,741	121,298	102,443
Rural Share (%)	DHS	Mean (SD)	58.22 (18.63)	58.31 (18.52)	58.11 (18.75)
		Median	57.95	57.95	57.95
		[Min, Max]	[15.07, 84.83]	[15.07, 84.83]	[15.07, 84.83]
		n	223,741	121,298	102,443
Predominant Religion	DHS	Christian	127,515 (57%)	69,615 (57%)	57,900 (57%)
		Muslim	79,755 (36%)	43,114 (36%)	36,641 (36%)
		Neither	16,471 (7%)	8,569 (7%)	7,902 (8%)
		n	223,741	121,298	102,443
Regime Type	Polity IV	Anocracy	144,584 (66%)	78,148 (66%)	66,436 (67%)
		Autocracy	41,979 (19%)	24,184 (20%)	17,795 (18%)
		Democracy	31,575 (14%)	16,406 (14%)	15,169 (15%)
		n	218,138	118,738	99,400
Prior Legal Minimum Age	WORLD	Mean (SD)	15.88 (1.20)	15.86 (1.17)	15.90 (1.25)
		Median	16.00	16.00	16.00
		[Min, Max]	[14, 18]	[14, 18]	[14, 18]
		n	223,741	121,298	102,443
Government Effectiveness	WB WGI	Mean (SD)	-0.68 (0.60)	-0.68 (0.58)	-0.69 (0.63)
		Median	-0.52	-0.52	-0.66
		[Min, Max]	[-1.65, 0.88]	[-1.65, 0.88]	[-1.65, 0.88]
		n	223,741	121,298	102,443

Descriptive statistics for the full sample and by reform eligibility for a bandwidth of 5 years on either side of the cutoff. All country-level variables are author-constructed, either merged in from outside the DHS and matched to each country's policy-introduction year (Polity IV score matched to the nearest available year for Regime Type) or built from the women of the five birth cohorts preceding the reform (informal unions, rural share, predominant religion). *Abbreviations:* World Bank (WB), World Development Indicators (WDI), Worldwide Governance Indicators (WGI), Standard Deviation (SD).

Table A4: Sample Countries: First Stage and Classification

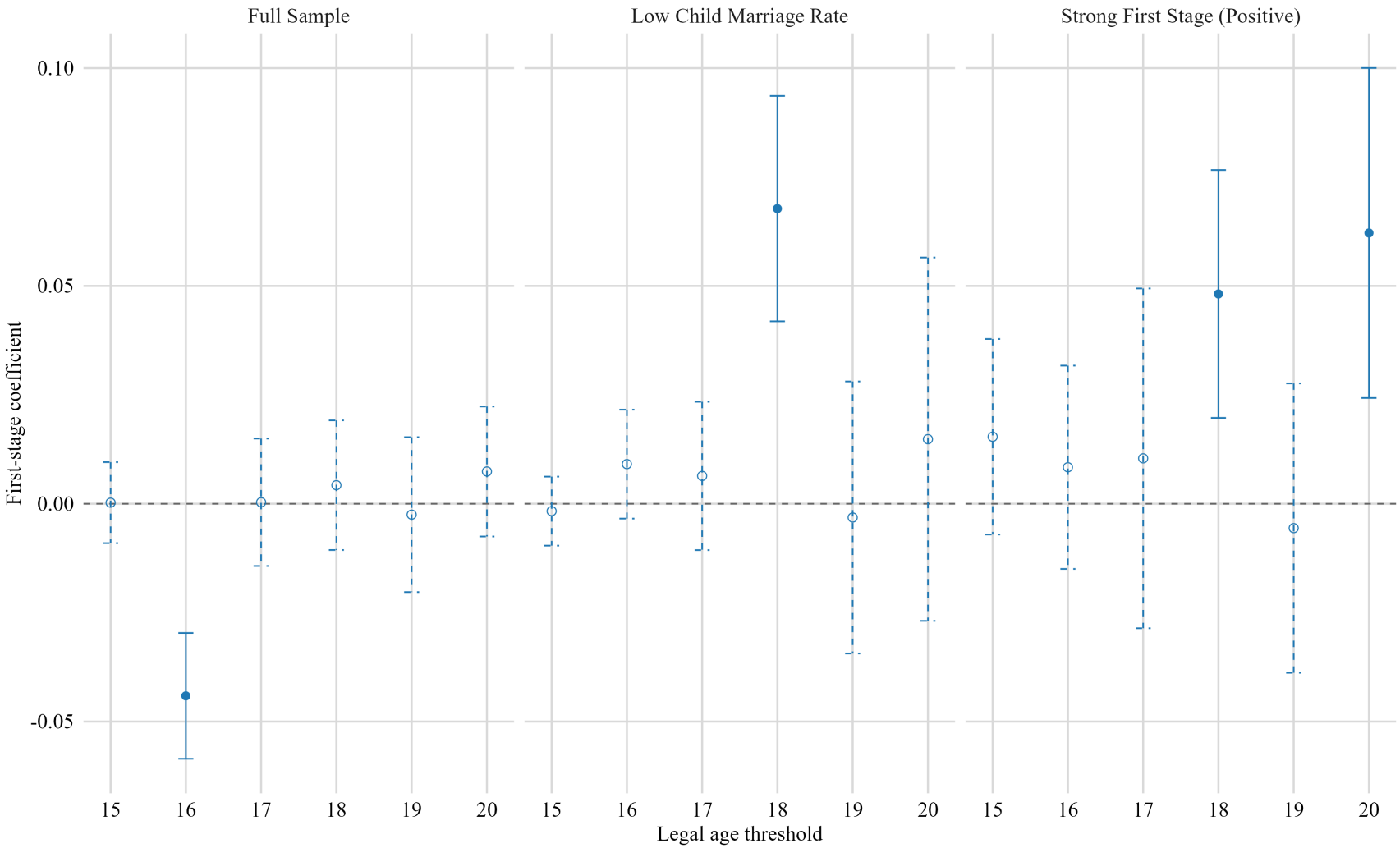
Country	Without Controls	With Controls	Child Marriage Rate	Informal Unions	Previous Min. Age	Rural Share	Religion	Regime	Gov. Effectiveness
Albania	0.035 (0.033)	0.034 (0.034)	Low	Low	16/17	Low	Muslim	Anocracy	High
Benin‡	0.023 (0.027)	0.052* (0.023)	Medium	High	14/15	Low	Christian	Democracy	High
Comoros	0.018 (0.064)	0.062 (0.068)	Medium	Low	None	High	Muslim	Anocracy	Low
DR Congo	-0.026 (0.035)	-0.020 (0.032)	Medium	High	14/15	Low	Christian	Autocracy	Low
Egypt	0.026 (0.059)	0.036 (0.058)	Low	Low	16/17	High	Muslim	Autocracy	High
Ethiopia	0.017 (0.036)	0.022 (0.035)	High	Low	14/15	High	Christian	Anocracy	Low
Gambia	0.009 (0.044)	0.008 (0.040)	Medium	Low	None	Low	Muslim	Democracy	Low
Ghana	-0.077* (0.032)	-0.064+ (0.036)	Medium	High	None	Low	Christian	Anocracy	High
Guinea	0.054 (0.061)	-0.030 (0.052)	High	Low	16/17	High	Muslim	Anocracy	Low
Jordan	0.015 (0.026)	-0.177*** (0.036)	Low	Low	14/15	Low	Muslim	Anocracy	High
Kenya	-0.040 (0.026)	-0.037 (0.027)	Medium	High	16/17	Low	Christian	Anocracy	High
Liberia	0.077 (0.049)	0.002 (0.081)	Medium	High	16/17	Low	Christian	Anocracy	Low
Madagascar	0.001 (0.040)	-0.099* (0.045)	High	High	14/15	High	Christian	Anocracy	High
Maldives	-0.036 (0.034)	-0.038 (0.034)	Low	Low	None	High	Muslim	–	High
Mozambique	-0.268**u (0.087)	-0.159* (0.078)	High	High	16/17	High	Christian	Autocracy	High
Namibia	-0.006 (0.019)	-0.006 (0.019)	Low	High	14/15	Low	Christian	Democracy	High
Nepal‡	-0.012 (0.029)	0.129*** (0.031)	High	Low	16/17	High	Neither	Anocracy	Low
Sierra Leone	0.090 (0.091)	0.052 (0.082)	High	High	None	High	Muslim	Autocracy	Low
Tajikistan‡	0.083*** (0.025)	0.083*** (0.025)	Low	Low	16/17	High	Muslim	Anocracy	Low
Togo	0.014 (0.065)	0.015 (0.060)	Medium	High	16/17	Low	Christian	Anocracy	Low

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on not marrying before age 18, estimated separately for each country with survey-year fixed effects, without and with controls for rural residence and religion. The remaining columns show the classification used for the country-level subsamples. Child marriage rate is low below 20%, medium 20 to 40% and high above 40%. Informal unions, rural share and government effectiveness are split at the cross-country median, previous min. age is the legal minimum age before the CMB, religion is the majority religion and regime follows the Polity score (Maldives has none). ‡ Countries with a significantly positive first stage at the 10% level with controls, which form the strong first stage group. u Unstable estimate (larger than half the standard deviation of the outcome), reported but not interpreted.

D Complete Results

D.1 Effect of CMBs on Not Marrying Before Age 18

Figure A1: First Stage: Sensitivity to the Legal Age Threshold (Ages 15–20)



Notes: Each point is the robust bias-corrected local-linear RDD coefficient (Calonico et al., 2014) for the probability of not marrying before age 18 shown on the x-axis, estimated separately at each threshold within each subsample. Vertical bars are 95% confidence intervals. All regressions control for rural residence, religion, and country and survey-year fixed effects, and cluster standard errors at the primary sampling unit level. Solid lines and filled markers denote significance at the 5% level, dashed lines and hollow markers denote non-significance. Panels report the full sample and the four subsamples with a significant first stage at the age-18 cutoff in the main specification.

Table A5: Effect of CMBs on Marrying in Adulthood by Subsample

	Without Controls						With Controls					
Subsample	Estimate (SE)	Mean	p-value	BH p	N	BW	Estimate (SE)	Mean	p-value	BH p	N	BW
Full Sample	0.019* (0.008)	0.667	0.017	0.117	174,870	53.0	0.004 (0.008)	0.665	0.574	0.674	174,994	54.0
Individual Level												
Rural	0.018+ (0.011)	0.586	0.097	0.374	89,477	47.2	0.015 (0.011)	0.586	0.159	0.330	89,467	49.0
Urban	-0.008 (0.010)	0.761	0.428	0.695	93,213	64.2	-0.012 (0.010)	0.763	0.221	0.425	86,201	60.8
Age 18-24	0.017 (0.017)	0.654	0.340	0.695	33,301	22.7	0.021 (0.014)	0.640	0.131	0.295	49,209	34.1
Age 25-34	0.009 (0.013)	0.691	0.474	0.695	55,212	35.0	0.017 (0.016)	0.693	0.300	0.540	37,587	24.0
Age 35+	-0.008 (0.038)	0.735	0.830	0.923	7,677	24.5	0.009 (0.038)	0.735	0.805	0.906	7,677	24.7
No Primary Education	-0.002 (0.022)	0.439	0.923	0.923	24,537	27.3	-0.008 (0.013)	0.441	0.553	0.674	67,512	75.2
Primary Education	-0.006 (0.008)	0.754	0.467	0.695	138,666	57.9	0.004 (0.008)	0.754	0.564	0.674	144,335	61.8
Christian	-0.013 (0.012)	0.693	0.265	0.596	72,166	52.3	-0.011 (0.012)	0.692	0.387	0.602	65,484	47.2
Muslim	0.016 (0.011)	0.693	0.144	0.485	81,702	55.7	0.021+ (0.012)	0.696	0.065	0.218	67,818	45.4
Country Level												
Low Child Marriage Rate	0.048*** (0.014)	0.828	<0.001	0.006	36,700	37.7	0.068*** (0.013)	0.825	<0.001	<0.001	40,726	41.4
Medium Child Marriage Rate	0.008 (0.016)	0.699	0.618	0.794	39,398	30.8	0.012 (0.015)	0.692	0.401	0.602	44,610	37.0
High Child Marriage Rate	-0.019 (0.017)	0.485	0.261	0.596	49,555	44.2	-0.021 (0.014)	0.479	0.119	0.292	72,176	68.3
Low Informal Unions	0.008 (0.010)	0.649	0.413	0.695	104,110	61.1	0.020* (0.010)	0.644	0.033	0.182	113,947	67.2
High Informal Unions	-0.008 (0.012)	0.684	0.489	0.695	80,402	49.1	-0.022* (0.010)	0.681	0.034	0.182	90,122	57.0
Previous Age 16/17	0.014 (0.011)	0.646	0.210	0.596	73,353	57.3	0.029* (0.013)	0.654	0.031	0.182	50,347	40.4
Previous Age 14/15	0.018 (0.014)	0.677	0.224	0.596	46,713	30.2	-0.010 (0.013)	0.674	0.472	0.660	52,816	34.7
No Previous Age	-0.161*** (0.024)	0.682	<0.001	<0.001	24,799	45.3	-0.037* (0.019)	0.682	0.049	0.218	24,225	44.5
Low Rural Share	0.003 (0.010)	0.740	0.730	0.857	100,888	55.4	0.002 (0.010)	0.742	0.865	0.934	95,969	54.2
High Rural Share	0.008 (0.015)	0.590	0.577	0.779	50,558	32.0	0.012 (0.013)	0.586	0.351	0.593	65,611	42.1

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Appendix Table A5 *(continued)*

	Without Controls						With Controls					
Subsample	Estimate (SE)	Mean	p-value	BH p	N	BW	Estimate (SE)	Mean	p-value	BH p	N	BW
Predominantly Christian	-0.004 (0.011)	0.660	0.729	0.857	88,523	48.2	-0.016 (0.010)	0.658	0.118	0.292	95,937	54.6
Predominantly Muslim	0.029^{+} (0.015)	0.734	0.063	0.287	33,778	27.7	0.032* (0.013)	0.738	0.016	0.182	46,067	37.9
Autocracy	-0.003 (0.027)	0.608	0.920	0.923	17,301	28.9	0.002 (0.023)	0.611	0.926	0.961	26,698	43.4
Anocracy	0.007 (0.010)	0.657	0.486	0.695	98,691	46.8	-0.000 (0.010)	0.656	0.983	0.983	105,623	51.7
Democracy	-0.090*** (0.021)	0.733	<0.001	<0.001	16,833	31.4	0.033^{+} (0.017)	0.731	0.061	0.218	17,463	32.7
Low Gov. Effectiveness	0.025^{+} (0.014)	0.599	0.064	0.287	57,197	37.9	0.026^{+} (0.015)	0.596	0.097	0.291	45,088	30.5
High Gov. Effectiveness	0.001 (0.013)	0.742	0.914	0.923	58,513	31.4	-0.007 (0.010)	0.736	0.489	0.660	90,382	50.4

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on not marrying before age 18, using Calonico et al. (2014). Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. BH p is the BH adjusted p-value within the family of the full sample and all 26 subsamples, computed separately for each specification. Mean refers to the weighted control group mean of the outcome within the estimation bandwidth. N is the number of effective observations within the MSE-optimal bandwidth on both sides of the cutoff. BW: Bandwidth (months).

D.1.1 Effect at Ages 16 and 17

Table A6: Effect of CMBs on Marrying by Age Threshold: 24-Country vs. 20-Country Sample

	24-Country Sample			20-Country Sample		
Age	Estimate (SE)	N	Control Mean	Estimate (SE)	N	Control Mean
15	0.010^{+} (0.005)	101,321	0.915	0.010^{+} (0.006)	97,562	0.913
16	0.015* (0.007)	106,126	0.843	0.010 (0.007)	118,758	0.838
17	0.015* (0.007)	137,217	0.757	0.014^{+} (0.008)	123,574	0.758
18	0.005 (0.008)	159,984	0.665	0.004 (0.008)	174,994	0.665
20	0.004 (0.009)	142,788	0.483	0.007 (0.008)	209,938	0.472
25	0.006 (0.011)	70,978	0.212	0.004 (0.011)	69,172	0.212
30	0.000 (0.012)	37,633	0.113	−0.010 (0.010)	42,420	0.113

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Robust bias-corrected RDD estimates (SE in parentheses) of the effect of CMB eligibility on not being married before age k, using Calonico et al. (2014). Treatment is fixed at the true age-18 eligibility cutoff, only the outcome and the restriction to women aged k and above change. All regressions include country and survey-year fixed effects and control for rural residence and religion. N is the number of effective observations within the MSE-optimal bandwidth. Control Mean is the weighted mean of the outcome on the untreated side within the bandwidth.

Table A7: Effect of CMBs on Marrying at or after 16: 24- vs. 20-Country Sample

	n = 24			n = 20		
Subsample	Estimate (SE)	N	Mean	Estimate (SE)	N	Mean
Full Sample	0.015* (0.007)	106,126	0.843	0.010 (0.007)	118,758	0.838
Individual Level						
Rural	0.023* (0.009)	74,704	0.788	0.029** (0.010)	58,287	0.790
Urban	-0.004 (0.009)	44,224	0.909	-0.008 (0.007)	82,644	0.902
Age 18-24	0.021* (0.010)	49,292	0.836	-0.001 (0.009)	61,483	0.828
Age 25-34	0.015 (0.010)	47,538	0.844	0.010 (0.011)	39,114	0.846
Age 35+	0.036 (0.022)	9,166	0.875	0.036+ (0.021)	11,302	0.874
No Primary Education	0.010 (0.015)	44,103	0.657	0.001 (0.017)	33,938	0.666
Primary Education	-0.010 (0.008)	55,301	0.904	0.009+ (0.005)	126,781	0.906
Christian	0.027** (0.010)	53,529	0.854	0.019+ (0.010)	57,804	0.851
Muslim	-0.004 (0.008)	60,373	0.856	-0.004 (0.008)	68,293	0.856
Country Level						
Low Child Marriage Rate	0.006 (0.007)	25,807	0.961	0.001 (0.005)	54,414	0.952
Medium Child Marriage Rate	0.013 (0.009)	64,051	0.865	-0.014 (0.012)	42,221	0.856
High Child Marriage Rate	0.012 (0.015)	45,324	0.702	0.005 (0.014)	47,118	0.711
Low Informal Unions	0.003 (0.007)	75,286	0.844	0.014+ (0.008)	69,661	0.828
High Informal Unions	0.007 (0.009)	69,708	0.837	0.013 (0.010)	55,792	0.850
Previous Age 16/17	-0.002 (0.008)	69,724	0.855	0.009 (0.010)	43,072	0.838
Previous Age 14/15	0.033** (0.011)	48,197	0.826	0.024* (0.010)	57,721	0.837
No Previous Age	-0.006 (0.013)	25,631	0.843	0.002 (0.016)	16,489	0.845
Low Rural Share	0.004 (0.007)	76,011	0.907	0.010 (0.007)	83,070	0.894
High Rural Share	0.016 (0.011)	63,393	0.780	0.001 (0.014)	34,876	0.784
Predominantly Christian	0.017* (0.008)	91,184	0.838	0.024* (0.011)	51,555	0.829
Predominantly Muslim	-0.002 (0.008)	58,933	0.871	-0.006 (0.007)	73,895	0.874
Autocracy	0.022 (0.020)	19,468	0.817	0.071** (0.027)	10,076	0.832
Anocracy	0.011 (0.007)	111,468	0.829	0.004 (0.007)	130,233	0.830
Democracy	0.016 (0.014)	17,620	0.875	0.018 (0.012)	22,974	0.873
Low Gov. Effectiveness	0.011 (0.010)	62,805	0.782	0.006 (0.011)	53,500	0.777
High Gov. Effectiveness	0.009 (0.007)	71,072	0.895	0.002 (0.006)	115,644	0.894

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Robust bias-corrected estimates (SE in parentheses) of CMB eligibility on not marrying before age 16, using Calonico et al. (2014). Treatment is fixed at the true eligibility cutoff throughout, and the sample is restricted to women aged 16 and above. All regressions include country and survey-year fixed effects and control for rural residence and religion. N is the number of effective observations within the MSE-optimal bandwidth, Mean the weighted control-group mean within the estimation bandwidth. The $n = 24$ sample adds Gabon, Kazakhstan, Chad and Zambia to the main $n = 20$ sample. The informal unions classification of the four added countries follows the same rule as for the main countries (Zambia and Kazakhstan low, Chad and Gabon high).

Table A8: Effect of CMBs on Marrying at or after 17: 24- vs. 20-Country Sample

	n = 24			n = 20		
Subsample	Estimate (SE)	N	Mean	Estimate (SE)	N	Mean
Full Sample	0.015* (0.007)	137,217	0.757	0.014+ (0.008)	123,574	0.758
Individual Level						
Rural	0.030*** (0.009)	109,100	0.693	0.027* (0.011)	73,232	0.691
Urban	-0.009 (0.008)	85,005	0.841	-0.011 (0.008)	82,607	0.837
Age 18-24	0.020* (0.010)	69,098	0.734	0.026* (0.012)	57,402	0.729
Age 25-34	0.019 (0.013)	42,204	0.777	0.016 (0.014)	34,947	0.776
Age 35+	0.023 (0.026)	10,891	0.805	0.004 (0.028)	8,761	0.809
No Primary Education	0.011 (0.015)	51,906	0.534	0.003 (0.016)	47,503	0.541
Primary Education	0.012+ (0.007)	112,802	0.841	0.024** (0.008)	80,202	0.841
Christian	0.017 (0.011)	54,714	0.772	0.016 (0.016)	29,156	0.781
Muslim	-0.002 (0.008)	97,110	0.779	-0.001 (0.009)	77,613	0.785
Country Level						
Low Child Marriage Rate	0.006 (0.009)	34,188	0.912	0.006 (0.009)	42,551	0.904
Medium Child Marriage Rate	0.026* (0.012)	55,780	0.780	0.040** (0.012)	55,686	0.773
High Child Marriage Rate	0.026+ (0.015)	54,622	0.577	0.018 (0.015)	54,461	0.584
Low Informal Unions	0.002 (0.008)	99,853	0.760	0.000 (0.008)	105,352	0.741
High Informal Unions	0.014 (0.011)	59,012	0.749	0.013 (0.011)	60,765	0.770
Previous Age 16/17	0.016 (0.010)	57,886	0.767	0.013 (0.011)	54,639	0.744
Previous Age 14/15	0.026* (0.012)	59,994	0.738	0.010 (0.009)	108,941	0.759
No Previous Age	-0.075*** (0.017)	23,404	0.766	0.110*** (0.019)	25,589	0.763
Low Rural Share	0.000 (0.009)	70,115	0.840	0.008 (0.010)	56,773	0.827
High Rural Share	0.026* (0.012)	66,504	0.680	0.014 (0.011)	73,993	0.680
Predominantly Christian	0.021+ (0.012)	59,950	0.751	0.012 (0.011)	75,094	0.744
Predominantly Muslim	0.007 (0.011)	44,833	0.803	-0.006 (0.010)	66,130	0.809
Autocracy	0.005 (0.023)	22,160	0.723	0.041 (0.029)	12,035	0.719
Anocracy	0.019+ (0.010)	83,601	0.744	0.015 (0.009)	85,765	0.745
Democracy	0.010 (0.016)	17,008	0.807	0.022 (0.017)	14,886	0.796
Low Gov. Effectiveness	0.023* (0.011)	65,152	0.685	0.021+ (0.012)	58,028	0.684
High Gov. Effectiveness	0.008 (0.009)	84,784	0.825	-0.003 (0.009)	79,119	0.824

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Robust bias-corrected estimates (SE in parentheses) of CMB eligibility on not marrying before age 17, using Calonico et al. (2014). Treatment is fixed at the true eligibility cutoff throughout, and the sample is restricted to women aged 17 and above. All regressions include country and survey-year fixed effects and control for rural residence and religion. N is the number of effective observations within the MSE-optimal bandwidth, Mean the weighted control-group mean within the estimation bandwidth. The $n = 24$ sample adds Gabon, Kazakhstan, Chad and Zambia to the main $n = 20$ sample. The informal unions classification of the four added countries follows the same rule as for the main countries (Zambia and Kazakhstan low, Chad and Gabon high).

D.2 Effect of CMBs on Life-Cycle Outcomes

D.2.1 Panel A: Fertility

Table A9: Effect of CMBs on Total Children

	Without Controls			With Controls		
Subsample	Estimate (SE)	BH p	N	Estimate (SE)	BH p	N
Full Sample	0.027 (0.030)	0.768	115,517	0.031 (0.023)	0.466	181,145
Individual Level						
Rural	0.103** (0.033)	0.021	94,394	0.054+ (0.029)	0.280	107,922
Urban	-0.014 (0.034)	0.840	78,766	-0.018 (0.037)	0.787	66,704
Age 18-24	-0.001 (0.033)	0.979	42,872	0.053* (0.025)	0.236	83,287
Age 25-34	0.029 (0.050)	0.835	60,010	0.075 (0.065)	0.466	33,627
Age 35+	-0.081 (0.168)	0.840	7,343	0.057 (0.165)	0.787	7,343
No Primary Education	0.045 (0.052)	0.768	51,591	0.023 (0.060)	0.787	35,919
Primary Education	0.003 (0.024)	0.922	146,501	0.012 (0.024)	0.787	143,043
Christian	0.075* (0.032)	0.084	85,451	0.071* (0.029)	0.211	88,590
Muslim	-0.053 (0.045)	0.733	46,823	-0.061 (0.046)	0.466	41,918
Country Level						
Low Child Marriage Rate	-0.049 (0.056)	0.768	24,169	-0.071 (0.048)	0.406	27,140
Medium Child Marriage Rate	0.007 (0.043)	0.922	51,379	0.015 (0.040)	0.787	64,373
High Child Marriage Rate	0.085 (0.052)	0.390	56,681	0.058 (0.046)	0.466	51,760
Low Informal Unions	-0.038 (0.042)	0.768	62,078	-0.003 (0.035)	0.932	90,067
High Informal Unions	0.054 (0.038)	0.532	65,085	0.053 (0.033)	0.362	78,099
Previous Age 16/17	-0.008 (0.039)	0.922	50,116	-0.028 (0.036)	0.719	56,721
Previous Age 14/15	0.138** (0.045)	0.021	65,994	0.129** (0.046)	0.137	54,493
No Previous Age	0.048 (0.065)	0.768	23,223	-0.077 (0.069)	0.466	23,848
Low Rural Share	-0.017 (0.037)	0.840	66,597	-0.016 (0.033)	0.787	84,279
High Rural Share	0.030 (0.039)	0.768	72,208	0.038 (0.031)	0.466	100,014
Predominantly Christian	0.167** (0.060)	0.034	24,925	0.055+ (0.029)	0.280	115,991
Predominantly Muslim	0.032 (0.038)	0.768	74,902	0.017 (0.037)	0.787	69,266
Autocracy	-0.021 (0.088)	0.922	14,388	0.015 (0.054)	0.806	32,395
Anocracy	0.021 (0.031)	0.781	118,476	0.020 (0.035)	0.787	80,844
Democracy	-0.936*** (0.112)	<0.001	19,798	0.176* (0.081)	0.236	8,578
Low Gov. Effectiveness	-0.017 (0.041)	0.840	74,419	-0.024 (0.038)	0.787	78,744
High Gov. Effectiveness	0.071* (0.028)	0.065	112,897	0.056+ (0.032)	0.318	78,010

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on total children, using Calonico et al. (2014). Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. BH p is the BH adjusted p-value within the family of the full sample and all 26 subsamples, computed separately for each specification. N is the total number of effective observations within the MSE-optimal bandwidth.

Table A10: Effect of CMBs on First Birth After Age 18

	Without Controls			With Controls		
Subsample	Estimate (SE)	BH p	N	Estimate (SE)	BH p	N
Full Sample	0.015* (0.007)	0.128	230,554	0.014 (0.009)	0.239	145,215
Individual Level						
Rural	0.028** (0.009)	0.009	114,807	0.026** (0.009)	0.020	121,101
Urban	0.006 (0.012)	0.871	83,779	0.006 (0.013)	0.770	57,331
Age 18-24	0.011 (0.014)	0.691	46,139	0.010 (0.014)	0.633	43,894
Age 25-34	0.003 (0.014)	0.924	58,897	0.003 (0.014)	0.906	61,885
Age 35+	-0.083** (0.032)	0.041	6,624	0.077* (0.031)	0.054	6,624
No Primary Education	0.028* (0.014)	0.140	56,758	0.018 (0.015)	0.355	47,588
Primary Education	0.013+ (0.008)	0.233	182,384	0.016* (0.008)	0.152	149,111
Christian	0.042*** (0.012)	0.009	60,509	0.040*** (0.011)	0.006	69,087
Muslim	-0.002 (0.012)	0.924	67,733	0.012 (0.009)	0.355	116,070
Country Level						
Low Child Marriage Rate	0.007 (0.026)	0.924	21,162	0.006 (0.016)	0.801	33,781
Medium Child Marriage Rate	0.017 (0.012)	0.352	63,951	0.023+ (0.013)	0.207	52,833
High Child Marriage Rate	-0.000 (0.015)	0.993	52,577	0.012 (0.013)	0.500	62,246
Low Informal Unions	-0.021 (0.016)	0.352	33,668	0.006 (0.012)	0.770	66,101
High Informal Unions	-0.034** (0.012)	0.026	64,134	0.029* (0.011)	0.054	60,290
Previous Age 16/17	0.022+ (0.011)	0.156	71,789	-0.006 (0.013)	0.779	47,776
Previous Age 14/15	0.010 (0.017)	0.834	64,885	0.020+ (0.012)	0.239	79,917
No Previous Age	0.003 (0.015)	0.924	38,211	0.028 (0.021)	0.325	17,281
Low Rural Share	0.014 (0.012)	0.459	72,838	0.017 (0.011)	0.239	72,500
High Rural Share	0.002 (0.011)	0.924	74,796	0.011 (0.011)	0.500	75,782
Predominantly Christian	0.039*** (0.011)	0.009	77,012	0.042*** (0.012)	0.006	66,186
Predominantly Muslim	-0.018 (0.013)	0.352	46,323	0.048** (0.017)	0.024	33,911
Autocracy	0.054+ (0.028)	0.156	14,785	0.038 (0.025)	0.239	18,892
Anocracy	0.008 (0.011)	0.775	83,382	0.000 (0.012)	0.990	64,562
Democracy	-0.007 (0.018)	0.920	26,492	0.052 (0.033)	0.239	8,111
Low Gov. Effectiveness	0.002 (0.014)	0.924	50,096	0.000 (0.014)	0.990	48,701
High Gov. Effectiveness	0.042** (0.013)	0.009	49,507	0.037** (0.012)	0.011	58,101

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on births after age 18, using Calonico et al. (2014). Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. BH p is the BH adjusted p-value within the family of the full sample and all 26 subsamples, computed separately for each specification. N is the total number of effective observations within the MSE-optimal bandwidth.

D.2.2 Panel B: Education and Employment

Table A11: Effect of CMBs on Completed Secondary Education

	Without Controls			With Controls		
Subsample	Estimate (SE)	BH p	N	Estimate (SE)	BH p	N
Full Sample	0.010 (0.007)	0.496	227,397	0.005 (0.006)	0.844	210,171
Individual Level						
Rural	-0.005 (0.006)	0.898	124,189	-0.003 (0.006)	0.844	135,134
Urban	0.016 (0.011)	0.496	96,283	0.018 (0.012)	0.518	77,003
Age 18-24	0.020^{+} (0.012)	0.496	66,753	0.009 (0.012)	0.844	59,047
Age 25-34	-0.002 (0.012)	0.898	74,023	-0.001 (0.011)	0.964	74,247
Age 35+	0.022 (0.040)	0.898	8,360	0.016 (0.039)	0.844	8,761
Christian	-0.002 (0.010)	0.898	100,149	-0.003 (0.010)	0.882	85,152
Muslim	0.003 (0.010)	0.898	106,407	0.010 (0.010)	0.844	91,329
Country Level						
Low Child Marriage Rate	0.057* (0.023)	0.327	23,898	0.041^{+} (0.021)	0.431	26,829
Medium Child Marriage Rate	0.004 (0.015)	0.898	50,118	-0.010 (0.015)	0.844	44,612
High Child Marriage Rate	-0.004 (0.010)	0.898	42,250	-0.002 (0.010)	0.956	52,525
Low Informal Unions	0.020^{+} (0.011)	0.496	78,058	0.027* (0.012)	0.348	59,829
High Informal Unions	-0.002 (0.011)	0.898	83,753	-0.007 (0.009)	0.844	86,834
Previous Age 16/17	-0.000 (0.013)	0.979	70,707	-0.010 (0.010)	0.844	86,459
Previous Age 14/15	0.016 (0.010)	0.496	72,859	0.017^{+} (0.010)	0.431	65,796
No Previous Age	-0.008 (0.017)	0.898	24,580	0.007 (0.016)	0.844	24,543
Low Rural Share	0.017 (0.013)	0.496	79,003	0.007 (0.009)	0.844	127,473
High Rural Share	-0.003 (0.009)	0.898	83,763	0.000 (0.009)	0.964	82,821
Predominantly Christian	-0.003 (0.009)	0.898	113,440	-0.001 (0.008)	0.962	124,561
Predominantly Muslim	0.019 (0.013)	0.496	60,152	0.023^{+} (0.013)	0.431	49,902
Autocracy	0.008 (0.021)	0.898	21,842	0.009 (0.017)	0.844	29,942
Anocracy	0.005 (0.009)	0.898	120,204	0.004 (0.007)	0.844	161,781
Democracy	-0.016 (0.021)	0.898	17,485	-0.078*** (0.020)	0.002	20,727
Low Gov. Effectiveness	-0.004 (0.011)	0.898	72,391	-0.006 (0.009)	0.844	80,120
High Gov. Effectiveness	0.016 (0.011)	0.496	89,353	0.015 (0.009)	0.431	105,602

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on completed secondary education, using Calonico et al. (2014). Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. BH p is the BH adjusted p-value within the family of the full sample and all 24 subsamples shown, computed separately for each specification. N is the total number of effective observations within the MSE-optimal bandwidth. The education-level subsamples are omitted as not meaningfully estimable, since the split variable and the outcome both measure educational attainment.

Table A12: Effect of CMBs on Worked Last Year

	Without Controls			With Controls		
Subsample	Estimate (SE)	BH p	N	Estimate (SE)	BH p	N
Full Sample	0.010 (0.008)	0.939	164,157	0.005 (0.008)	0.801	160,957
Individual Level						
Rural	0.000 (0.009)	0.988	134,879	0.001 (0.009)	0.908	114,380
Urban	0.011 (0.014)	0.977	63,157	0.006 (0.014)	0.850	60,827
Age 18-24	0.003 (0.016)	0.977	49,327	-0.007 (0.013)	0.824	72,695
Age 25-34	0.004 (0.015)	0.977	45,588	0.015 (0.013)	0.677	58,215
Age 35+	0.020 (0.033)	0.977	8,359	-0.039 (0.036)	0.689	7,342
No Primary Education	0.009 (0.012)	0.977	62,404	0.009 (0.013)	0.801	49,312
Primary Education	0.003 (0.011)	0.977	97,306	0.004 (0.010)	0.860	105,183
Christian	0.010 (0.014)	0.977	56,693	0.011 (0.014)	0.801	57,981
Muslim	-0.008 (0.013)	0.977	70,292	-0.008 (0.013)	0.801	67,646
Country Level						
Low Child Marriage Rate	0.075*** (0.022)	0.016	29,300	0.005 (0.018)	0.901	31,807
Medium Child Marriage Rate	-0.004 (0.016)	0.977	56,656	0.072* (0.031)	0.306	16,815
High Child Marriage Rate	-0.008 (0.014)	0.977	56,132	-0.026 (0.020)	0.677	25,978
Low Informal Unions	-0.008 (0.011)	0.977	89,121	-0.002 (0.010)	0.901	123,295
High Informal Unions	0.024+ (0.012)	0.516	73,380	0.024+ (0.012)	0.468	70,662
Previous Age 16/17	-0.001 (0.016)	0.977	44,399	-0.018 (0.014)	0.677	54,253
Previous Age 14/15	-0.001 (0.013)	0.977	66,781	-0.018 (0.015)	0.677	54,564
No Previous Age	-0.009 (0.019)	0.977	31,714	0.026 (0.018)	0.677	31,674
Low Rural Share	0.017 (0.013)	0.939	71,344	0.015 (0.013)	0.677	68,722
High Rural Share	-0.001 (0.012)	0.977	78,892	-0.011 (0.011)	0.772	88,778
Predominantly Christian	0.014 (0.012)	0.939	77,390	0.008 (0.013)	0.801	71,516
Predominantly Muslim	-0.028 (0.027)	0.970	15,656	-0.023 (0.026)	0.772	16,781
Autocracy	0.108** (0.040)	0.084	17,294	-0.002 (0.025)	0.937	20,160
Anocracy	0.003 (0.012)	0.977	79,362	0.002 (0.012)	0.901	80,770
Democracy	0.013 (0.021)	0.977	21,440	-0.094*** (0.024)	0.003	17,458
Low Gov. Effectiveness	-0.003 (0.013)	0.977	77,959	-0.008 (0.012)	0.801	82,033
High Gov. Effectiveness	0.018 (0.013)	0.939	54,965	0.017 (0.012)	0.677	61,527

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on worked last year, using Calonico et al. (2014). Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. BH p is the BH adjusted p-value within the family of the full sample and all 26 subsamples, computed separately for each specification. N is the total number of effective observations within the MSE-optimal bandwidth.

D.2.3 Panel C: Decision-Making & IPV

Table A13: Effect of CMBs on Decision-Making

	Without Controls			With Controls		
Subsample	Estimate (SE)	BH p	N	Estimate (SE)	BH p	N
Full Sample	0.003 (0.006)	0.755	107,196	0.002 (0.006)	0.894	103,015
Individual Level						
Rural	0.000 (0.007)	0.973	78,546	0.000 (0.007)	0.964	78,577
Urban	0.005 (0.009)	0.755	42,052	0.004 (0.009)	0.894	41,425
Age 18-24	0.005 (0.010)	0.761	38,662	0.013 (0.010)	0.706	36,324
Age 25-34	-0.005 (0.008)	0.755	45,526	-0.001 (0.008)	0.944	49,214
Age 35+	-0.074^{+} (0.039)	0.376	3,323	-0.067^{+} (0.039)	0.706	3,323
No Primary Education	-0.007 (0.009)	0.755	44,685	-0.010 (0.008)	0.706	44,743
Primary Education	0.007 (0.007)	0.743	62,100	0.007 (0.007)	0.706	59,592
Christian	0.014 (0.010)	0.655	34,640	0.014 (0.011)	0.706	32,267
Muslim	0.008 (0.007)	0.720	54,718	0.007 (0.007)	0.706	56,990
Country Level						
Low Child Marriage Rate	0.011 (0.010)	0.720	26,038	0.009 (0.009)	0.706	37,050
Medium Child Marriage Rate	0.039* (0.015)	0.135	17,900	0.006 (0.014)	0.894	19,039
High Child Marriage Rate	-0.022 (0.014)	0.503	19,107	-0.004^{a} (0.010)	0.894	33,735
Low Informal Unions	-0.000 (0.007)	0.973	75,740	-0.000 (0.007)	0.964	80,397
High Informal Unions	0.010 (0.009)	0.720	42,216	0.012 (0.011)	0.706	33,207
Previous Age 16/17	-0.002 (0.010)	0.956	35,802	0.008 (0.009)	0.739	45,992
Previous Age 14/15	0.008 (0.009)	0.755	38,850	-0.005 (0.009)	0.884	37,010
No Previous Age	0.026^{+} (0.015)	0.498	14,309	0.024 (0.015)	0.706	14,284
Low Rural Share	-0.004 (0.012)	0.895	22,760	0.002 (0.008)	0.894	49,228
High Rural Share	0.004 (0.008)	0.761	57,583	0.005 (0.007)	0.863	65,465
Predominantly Christian	0.008 (0.010)	0.755	40,900	-0.002 (0.008)	0.944	65,607
Predominantly Muslim	0.030** (0.010)	0.051	31,631	0.012 (0.009)	0.706	38,114
Autocracy	0.003 (0.024)	0.962	9,114	0.016 (0.017)	0.739	17,719
Anocracy	0.006 (0.008)	0.755	55,238	0.005 (0.008)	0.863	56,909
Democracy	-0.048* (0.021)	0.192	5,674	-0.067*** (0.014)	<0.001	12,873
Low Gov. Effectiveness	-0.005 (0.007)	0.755	67,587	-0.002 (0.007)	0.894	65,867

High Gov. Effectiveness	0.010 (0.008)	0.720	44,029	0.013 (0.009)	0.706	41,375

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on decision-making, using Calonico et al. (2014). Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. BH p is the BH adjusted p-value within the family of the full sample and all 26 subsamples, computed separately for each specification. N is the total number of effective observations within the MSE-optimal bandwidth. [a] Estimate after dropping country-survey cells observed on one side of the cutoff only, used because the estimate on the full sample was implausible (larger than half the outcome's standard deviation).

Table A14: Effect of CMBs on IPV Justification

	Without Controls			With Controls		
Subsample	Estimate (SE)	BH p	N	Estimate (SE)	BH p	N
Full Sample	-0.022*** (0.006)	0.001	111,521	-0.021*** (0.006)	0.004	101,330
Individual Level						
Rural	-0.032*** (0.008)	<0.001	54,663	-0.030*** (0.008)	0.004	53,623
Urban	-0.005 (0.006)	0.744	86,573	-0.005 (0.007)	0.705	78,533
Age 18-24	-0.041*** (0.009)	<0.001	55,521	-0.016* (0.007)	0.057	87,856
Age 25-34	-0.000 (0.007)	0.953	75,553	0.024* (0.010)	0.057	35,636
Age 35+	0.011 (0.017)	0.745	9,588	0.011 (0.016)	0.707	10,470
No Primary Education	0.006 (0.011)	0.745	42,364	0.008 (0.010)	0.660	58,304
Primary Education	-0.019** (0.006)	0.007	77,852	-0.020** (0.006)	0.008	79,020
Christian	0.006 (0.006)	0.634	99,525	0.006 (0.006)	0.578	111,849
Muslim	-0.021* (0.009)	0.054	51,918	-0.021* (0.009)	0.057	53,421
Country Level						
Low Child Marriage Rate	-0.052*** (0.012)	<0.001	23,215	-0.040*** (0.009)	<0.001	32,794
Medium Child Marriage Rate	-0.001 (0.007)	0.953	83,159	0.002 (0.007)	0.869	87,694
High Child Marriage Rate	-0.032** (0.010)	0.006	47,719	0.015* (0.007)	0.107	75,966
Low Informal Unions	-0.009 (0.007)	0.409	80,408	-0.018* (0.008)	0.057	62,058
High Informal Unions	0.004 (0.006)	0.744	104,240	-0.001 (0.005)	0.869	122,355
Previous Age 16/17	-0.015* (0.007)	0.066	74,920	-0.002 (0.006)	0.869	113,722
Previous Age 14/15	0.004 (0.006)	0.744	112,939	0.003 (0.006)	0.739	109,264
No Previous Age	-0.002 (0.010)	0.904	41,090	0.725***[u] (0.074)	–	28,973
Low Rural Share	-0.003 (0.006)	0.749	107,972	-0.001 (0.006)	0.869	106,434
High Rural Share	-0.003 (0.006)	0.745	107,434	0.001 (0.006)	0.869	106,229
Predominantly Christian	0.013* (0.006)	0.058	118,237	0.006 (0.006)	0.578	105,900
Predominantly Muslim	-0.025** (0.009)	0.024	43,161	-0.026** (0.009)	0.012	47,920
Autocracy	-0.135*** (0.035)	<0.001	11,720	-0.015 (0.015)	0.578	28,387
Anocracy	-0.026*** (0.007)	0.001	60,539	-0.025*** (0.007)	0.005	57,155
Democracy	-0.050*** (0.014)	0.001	19,304	0.005 (0.017)	0.869	9,725
Low Gov. Effectiveness	-0.005 (0.009)	0.745	74,419	-0.008 (0.009)	0.660	74,245
High Gov. Effectiveness	-0.002 (0.004)	0.749	144,096	-0.003 (0.005)	0.707	99,672

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on IPV justification, using Calonico et al. (2014). Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. BH p is the BH adjusted p-value within the family of the full sample and all 26 subsamples, computed separately for each specification. N is the total number of effective observations within the MSE-optimal bandwidth. [u] Unstable estimate (still larger than half the standard deviation), reported but excluded from the BH correction.

Table A15: Effect of CMBs on IPV Experience

	Without Controls			With Controls		
Subsample	Estimate (SE)	BH p	N	Estimate (SE)	BH p	N
Full Sample	-0.005 (0.013)	0.894	59,977	-0.003 (0.014)	0.980	54,001
Individual Level						
Rural	-0.018 (0.017)	0.894	35,808	-0.023 (0.019)	0.930	27,237
Urban	0.014 (0.023)	0.894	23,041	0.018 (0.023)	0.930	21,961
Age 18-24	0.060^{+} (0.035)	0.894	9,951	-0.015 (0.032)	0.950	10,439
Age 25-34	-0.010 (0.024)	0.894	19,178	0.013 (0.019)	0.930	29,557
Age 35+	0.027 (0.061)	0.894	3,139	0.033 (0.061)	0.930	3,139
No Primary Education	-0.035 (0.031)	0.894	10,696	-0.040 (0.030)	0.930	11,004
Primary Education	0.007 (0.015)	0.894	48,855	0.009 (0.016)	0.930	42,013
Christian	-0.004 (0.029)	0.894	16,430	-0.061* (0.030)	0.930	16,115
Muslim	0.004 (0.019)	0.894	27,318	0.004 (0.019)	0.980	26,959
Country Level						
Low Child Marriage Rate	0.013 (0.022)	0.894	22,738	0.014 (0.022)	0.930	22,996
Medium Child Marriage Rate	-0.006 (0.028)	0.894	15,050	0.001 (0.026)	0.989	16,263
High Child Marriage Rate	-0.044 (0.027)	0.894	13,390	-0.045 (0.027)	0.930	12,493
Low Informal Unions	0.002 (0.018)	0.894	28,419	0.002 (0.018)	0.980	28,021
High Informal Unions	-0.007 (0.030)	0.894	16,600	0.025 (0.044)	0.930	7,180
Previous Age 16/17	-0.017 (0.028)	0.894	14,237	-0.010 (0.027)	0.980	13,136
Previous Age 14/15	0.030 (0.027)	0.894	17,115	0.027 (0.026)	0.930	18,411
No Previous Age	-0.033 (0.034)	0.894	6,716	-0.034 (0.035)	0.930	6,462
Low Rural Share	0.006 (0.023)	0.894	24,787	0.008 (0.025)	0.980	19,998
High Rural Share	-0.019 (0.019)	0.894	25,293	-0.018 (0.018)	0.930	26,315
Predominantly Christian	-0.018 (0.023)	0.894	26,666	0.050 (0.045)	0.930	6,744
Predominantly Muslim	0.005 (0.019)	0.894	25,984	0.005 (0.019)	0.980	25,975
Autocracy	-0.020 (0.034)	0.894	11,229	-0.020 (0.034)	0.930	10,809
Anocracy	-0.004 (0.017)	0.894	33,977	0.000 (0.024)	0.989	17,529
Democracy	0.021 (0.040)	0.894	5,122	0.023 (0.040)	0.930	5,278
Low Gov. Effectiveness	-0.007 (0.017)	0.894	30,221	-0.002 (0.018)	0.980	27,251
High Gov. Effectiveness	-0.015 (0.025)	0.894	20,548	-0.013 (0.024)	0.930	22,318

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on the probability of experiencing any form of IPV, using Calonico et al. (2014). Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. BH p is the BH adjusted p-value within the family of the full sample and all 26 subsamples, computed separately for each specification. N is the total number of effective observations within the MSE-optimal bandwidth.

Table A16: Effect of CMBs on IPV Experience: Emotional

	Without Controls			With Controls		
Subsample	Estimate (SE)	BH p	N	Estimate (SE)	BH p	N
Full Sample	-0.009 (0.014)	0.889	47,054	-0.006 (0.013)	0.986	47,407
Individual Level						
Rural	-0.020 (0.016)	0.889	28,825	-0.015 (0.016)	0.986	27,958
Urban	0.005 (0.021)	0.896	22,434	0.015 (0.021)	0.986	21,629
Age 18-24	-0.028 (0.032)	0.889	8,944	-0.011 (0.034)	0.986	7,557
Age 25-34	-0.026 (0.022)	0.889	17,846	-0.006 (0.021)	0.986	18,610
Age 35+	0.024 (0.054)	0.889	3,274	0.032 (0.055)	0.986	3,139
No Primary Education	-0.019 (0.026)	0.889	12,087	-0.011 (0.025)	0.986	11,587
Primary Education	-0.005 (0.015)	0.889	37,362	-0.000 (0.014)	0.986	40,897
Christian	-0.033 (0.028)	0.889	16,430	0.008 (0.026)	0.986	18,125
Muslim	-0.006 (0.018)	0.889	21,081	-0.006 (0.018)	0.986	21,531
Country Level						
Low Child Marriage Rate	-0.000 (0.019)	0.995	19,144	-0.000 (0.020)	0.986	19,143
Medium Child Marriage Rate	0.003 (0.036)	0.976	7,931	-0.013 (0.024)	0.986	17,254
High Child Marriage Rate	-0.022 (0.025)	0.889	12,318	-0.021 (0.026)	0.986	11,750
Low Informal Unions	-0.005 (0.017)	0.896	21,721	-0.005 (0.017)	0.986	21,320
High Informal Unions	-0.025 (0.043)	0.889	7,576	-0.004 (0.038)	0.986	8,805
Previous Age 16/17	-0.120*** (0.024)	<0.001	17,157	-0.001 (0.019)	0.986	18,306
Previous Age 14/15	0.012 (0.022)	0.889	23,590	0.012 (0.022)	0.986	24,059
No Previous Age	-0.017 (0.029)	0.889	7,056	-0.020 (0.030)	0.986	6,824
Low Rural Share	0.005 (0.022)	0.898	24,305	0.013 (0.024)	0.986	19,020
High Rural Share	-0.026 (0.017)	0.889	18,535	-0.027 (0.018)	0.986	17,868
Predominantly Christian	-0.026 (0.031)	0.889	14,222	-0.003 (0.028)	0.986	16,001
Predominantly Muslim	-0.009 (0.019)	0.889	17,715	-0.010 (0.019)	0.986	17,706
Autocracy	-0.030 (0.034)	0.889	10,621	-0.032 (0.034)	0.986	10,667
Anocracy	-0.029 (0.020)	0.889	19,872	-0.039 (0.028)	0.986	9,204
Democracy	0.038 (0.036)	0.889	5,280	0.032 (0.036)	0.986	5,421
Low Gov. Effectiveness	-0.019 (0.019)	0.889	20,920	-0.004 (0.018)	0.986	20,579
High Gov. Effectiveness	-0.013 (0.035)	0.889	10,241	-0.002 (0.021)	0.986	24,852

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on IPV experience: emotional, using Calonico et al. (2014). Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. BH p is the BH adjusted p-value within the family of the full sample and all 26 subsamples, computed separately for each specification. N is the total number of effective observations within the MSE-optimal bandwidth.

Table A17: Effect of CMBs on IPV Experience: Physical

	Without Controls			With Controls		
Subsample	Estimate (SE)	BH p	N	Estimate (SE)	BH p	N
Full Sample	-0.003 (0.013)	0.982	52,260	0.001 (0.012)	0.926	55,545
Individual Level						
Rural	-0.024 (0.017)	0.982	28,076	-0.021 (0.017)	0.861	26,849
Urban	0.013 (0.018)	0.982	28,446	0.014 (0.017)	0.861	29,427
Age 18-24	0.024 (0.031)	0.982	10,302	0.041 (0.028)	0.861	11,651
Age 25-34	-0.022 (0.021)	0.982	20,103	-0.015 (0.021)	0.861	19,028
Age 35+	-0.010 (0.059)	0.982	2,596	-0.016 (0.059)	0.926	2,596
No Primary Education	-0.019 (0.028)	0.982	11,480	-0.022 (0.026)	0.861	12,151
Primary Education	0.006 (0.013)	0.982	51,069	0.010 (0.014)	0.861	44,798
Christian	0.001 (0.027)	0.996	15,779	0.005 (0.027)	0.926	15,779
Muslim	0.010 (0.016)	0.982	29,080	0.010 (0.016)	0.892	29,080
Country Level						
Low Child Marriage Rate	0.002 (0.020)	0.996	20,938	0.002 (0.020)	0.926	21,250
Medium Child Marriage Rate	0.012 (0.026)	0.982	14,283	0.035 (0.029)	0.861	10,365
High Child Marriage Rate	-0.052* (0.027)	0.982	11,158	-0.055* (0.027)	0.530	10,648
Low Informal Unions	0.004 (0.014)	0.982	34,663	0.004 (0.014)	0.926	34,658
High Informal Unions	0.008 (0.033)	0.982	12,293	-0.205*** (0.039)	<0.001	19,557
Previous Age 16/17	-0.000 (0.024)	0.996	15,326	0.028 (0.025)	0.861	12,190
Previous Age 14/15	0.014 (0.022)	0.982	21,415	0.009 (0.023)	0.926	19,452
No Previous Age	-0.007 (0.026)	0.982	9,551	-0.006 (0.026)	0.926	9,417
Low Rural Share	0.007 (0.021)	0.982	24,787	0.010 (0.023)	0.926	18,112
High Rural Share	-0.013 (0.016)	0.982	28,522	-0.013 (0.016)	0.861	28,233
Predominantly Christian	-0.014 (0.026)	0.982	23,167	-0.022 (0.024)	0.861	19,515
Predominantly Muslim	0.004 (0.018)	0.982	21,534	0.004 (0.018)	0.926	21,525
Autocracy	0.008 (0.035)	0.982	10,341	0.005 (0.036)	0.926	10,097
Anocracy	-0.008 (0.014)	0.982	40,653	-0.007 (0.018)	0.926	24,055
Democracy	0.033 (0.040)	0.982	3,632	0.031 (0.040)	0.861	3,688
Low Gov. Effectiveness	0.008 (0.019)	0.982	19,695	0.027 (0.018)	0.861	19,732
High Gov. Effectiveness	-0.020 (0.027)	0.982	15,316	-0.013 (0.018)	0.861	28,315

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on IPV experience: physical, using Calonico et al. (2014). Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. BH p is the BH adjusted p-value within the family of the full sample and all 26 subsamples, computed separately for each specification. N is the total number of effective observations within the MSE-optimal bandwidth.

Table A18: Effect of CMBs on IPV Experience: Sexual

	Without Controls			With Controls		
Subsample	Estimate (SE)	BH p	N	Estimate (SE)	BH p	N
Full Sample	0.008 (0.008)	0.800	47,051	0.009 (0.008)	0.649	41,672
Individual Level						
Rural	0.009 (0.009)	0.800	31,020	0.008 (0.010)	0.649	28,756
Urban	0.007 (0.012)	0.838	19,724	0.011 (0.012)	0.649	18,562
Age 18-24	0.017 (0.014)	0.800	14,101	0.021 (0.017)	0.649	9,413
Age 25-34	0.007 (0.011)	0.838	22,608	0.005 (0.012)	0.889	19,513
Age 35+	-0.014 (0.034)	0.838	3,139	-0.014 (0.035)	0.889	3,005
No Primary Education	0.012 (0.017)	0.800	11,751	0.015 (0.017)	0.649	10,590
Primary Education	0.007 (0.009)	0.800	33,714	0.008 (0.009)	0.649	33,437
Christian	0.013 (0.018)	0.800	15,445	-0.003 (0.021)	0.925	11,067
Muslim	0.008 (0.009)	0.800	19,928	0.008 (0.009)	0.649	19,928
Country Level						
Low Child Marriage Rate	0.013 (0.011)	0.800	14,843	0.013 (0.011)	0.649	14,843
Medium Child Marriage Rate	0.007 (0.016)	0.838	16,170	0.012 (0.015)	0.649	17,254
High Child Marriage Rate	-0.001 (0.017)	0.960	10,649	-0.001 (0.018)	0.950	10,285
Low Informal Unions	0.010 (0.009)	0.800	20,954	0.010 (0.009)	0.649	20,949
High Informal Unions	-0.008 (0.020)	0.838	12,737	0.010 (0.017)	0.804	15,244
Previous Age 16/17	0.008 (0.016)	0.838	10,569	0.026* (0.011)	0.201	19,278
Previous Age 14/15	0.020 (0.016)	0.800	16,773	0.020 (0.016)	0.649	16,707
No Previous Age	-0.015 (0.015)	0.800	7,184	-0.015 (0.015)	0.649	7,173
Low Rural Share	0.016 (0.013)	0.800	26,880	0.014 (0.015)	0.649	17,270
High Rural Share	-0.003 (0.010)	0.888	16,942	-0.002 (0.010)	0.910	16,934
Predominantly Christian	-0.002 (0.019)	0.956	14,222	-0.005 (0.020)	0.910	11,314
Predominantly Muslim	0.012 (0.010)	0.800	17,383	0.009 (0.010)	0.649	17,374
Autocracy	-0.002 (0.024)	0.956	8,184	-0.005 (0.024)	0.910	7,976
Anocracy	0.008 (0.010)	0.800	28,294	0.012 (0.008)	0.649	40,259
Democracy	0.016 (0.022)	0.800	5,057	0.012 (0.021)	0.804	5,278
Low Gov. Effectiveness	0.011 (0.011)	0.800	20,488	0.036** (0.012)	0.075	18,601
High Gov. Effectiveness	0.003 (0.011)	0.888	24,899	0.003 (0.011)	0.910	23,681

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on IPV experience: sexual, using Calonico et al. (2014). Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. BH p is the BH adjusted p-value within the family of the full sample and all 26 subsamples, computed separately for each specification. N is the total number of effective observations within the MSE-optimal bandwidth.

D.2.4 Panel D: Marriage Market Outcomes

Table A19: Effect of CMBs on Husband Education Level

	Without Controls			With Controls		
Subsample	Estimate (SE)	BH p	N	Estimate (SE)	BH p	N
Full Sample	0.054** (0.018)	0.014	115,442	0.051** (0.017)	0.013	117,729
Individual Level						
Rural	0.016 (0.023)	0.584	56,499	0.043* (0.019)	0.045	86,109
Urban	0.066** (0.025)	0.019	58,239	0.072** (0.027)	0.018	47,457
Age 18-24	0.080** (0.028)	0.015	37,408	0.135*** (0.031)	<0.001	29,404
Age 25-34	0.002 (0.034)	0.946	37,936	-0.007 (0.028)	0.842	46,132
Age 35+	0.133+ (0.072)	0.111	9,058	0.172* (0.078)	0.049	6,939
No Primary Education	0.047 (0.034)	0.221	27,395	0.039 (0.028)	0.262	36,966
Primary Education	0.066** (0.022)	0.014	63,379	0.056** (0.018)	0.013	78,131
Christian	0.026 (0.026)	0.405	51,230	0.022 (0.024)	0.465	55,261
Muslim	0.084*** (0.024)	0.012	59,880	0.084*** (0.023)	0.004	56,755
Country Level						
Low Child Marriage Rate	0.096* (0.039)	0.027	18,283	0.202** (0.074)	0.018	18,279
Medium Child Marriage Rate	0.019 (0.040)	0.722	31,565	0.016 (0.029)	0.695	49,666
High Child Marriage Rate	-0.123** (0.044)	0.015	32,593	-0.059 (0.090)	0.626	8,956
Low Informal Unions	0.067** (0.024)	0.015	71,977	0.071*** (0.020)	0.004	82,314
High Informal Unions	0.080** (0.025)	0.012	59,928	0.010 (0.024)	0.761	53,733
Previous Age 16/17	0.083** (0.026)	0.012	54,715	0.029 (0.024)	0.349	56,652
Previous Age 14/15	0.082* (0.038)	0.051	23,403	0.062* (0.025)	0.030	44,244
No Previous Age	0.014 (0.050)	0.818	18,058	-0.019 (0.058)	0.797	10,511
Low Rural Share	0.039 (0.026)	0.194	53,687	0.057* (0.023)	0.030	57,833
High Rural Share	0.041 (0.026)	0.180	57,012	0.029 (0.026)	0.371	48,601
Predominantly Christian	0.036 (0.025)	0.215	57,226	0.001 (0.022)	0.963	68,488
Predominantly Muslim	0.078* (0.032)	0.031	54,115	0.071* (0.031)	0.040	36,456
Autocracy	0.037 (0.064)	0.663	10,601	-0.051 (0.052)	0.449	11,367
Anocracy	0.072** (0.024)	0.014	70,050	0.063** (0.022)	0.013	69,856
Democracy	0.013 (0.048)	0.818	15,157	-0.196** (0.062)	0.010	15,796
Low Gov. Effectiveness	0.069** (0.025)	0.015	70,860	0.057* (0.025)	0.040	66,404
High Gov. Effectiveness	0.058** (0.022)	0.019	66,459	-0.074** (0.025)	0.013	57,642

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on husband education level, using Calonico et al. (2014). Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. BH p is the BH adjusted p-value within the family of the full sample and all 26 subsamples, computed separately for each specification. N is the total number of effective observations within the MSE-optimal bandwidth.

Table A20: Effect of CMBs on Husband Education by Level

	Without Controls			With Controls		
Husband Education	Estimate (SE)	BH p	N	Estimate (SE)	BH p	N
Full Sample						
None	-0.010 (0.007)	0.554	133,770	-0.008 (0.007)	0.676	134,255
Primary	-0.029*** (0.008)	0.002	106,388	-0.021** (0.007)	0.014	124,658
Secondary	0.015+ (0.008)	0.292	156,598	0.020* (0.010)	0.179	104,783
Higher	0.014* (0.006)	0.060	164,199	0.014* (0.006)	0.094	154,216
Low Child Marriage Rate						
None	-0.006 (0.009)	0.607	18,283	-0.006 (0.008)	0.676	23,398
Primary	-0.015 (0.017)	0.442	22,708	-0.032+ (0.017)	0.112	20,515
Secondary	-0.005 (0.027)	0.874	19,020	0.000 (0.028)	0.993	16,784
Higher	0.045* (0.018)	0.040	33,112	0.045* (0.021)	0.094	22,701
Strong First Stage (Positive)						
None	-0.081*** (0.014)	–	32,744	-0.090***[a] (0.014)	–	29,414
Primary	-0.014 (0.013)	–	28,947	-0.015 (0.013)	–	28,940
Secondary	0.023 (0.018)	–	23,499	0.021 (0.017)	–	24,540
Higher	0.027*[a] (0.011)	–	36,268	0.027* (0.012)	–	35,155

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on the probability that the husband has the respective education level (none, primary, secondary or higher), using Calonico et al. (2014). Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. BH p is the BH adjusted p-value within the family of the full sample and all 26 subsamples, computed separately for each specification and outcome. The strong first stage countries are not part of this family, so no BH p is reported for them. N is the total number of effective observations within the MSE-optimal bandwidth. [a] Estimate after dropping country-survey cells observed on one side of the cutoff only, used because the estimate on the full sample was implausible (larger than half the outcome's standard deviation).

Table A21: Effect of CMBs on Spousal Age Gap ≥ 5 yrs

	Without Controls			With Controls		
Subsample	Estimate (SE)	BH p	N	Estimate (SE)	BH p	N
Full Sample	-0.008 (0.010)	0.716	120,190	-0.011 (0.010)	0.717	127,787
Individual Level						
Rural	-0.017 (0.012)	0.612	71,248	-0.021^{+} (0.012)	0.509	78,429
Urban	0.007 (0.015)	0.818	57,710	0.005 (0.015)	0.865	53,506
Age 18-24	0.004 (0.023)	0.916	21,603	-0.011 (0.018)	0.775	33,641
Age 25-34	0.030 (0.020)	0.612	40,455	-0.008 (0.016)	0.795	44,622
Age 35+	0.016 (0.048)	0.895	6,650	-0.006 (0.048)	0.970	6,650
No Primary Education	-0.025 (0.016)	0.612	37,893	-0.025 (0.016)	0.509	37,407
Primary Education	-0.006 (0.011)	0.805	107,252	-0.005 (0.011)	0.818	99,503
Christian	-0.019 (0.017)	0.618	42,988	-0.017 (0.017)	0.717	42,131
Muslim	-0.012 (0.012)	0.716	75,043	-0.010 (0.013)	0.775	65,023
Country Level						
Low Child Marriage Rate	0.023 (0.020)	0.618	34,122	0.008 (0.026)	0.871	18,400
Medium Child Marriage Rate	0.017 (0.016)	0.677	50,463	-0.009 (0.017)	0.795	34,746
High Child Marriage Rate	0.012 (0.019)	0.773	29,371	-0.020 (0.025)	0.775	18,122
Low Informal Unions	-0.010 (0.014)	0.736	65,243	-0.009 (0.013)	0.775	67,498
High Informal Unions	-0.003 (0.014)	0.916	56,695	-0.007 (0.014)	0.795	53,692
Previous Age 16/17	-0.020 (0.014)	0.612	56,978	-0.025^{+} (0.015)	0.509	45,980
Previous Age 14/15	0.000 (0.014)	0.990	67,885	0.000 (0.013)	0.976	75,965
No Previous Age	-0.003 (0.022)	0.916	19,965	0.002 (0.022)	0.976	19,635
Low Rural Share	0.019 (0.013)	0.612	78,712	0.018 (0.012)	0.509	78,786
High Rural Share	-0.070*** (0.016)	<0.001	47,682	-0.024 (0.016)	0.509	46,336
Predominantly Christian	-0.023^{+} (0.012)	0.594	81,014	-0.024* (0.012)	0.509	71,458
Predominantly Muslim	-0.011 (0.015)	0.736	51,459	-0.010 (0.014)	0.775	52,468
Autocracy	-0.008 (0.041)	0.916	9,081	-0.027 (0.025)	0.717	20,831
Anocracy	-0.009 (0.013)	0.736	71,338	-0.013 (0.013)	0.717	70,966
Democracy	0.059* (0.025)	0.241	10,683	0.081** (0.025)	0.028	11,053
Low Gov. Effectiveness	-0.019 (0.015)	0.618	50,389	-0.021 (0.015)	0.509	50,147
High Gov. Effectiveness	0.011 (0.012)	0.716	78,815	0.012 (0.013)	0.717	71,146

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on spousal age gap ≥ 5 yrs, using Calonico et al. (2014). Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. BH p is the BH adjusted p-value within the family of the full sample and all 26 subsamples, computed separately for each specification. N is the total number of effective observations within the MSE-optimal bandwidth.

D.2.5 Fuzzy 2SLS Results for Significant Subsamples

Table A22: Fuzzy RDD Effects of Not Marrying Before Age 18: Low Child Marriage Rate

	Without Controls			With Controls		
Outcome	Estimate (SE)	F	N	Estimate (SE)	F	N
Births After Age 18	–	6.5	29,941	–	6.0	45,482
Total Children	–	2.2	29,072	–	0.2	37,665
Completed Secondary Education	-15.013*** (1.486)	139.4	37,244	0.574 (0.516)	10.9	27,763
Worked Last Year	–	8.1	20,794	–	6.0	34,752
Decision-Making	–	9.1	37,049	–	8.3	33,857
IPV Justification	–	0.3	30,812	-0.685+ (0.383)	12.4	34,724
IPV Experience	0.215 (0.463)	10.6	22,231	0.214 (0.465)	10.3	21,753
IPV Experience: Emotional	–	9.3	20,130	–	9.6	20,937
IPV Experience: Physical	0.071 (0.400)	10.6	22,231	0.056 (0.402)	10.3	21,753
IPV Experience: Sexual	–	4.9	17,302	–	5.5	17,566
Husband Education Level	–	5.6	18,278	0.675 (0.986)	15.8	25,658
Spousal Age Gap ≥ 5 yrs	2.410*** (0.556)	25.4	35,560	–	5.9	37,016

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Fuzzy RDD estimates (SE in parentheses) of the effect of not marrying before age 18 on each outcome for compliers in countries with a low child marriage rate, using Calonico et al. (2014), with eligibility as the instrument. F is the first-stage F-statistic at the fuzzy bandwidth of each outcome, and estimates are reported only where it is at least 10. Because the first stage is small, the fuzzy estimates scale the reduced form by a small number and are inflated, and they should be read as indicative. The first stage of this subsample, the effect of eligibility on marital timing, has an F-statistic of 11.7 without controls and 26.3 with controls. Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion. N is the total number of effective observations.

Table A23: Fuzzy RDD Effects of Not Marrying Before Age 18: Strong First Stage Countries

	Without Controls			With Controls		
Outcome	Estimate (SE)	F	N	Estimate (SE)	F	N
Births After Age 18	-0.133 (0.276)	12.4	30,883	–	6.5	27,951
Total Children	-0.13 (0.74)	11.4	28,872	-0.01 (0.68)	13.2	34,479
Completed Secondary Education	0.056 (0.230)	30.3	46,483	-1.173*** (0.252)	79.0	46,470
Worked Last Year	-0.006 (0.257)	17.6	41,043	-0.036 (0.271)	17.0	39,117
Decision-Making	–	4.9	25,447	-0.286 (0.195)	10.1	32,203
IPV Justification	-0.564* (0.233)	13.7	30,883	-0.570* (0.245)	12.1	31,573
IPV Experience	–	0.6	10,275	–	0.6	10,592
IPV Experience: Emotional	–	0.4	8,652	–	0.3	8,819
IPV Experience: Physical	–	1.1	12,845	–	1.4	13,019
IPV Experience: Sexual	–	0.4	10,047	–	0.5	10,274
Husband Education Level	–	3.2	28,332	1.630 (1.151)	10.3	31,696
Spousal Age Gap ≥ 5 yrs	–	0.7	32,501	-0.562 (0.547)	17.7	35,443

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Fuzzy RDD estimates (SE in parentheses) of the effect of not marrying before age 18 on each outcome for compliers in the countries with a strong positive first stage, Benin, Nepal and Tajikistan, using Calonico et al. (2014), with eligibility as the instrument. These countries are selected on a significantly positive effect of eligibility on marital timing in the specification with controls, so the p-values are conditional on this selection. F is the first-stage F-statistic at the fuzzy bandwidth of each outcome, and estimates are reported only where it is at least 10. Because the first stage is small, the fuzzy estimates scale the reduced form by a small number and are inflated, and they should be read as indicative. The first stage of this subsample, the effect of eligibility on marital timing, has an F-statistic of 15.3 without controls and 11.0 with controls. Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion. N is the total number of effective observations.

D.3 Robustness

Table A24: Effect of CMBs on Marrying in Adulthood: Ever-Married Women

Subsample	Estimate (SE)	Mean	p-value	BH p	N	BW
Full Sample	0.012 (0.008)	0.584	0.139	0.463	160,508	61.6
Individual Level						
Rural	0.023^{+} (0.012)	0.506	0.051	0.274	74,707	47.6
Urban	-0.007 (0.011)	0.681	0.501	0.627	86,227	81.9
Age 18-24	0.045* (0.018)	0.434	0.013	0.177	30,440	32.1
Age 25-34	0.014 (0.014)	0.659	0.315	0.592	52,589	37.2
Age 35+	-0.030 (0.039)	0.714	0.439	0.596	8,418	27.0
No Primary Education	0.009 (0.015)	0.380	0.549	0.627	49,482	60.3
Primary Education	0.021^{+} (0.011)	0.686	0.064	0.287	74,016	40.4
Christian	0.007 (0.016)	0.544	0.686	0.741	40,984	42.6
Muslim	0.016 (0.012)	0.659	0.188	0.463	66,899	51.5
Country Level						
Low Child Marriage Rate	0.030* (0.012)	0.781	0.013	0.177	57,669	68.9
Medium Child Marriage Rate	0.014 (0.018)	0.577	0.442	0.596	34,652	37.5
High Child Marriage Rate	-0.005 (0.018)	0.378	0.799	0.799	37,755	42.8
Low Informal Unions	0.026* (0.011)	0.609	0.020	0.177	83,301	55.1
High Informal Unions	-0.004 (0.015)	0.547	0.797	0.799	42,590	37.7
Previous Age 16/17	0.027^{+} (0.016)	0.564	0.092	0.356	41,247	42.8
Previous Age 14/15	0.020 (0.015)	0.591	0.188	0.463	45,245	36.9
No Previous Age	-0.012 (0.019)	0.614	0.526	0.627	22,446	50.8
Low Rural Share	0.012 (0.010)	0.650	0.251	0.565	101,143	73.3
High Rural Share	0.011 (0.013)	0.507	0.396	0.596	58,323	45.7
Predominantly Christian	0.008 (0.014)	0.515	0.558	0.627	51,451	39.6
Predominantly Muslim	0.013 (0.013)	0.688	0.329	0.592	53,010	48.6
Autocracy	0.049 (0.037)	0.488	0.189	0.463	11,219	23.4
Anocracy	0.019* (0.009)	0.575	0.046	0.274	125,882	77.8
Democracy	0.022 (0.022)	0.622	0.310	0.592	14,456	37.9
Low Gov. Effectiveness	0.013 (0.015)	0.483	0.370	0.596	50,747	42.0
High Gov. Effectiveness	0.010 (0.012)	0.676	0.431	0.596	62,517	42.1

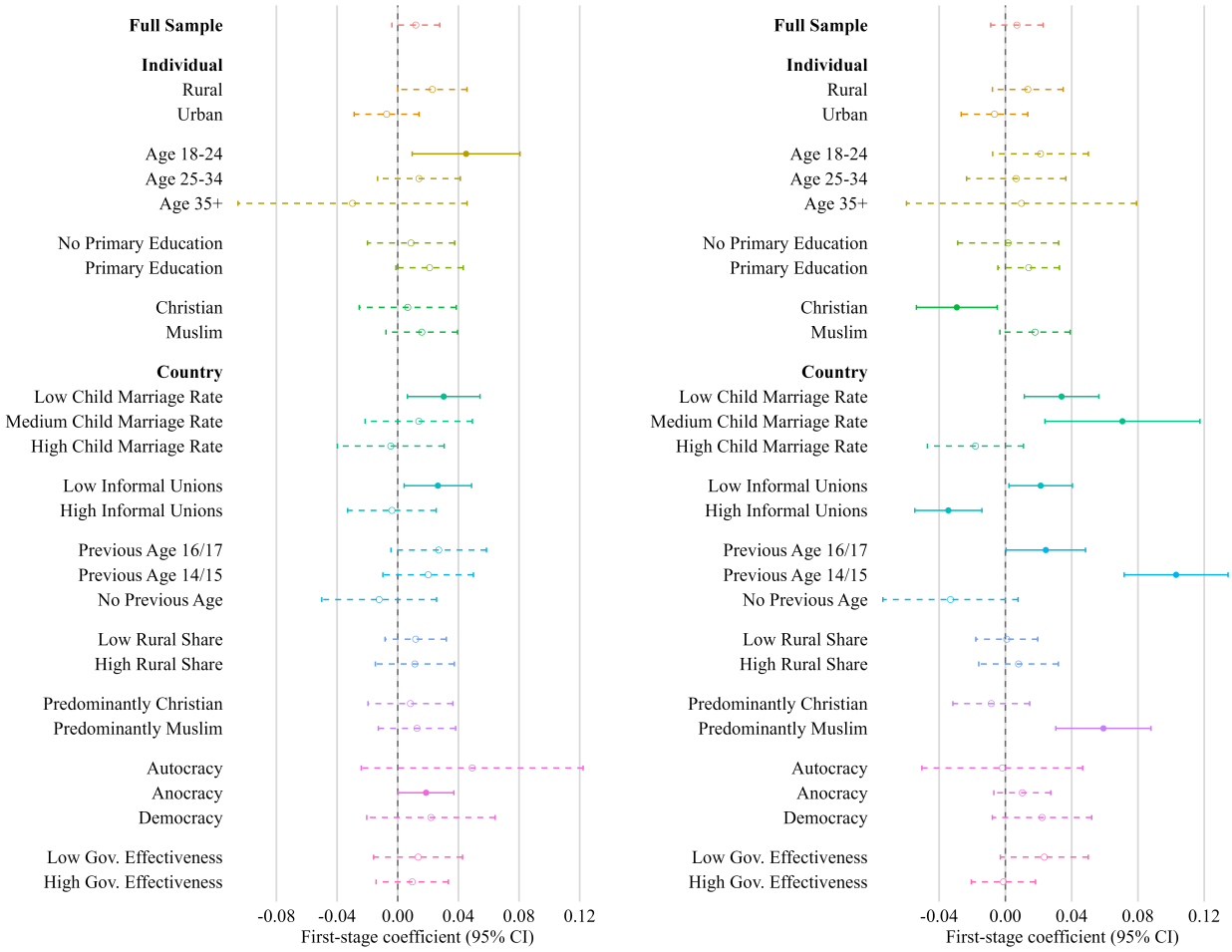
Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on not marrying before age 18, using Calonico et al. (2014), estimated for women who were ever married. All regressions include country and survey-year fixed effects and control for rural residence and religion. BH p is the BH adjusted p-value within the family of the full sample and all 26 subsamples. Mean refers to the weighted control group mean of the outcome within the estimation bandwidth. N is the number of effective observations within the MSE-optimal bandwidth on both sides of the cutoff. BW: Bandwidth (months).

Table A25: Effect of CMBs on Marrying in Adulthood: Excluding Pregnancy-Driven Marriages

Subsample	Estimate (SE)	Mean	p-value	BH p	N	BW
Full Sample	0.007 (0.008)	0.677	0.388	0.582	148,290	51.0
Individual Level						
Rural	0.014 (0.011)	0.600	0.214	0.374	85,317	52.7
Urban	-0.007 (0.010)	0.770	0.520	0.669	82,528	63.1
Age 18-24	0.021 (0.015)	0.664	0.150	0.289	39,833	30.6
Age 25-34	0.007 (0.015)	0.694	0.670	0.822	40,584	29.7
Age 35+	0.010 (0.035)	0.726	0.783	0.920	9,670	31.3
No Primary Education	0.002 (0.016)	0.447	0.914	0.940	50,543	64.8
Primary Education	0.014 (0.009)	0.771	0.139	0.289	85,657	39.1
Christian	-0.029* (0.012)	0.708	0.018	0.083	60,112	50.4
Muslim	0.018+ (0.011)	0.705	0.098	0.265	73,895	54.4
Country Level						
Low Child Marriage Rate	0.034** (0.011)	0.824	0.003	0.017	48,122	51.4
Medium Child Marriage Rate	0.071** (0.024)	0.718	0.003	0.017	16,919	16.9
High Child Marriage Rate	-0.018 (0.015)	0.492	0.222	0.374	60,886	62.1
Low Informal Unions	0.021* (0.010)	0.647	0.029	0.113	107,613	66.4
High Informal Unions	-0.034*** (0.010)	0.710	<0.001	0.008	74,969	57.8
Previous Age 16/17	0.024* (0.012)	0.652	0.046	0.156	52,918	46.8
Previous Age 14/15	0.103*** (0.016)	0.686	<0.001	<0.001	41,682	30.6
No Previous Age	-0.033 (0.021)	0.706	0.111	0.273	18,855	40.5
Low Rural Share	0.001 (0.010)	0.756	0.931	0.940	96,945	63.3
High Rural Share	0.008 (0.012)	0.591	0.517	0.669	67,670	47.8
Predominantly Christian	-0.009 (0.012)	0.674	0.472	0.669	71,103	46.5
Predominantly Muslim	0.059*** (0.015)	0.742	<0.001	<0.001	36,228	31.0
Autocracy	-0.002 (0.025)	0.633	0.940	0.940	24,218	45.4
Anocracy	0.010 (0.009)	0.657	0.243	0.387	131,506	71.6
Democracy	0.022 (0.015)	0.754	0.148	0.289	21,217	48.9
Low Gov. Effectiveness	0.023+ (0.014)	0.597	0.083	0.248	55,629	40.8
High Gov. Effectiveness	-0.001 (0.010)	0.755	0.903	0.940	82,480	52.4

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on not marrying before age 18, using Calonico et al. (2014), estimated excluding marriages motivated by pregnancy. All regressions include country and survey-year fixed effects and control for rural residence and religion. BH p is the BH adjusted p-value within the family of the full sample and all 26 subsamples. Mean refers to the weighted control group mean of the outcome within the estimation bandwidth. N is the number of effective observations within the MSE-optimal bandwidth on both sides of the cutoff. BW: Bandwidth (months).

Figure A2: Ever-Married Women and Excluding Pregnancy-Driven



(a) Ever-Married Women **(b) Excluding Pregnancy-Driven Marriages**

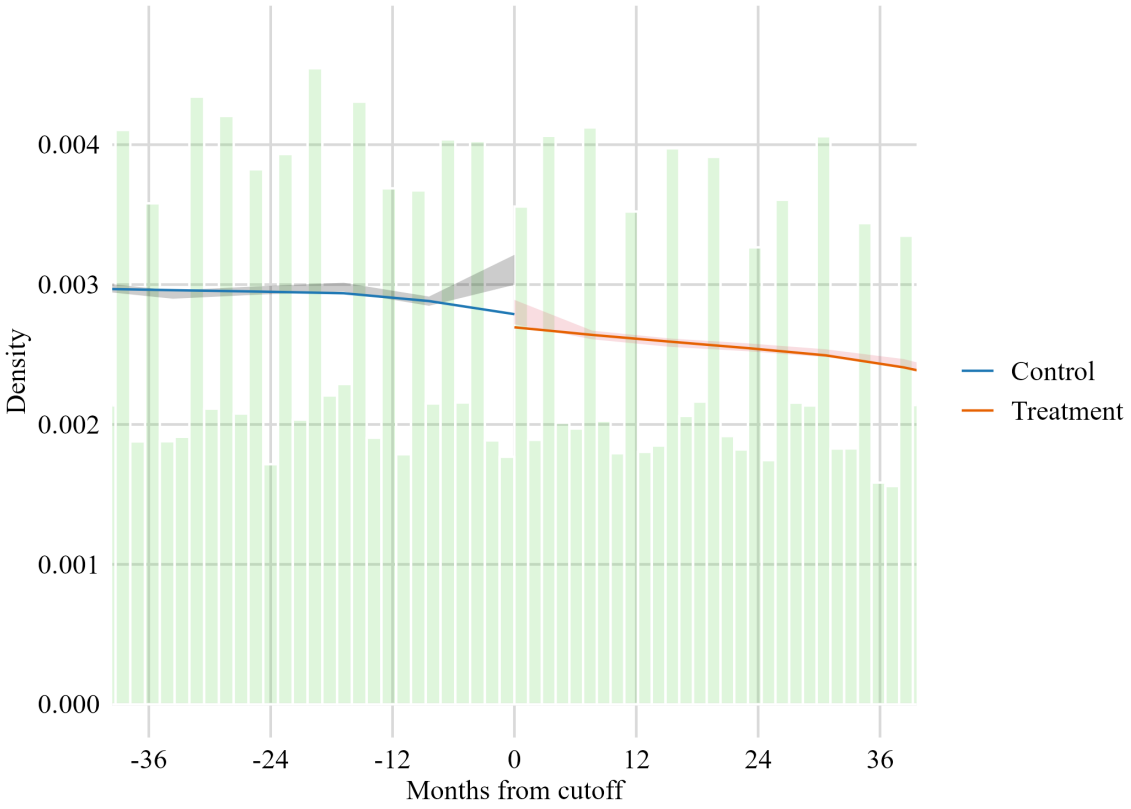
Notes: Both figures feature only subsamples that are statistically significant at the 10% level, in panel (a) for women that were ever married and in panel (b) excluding marriages that were motivated by pregnancy. Each point is the robust bias-corrected local-linear RDD coefficient (Calonico et al., 2014) for the probability of marrying at or after the legal minimum age (i.e., not marrying before age 18) at the reform cutoff, estimated separately within each subsample. Horizontal bars are 95% confidence intervals. All regressions control for rural residence, religion, and country and survey-year fixed effects, and cluster standard errors at the primary sampling unit level. Solid lines and filled markers denote significance at the 5% level, dashed lines and hollow markers denote significance only at the 10% level.

Table A26: Manipulation Test: Pooled Density Test

Robust T	p-value	N (Effective, Left / Right)	N (Raw, Left / Right)
-4.26	<0.001	50,050 / 42,898	475,338 / 211,717

Notes: Cattaneo et al. (2020)'s manipulation test on the running variable at the reform cutoff, pooled across all countries. N (Effective) is the number of observations used by the local polynomial density estimator on each side of the cutoff; N (Raw) is the full sample on each side.

Figure A3: Manipulation Test: Density of the Running Variable at the Cutoff (Pooled Sample)



Notes: Estimated density of the running variable (months from the reform cutoff) on either side of the cutoff, pooled across all countries, using Cattaneo et al. (2020)'s local polynomial density estimator. Shaded regions are 95% confidence bands. A discontinuity in the density at the cutoff would indicate manipulation or sorting of the running variable; Appendix Table A26 reports the corresponding formal test.

Table A27: Manipulation Test: Country-Level Heaping Collision Check

Country	Cutoff Month	Heaping Ratio	t_{jk}	p-value (BH)	Flagged
Benin	Sep	0.86	-8.70	<0.001	Yes
Ghana	Jan	1.30	-8.25	<0.001	Yes
Comoros	Jan	1.30	5.73	<0.001	Yes
Guinea	Sep	0.86	-4.53	<0.001	Yes
Tajikistan	Jan	1.30	3.98	<0.001	Yes
Sierra Leone	Sep	0.86	-2.88	0.013	Yes
Madagascar	Aug	0.97	-2.64	0.024	Yes
Mozambique	Dec	0.88	2.31	0.052	Yes
Liberia	Feb	1.05	2.07	0.086	Yes
DR Congo	Jan	1.30	1.78	0.150	No
Namibia	Jul	1.00	-1.64	0.185	No
Nepal	Sep	0.86	-1.54	0.206	No
Kenya	May	1.08	-1.33	0.280	No
Jordan	Apr	1.11	-0.66	0.669	No
Albania	May	1.08	0.64	0.669	No
Ethiopia	Jul	1.00	-0.58	0.669	No
Egypt	Jun	1.04	0.57	0.669	No
Maldives	Jan	1.30	-0.48	0.699	No
Gambia	Jul	1.00	-0.31	0.755	No
Togo	Jul	1.00	-0.31	0.755	No

Notes: Per-country Cattaneo et al. (2020) density test at each country's own reform cutoff. Cutoff Month is the calendar month the country's cutoff falls on. Heaping Ratio is the ratio of births in that calendar month to the count expected under a uniform distribution across months, pooled across the full sample. p-values are Benjamini-Hochberg adjusted across the 20 countries; Flagged indicates significance at the 10% level after adjustment.

Table A28: Balance of Covariates at CMB Cutoff

Covariate	Estimate (SE)	p-value	N (Left)	N (Right)	BW
Rural	0.0042 (0.0106)	0.692	129,589	107,794	64.6
Christian	-0.0162 (0.0105)	0.124	96,462	84,683	49.1
Muslim	0.0187^{+} (0.0111)	0.091	74,060	66,955	38.7

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Each row is a separate robust bias-corrected local-linear RD estimate (Calonico et al., 2014) of the listed covariate on the running variable at the reform cutoff, for the full sample, with no additional controls. Standard errors clustered at the primary sampling unit level in parentheses. BW: Bandwidth (months).

Table A29: Effect of CMBs on Age at First Marriage, Months

	Without Controls			With Controls		
Subsample	Estimate (SE)	N	BW	Estimate (SE)	N	BW
Full Sample	3.448*** (1.045)	76,355	34.5	1.111 (0.904)	103,965	48.5
Individual Level						
Rural	2.506** (0.941)	70,099	55.1	1.905* (0.902)	75,378	61.2
Urban	1.508 (2.038)	26,542	28.2	-0.474 (1.845)	30,728	33.0
Age 18-24	3.235* (1.276)	19,861	24.5	3.671** (1.313)	18,220	23.0
Age 25-34	-2.509 (1.879)	33,666	28.2	1.068 (1.735)	37,749	32.7
Age 35+	-4.705 (7.542)	6,550	29.4	4.599 (7.556)	6,232	28.6
No Primary Education	-0.494 (1.403)	33,936	59.2	-0.708 (1.379)	34,293	61.6
Primary Education	2.613* (1.196)	59,563	36.4	2.224+ (1.302)	49,100	30.9
Christian	0.230 (1.925)	27,367	33.0	2.163 (2.008)	22,945	28.5
Muslim	1.687 (1.274)	50,090	41.6	0.451 (1.299)	47,462	39.8
Country Level						
Low Child Marriage Rate	3.552+ (1.956)	21,756	26.0	5.509** (1.762)	26,732	31.2
Medium Child Marriage Rate	-0.827 (1.641)	46,781	48.6	-2.426 (2.024)	27,220	29.9
High Child Marriage Rate	1.725 (3.022)	6,513	16.7	2.841 (1.882)	13,667	35.8
Low Informal Unions	0.213 (1.522)	37,051	36.1	-1.996 (1.542)	36,038	36.0
High Informal Unions	0.713 (1.155)	64,030	54.4	2.101 (1.302)	50,704	44.1
Previous Age 16/17	-3.874* (1.625)	22,364	29.4	3.298* (1.497)	19,491	27.8
Previous Age 14/15	3.563 (2.347)	19,044	18.9	1.659 (1.522)	37,790	38.9
No Previous Age	-6.137+ (3.184)	8,952	19.1	-3.603 (2.708)	12,718	28.1
Low Rural Share	0.829 (1.550)	42,264	29.8	2.283 (1.430)	49,315	35.3
High Rural Share	1.606 (1.434)	28,812	37.5	-0.471 (1.307)	32,803	42.3
Predominantly Christian	-0.540 (1.650)	34,142	30.6	1.366 (1.457)	39,718	37.9
Predominantly Muslim	0.640 (1.850)	24,890	22.5	1.142 (1.405)	40,908	37.6
Autocracy	78.152***[u] (10.882)	9,251	19.3	3.047 (3.357)	5,719	13.7
Anocracy	1.217 (1.525)	41,311	34.0	0.425 (1.134)	72,116	62.6
Democracy	-28.970***[u] (4.070)	12,795	32.8	-0.090 (3.052)	7,824	19.0
Low Gov. Effectiveness	191.034***[u] (18.187)	24,239	33.4	4.061* (1.794)	21,731	31.3
High Gov. Effectiveness	1.421 (1.263)	60,090	40.8	0.145 (1.254)	61,124	41.4

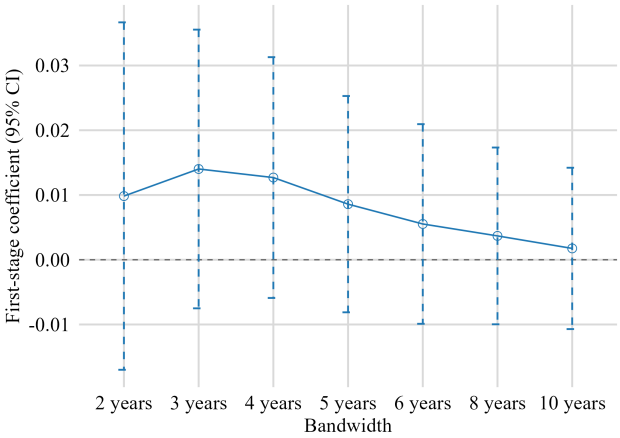
Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on age at first marriage in months, using Calonico et al. (2014). The outcome is constructed directly from month-level marriage- and birth-date components, not by rescaling the years variable. Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. Since age at first marriage is only observed for ever-married respondents, the estimation sample is restricted relative to the binary first stage and is not directly comparable to it. N is the total number of effective observations within the MSE-optimal bandwidth. BW: Bandwidth (months). [u] Unstable estimate (still larger than half the standard deviation), reported but not interpreted.

Table A30: Effect of CMBs on Marrying in Adulthood: Annual Running Variable

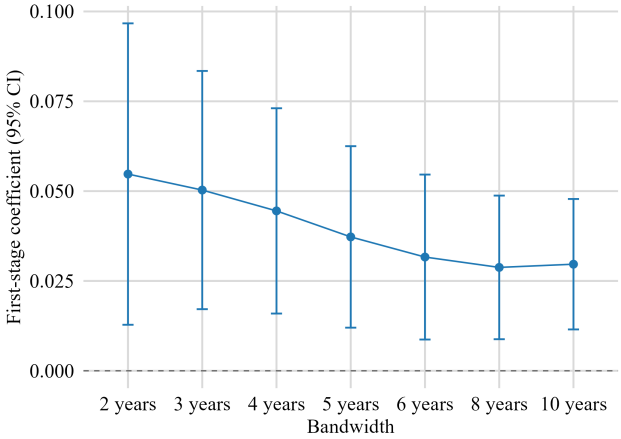
	Without Controls			With Controls		
Subsample	Estimate (SE)	N	BW	Estimate (SE)	N	BW
Full Sample	-0.003 (0.008)	217,343	5.5	0.002 (0.008)	176,983	4.5
Individual Level						
Rural	0.014 (0.010)	122,263	5.3	0.003 (0.012)	99,522	4.2
Urban	-0.009 (0.010)	95,080	5.2	-0.008 (0.010)	93,367	5.2
Age 18-24	-0.022 (0.018)	59,704	3.3	-0.003 (0.015)	58,664	3.8
Age 25-34	0.009 (0.014)	66,183	3.4	0.009 (0.014)	64,783	3.7
Age 35+	0.077** (0.028)	14,873	3.4	0.043 (0.028)	14,853	3.4
No Primary Education	-0.236*** (0.018)	60,773	5.3	-0.004 (0.014)	59,879	5.8
Primary Education	-0.002 (0.008)	156,518	5.0	-0.000 (0.008)	127,945	4.6
Christian	-0.018 (0.013)	74,332	4.1	-0.032* (0.013)	57,798	3.9
Muslim	0.028+ (0.016)	62,171	3.2	0.015 (0.013)	79,529	4.1
Country Level						
Low Child Marriage Rate	-0.021 (0.016)	40,940	3.5	0.004 (0.015)	40,925	3.6
Medium Child Marriage Rate	-0.024 (0.018)	53,017	4.0	-8.095***u (0.752)	50,601	3.9
High Child Marriage Rate	0.017 (0.023)	46,803	3.1	0.011 (0.017)	58,806	4.2
Low Informal Unions	0.023 (0.019)	50,850	2.8	0.015 (0.010)	129,391	6.2
High Informal Unions	-0.017 (0.011)	88,230	4.9	-0.016 (0.013)	65,963	4.0
Previous Age 16/17	0.031+ (0.016)	54,250	3.6	0.052*** (0.014)	66,559	4.2
Previous Age 14/15	-0.000 (0.014)	63,705	3.8	0.002 (0.014)	63,537	3.8
No Previous Age	-0.061** (0.019)	35,355	5.1	-0.038* (0.016)	35,311	5.6
Low Rural Share	-0.008 (0.013)	98,097	4.1	-0.003 (0.013)	73,471	3.7
High Rural Share	0.020 (0.014)	82,149	4.1	0.018 (0.014)	82,098	4.1
Predominantly Christian	-0.025* (0.012)	98,561	4.5	-0.021+ (0.011)	95,345	4.3
Predominantly Muslim	0.137*** (0.016)	66,732	4.7	0.018 (0.012)	66,685	4.9
Autocracy	0.450***u (0.017)	10,771	2.0	0.002 (0.027)	25,539	3.6
Anocracy	0.022* (0.010)	135,704	5.4	0.010 (0.009)	131,983	5.5
Democracy	0.027 (0.023)	22,121	3.2	-0.054** (0.020)	22,094	3.8
Low Gov. Effectiveness	0.024 (0.016)	64,012	3.7	0.020 (0.016)	62,133	3.5
High Gov. Effectiveness	-0.015 (0.010)	97,762	4.4	-0.006 (0.011)	97,001	4.1
Strong First Stage (Positive)	0.118*** (0.017)	46,172	5.3	0.020 (0.016)	46,157	5.1

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on not marrying before age 18, using Calonico et al. (2014), with the running variable collapsed to years since the reform in place of the baseline monthly measure. Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. N is the total number of effective observations within the bandwidth. BW: Bandwidth (years). [u] Unstable estimate (still larger than half the standard deviation), reported but not interpreted.

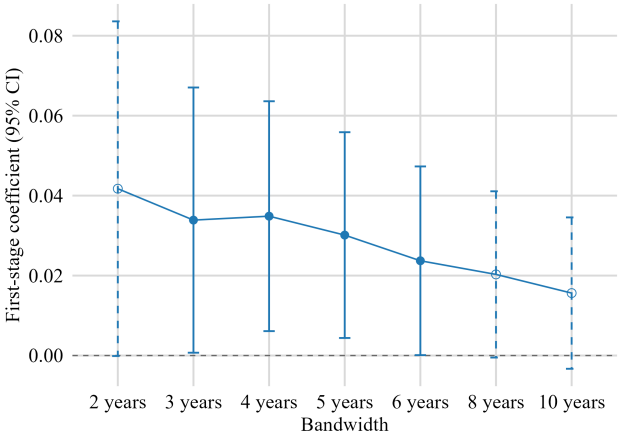
Figure A4: First Stage Across Fixed Bandwidths: Full Sample and Headline Subsamples



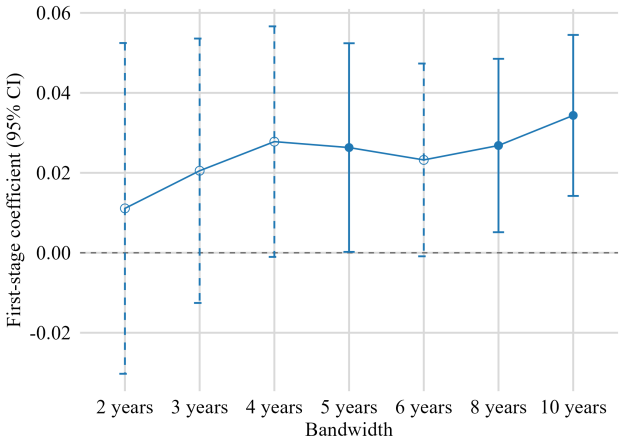
(a) Full Sample



(b) Low Child Marriage Rate



(c) Predominantly Muslim



(d) Previous Age 16/17

Notes: Each point is the robust bias-corrected first-stage RDD coefficient (Calonico et al., 2014), estimated at the fixed bandwidth (h = b, not data-driven) shown on the x-axis, in place of the baseline MSE-optimal bandwidth, with 95% confidence intervals. Solid lines and filled markers denote significance at the 5% level, dashed lines and hollow markers denote non-significance. Appendix Table A31 reports the full comparison across all subsamples.

Table A31: First Stage Across Fixed Bandwidths

Subsample	2 yrs	3 yrs	4 yrs	5 yrs	6 yrs	8 yrs	10 yrs
Without Controls							
Full Sample	0.012	0.015	0.014	0.011	0.008	0.007	0.004
Individual Level							
Rural	0.024	0.034*	0.025*	0.018	0.018+	0.016+	0.012
Urban	0.010	0.000	0.007	0.004	-0.001	-0.006	-0.007
Age 18-24	0.017	0.009	0.012	0.006	0.004	0.006	0.007
Age 25-34	0.032	0.018	0.019	0.016	0.016	0.007	0.001
Age 35+	-0.048	-0.021	-0.017	0.003	0.003	-0.001	0.008
No Primary Education	-0.013	-0.003	-0.005	-0.009	-0.010	-0.004	-0.003
Primary Education	0.020	0.021+	0.021*	0.017+	0.014	0.009	0.006
Christian	-0.019	-0.010	-0.005	-0.010	-0.010	-0.013	-0.013
Muslim	0.037+	0.038*	0.036*	0.029*	0.021+	0.016	0.011
Country Level							
Low Child Marriage Rate	0.052*	0.056**	0.047**	0.038**	0.032**	0.028**	0.025**
Medium Child Marriage Rate	0.009	0.043*	0.017	0.013	0.011	0.008	0.007
High Child Marriage Rate	-0.033	-0.021	-0.019	-0.000	-0.025	-0.026+	-0.030*
Low Informal Unions	0.029	0.032*	0.032*	0.025*	0.020+	0.019*	0.018*
High Informal Unions	-0.007	-0.003	-0.003	-0.005	-0.006	-0.011	-0.015+
Previous Age 16/17	0.022	0.020	0.035*	0.028*	0.025*	0.023*	0.028**
Previous Age 14/15	0.014	0.027	0.017	0.014	0.012	0.002	-0.008
No Previous Age	-0.017	-0.026	-0.027	-0.036+	-0.026	-0.025	-0.022
Low Rural Share	0.017	0.020	0.020	0.015	0.011	0.007	0.002
High Rural Share	0.009	0.012	0.009	0.006	0.004	0.004	0.000
Predominantly Christian	-0.017	0.000	-0.001	-0.004	-0.003	-0.010	-0.013
Predominantly Muslim	0.049*	0.032+	0.037*	0.031*	0.025*	0.020+	0.015
Autocracy	0.241***	0.081*	-0.003	0.002	-0.006	-0.018	-0.032
Anocracy	0.013	0.014	0.013	0.008	0.005	0.006	0.008
Democracy	0.022	0.036+	0.040*	0.034*	0.034*	0.022	0.018
Low Gov. Effectiveness	0.038+	0.028	0.030*	0.025+	0.021+	0.019+	0.014
High Gov. Effectiveness	-0.009	0.003	0.002	-0.002	-0.003	-0.004	-0.006
Strong First Stage (Positive)	0.052*	0.059**	0.045*	0.045**	0.052***	0.056***	0.050***
With Controls							
Full Sample	0.010	0.014	0.013	0.009	0.006	0.004	0.002
Individual Level							
Rural	0.021	0.031*	0.023+	0.017	0.016	0.013	0.010

(Continued on next page)

Appendix Table A31 *(continued)*

Subsample	2 yrs	3 yrs	4 yrs	5 yrs	6 yrs	8 yrs	10 yrs
Urban	0.004	-0.004	0.002	-0.001	-0.005	-0.009	-0.010
Age 18-24	0.012	0.023	0.015	0.007	0.005	0.002	0.002
Age 25-34	0.022	0.008	0.015	0.011	0.011	0.006	0.001
Age 35+	-0.050	-0.011	-0.002	-0.008	-0.008	-0.012	-0.007
No Primary Education	-0.009	0.000	-0.003	-0.007	-0.008	-0.004	-0.003
Primary Education	0.014	0.017	0.017	0.012	0.010	0.005	0.003
Christian	-0.012	-0.006	-0.007	-0.009	-0.011	-0.013	-0.013
Muslim	0.037^{+}	0.036*	0.036**	0.029*	0.021^{+}	0.017^{+}	0.012
Country Level							
Low Child Marriage Rate	0.055*	0.050**	0.044**	0.037**	0.032**	0.029**	0.030**
Medium Child Marriage Rate	0.011	0.013	0.011	0.008	0.006	0.003	0.001
High Child Marriage Rate	-0.037	-0.022	-0.021	-0.022	-0.023	-0.025^{+}	-0.023^{+}
Low Informal Unions	0.028	0.032*	0.033*	0.025*	0.021^{+}	0.020*	0.018*
High Informal Unions	-0.007	-0.003	-0.004	-0.012	-0.014	-0.019^{+}	-0.022*
Previous Age 16/17	0.011	0.021	0.028^{+}	0.026*	0.023^{+}	0.027*	0.034***
Previous Age 14/15	0.013	0.020	0.014	0.012	0.009	0.000	-0.008
No Previous Age	-0.003	-0.011	-0.025	-0.024	-0.034^{+}	-0.032*	-0.030*
Low Rural Share	0.016	0.017	0.015	0.009	0.005	0.001	-0.002
High Rural Share	-0.005	0.014	0.011	0.008	0.005	0.005	0.004
Predominantly Christian	-0.013	-0.002	-0.002	-0.010	-0.012	-0.017^{+}	-0.019*
Predominantly Muslim	0.042^{+}	0.034*	0.035*	0.030*	0.024*	0.020^{+}	0.016
Autocracy	0.041	0.034	0.029	0.015	0.001	0.002	-0.011
Anocracy	0.007	0.008	0.009	0.006	0.003	0.004	0.004
Democracy	0.017	0.033	0.035^{+}	0.028^{+}	0.026^{+}	0.017	0.011
Low Gov. Effectiveness	0.034	0.029^{+}	0.026^{+}	0.023^{+}	0.020	0.018^{+}	0.015
High Gov. Effectiveness	-0.007	0.003	0.001	-0.004	-0.006	-0.008	-0.007
Strong First Stage (Positive)	0.054*	0.058**	0.040*	0.041**	0.048***	0.054***	0.049***

Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Each cell is a separate robust bias-corrected first-stage RDD coefficient (Calonico et al., 2014), estimated at the fixed bandwidth (h = b, not data-driven) shown in the column header, in place of the baseline MSE-optimal bandwidth. Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. Standard errors are omitted for space and available on request.

Table A32: Effect of CMBs on Marrying in Adulthood (CER-Optimal Bandwidth)

	Without Controls			With Controls		
Subsample	Estimate (SE)	N	BW	Estimate (SE)	N	BW
Full Sample	0.015^{+} (0.009)	103,316	30.4	0.012 (0.009)	104,965	31.0
Individual Level						
Rural	0.023^{+} (0.012)	51,421	27.9	0.025* (0.012)	54,814	29.0
Urban	0.001 (0.011)	56,462	38.4	-0.000 (0.011)	52,794	36.4
Age 18-24	0.019 (0.020)	20,041	13.3	0.015 (0.016)	29,978	20.0
Age 25-34	0.019 (0.015)	33,043	20.4	0.018 (0.019)	22,074	14.1
Age 35+	-0.035 (0.046)	3,859	14.6	-0.038 (0.045)	3,859	14.8
No Primary Education	-0.014 (0.027)	14,492	16.1	-0.005 (0.015)	40,957	44.6
Primary Education	0.017^{+} (0.009)	82,488	33.3	0.014^{+} (0.009)	85,506	35.7
Christian	-0.025^{+} (0.013)	44,246	31.1	-0.009 (0.014)	39,695	28.0
Muslim	0.029* (0.012)	48,425	32.9	0.032* (0.013)	38,974	26.8
Country Level						
Low Child Marriage Rate	-0.004 (0.017)	22,197	22.6	0.041** (0.015)	24,157	24.9
Medium Child Marriage Rate	0.003 (0.019)	23,725	18.6	-0.072*** (0.018)	27,488	22.4
High Child Marriage Rate	-0.055** (0.020)	30,045	27.2	-0.020 (0.015)	47,137	42.0
Low Informal Unions	0.028* (0.011)	62,639	36.2	0.021^{+} (0.011)	67,954	39.9
High Informal Unions	0.008 (0.013)	49,670	29.3	-0.012 (0.011)	55,484	34.1
Previous Age 16/17	0.021^{+} (0.013)	45,198	34.3	0.032* (0.015)	30,771	24.2
Previous Age 14/15	0.022 (0.017)	28,183	18.1	0.019 (0.016)	31,221	20.8
No Previous Age	-0.020 (0.022)	15,300	28.8	-0.039^{+} (0.021)	15,271	28.3
Low Rural Share	0.002 (0.011)	59,867	32.7	0.014 (0.011)	58,001	32.1
High Rural Share	0.021 (0.017)	30,146	19.2	0.011 (0.014)	38,932	25.3
Predominantly Christian	0.010 (0.013)	53,044	28.6	-0.008 (0.012)	59,172	32.4
Predominantly Muslim	0.040* (0.018)	20,271	16.6	0.034* (0.015)	28,178	22.8
Autocracy	-0.013 (0.032)	9,918	17.9	0.002 (0.026)	15,713	27.0
Anocracy	0.014 (0.012)	58,380	27.5	0.009 (0.011)	63,530	30.4
Democracy	0.215*** (0.030)	10,704	20.0	-0.840^{***u} (0.090)	11,340	20.8
Low Gov. Effectiveness	0.033* (0.016)	34,917	23.1	0.021 (0.017)	26,227	18.6
High Gov. Effectiveness	-0.003 (0.014)	35,415	18.3	-0.002 (0.011)	54,699	29.5
Strong First Stage (Positive)	0.051** (0.016)	24,708	34.5	0.046** (0.016)	21,905	30.8

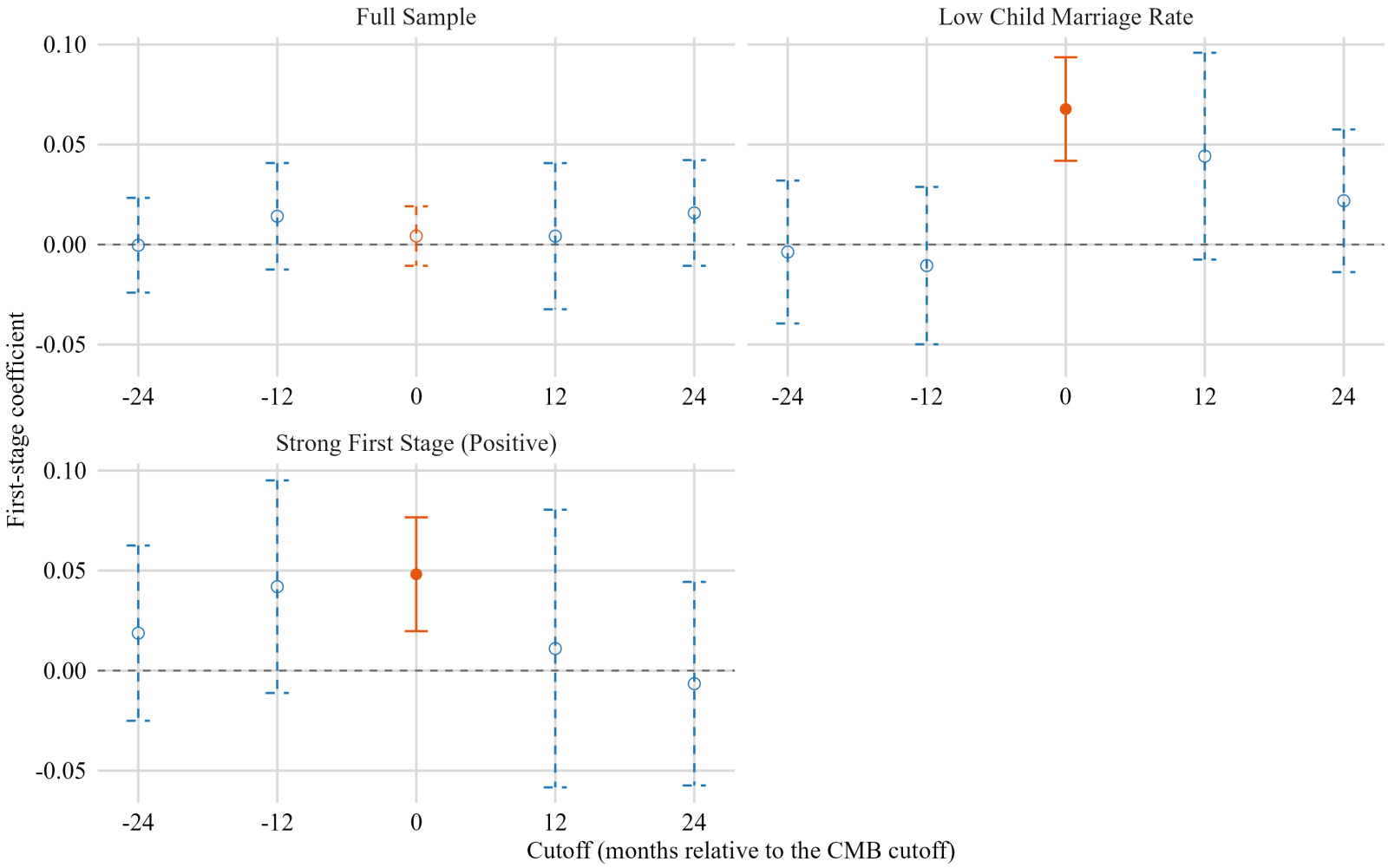
Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on not marrying before age 18, using Calonico et al. (2014), with the CER-optimal bandwidth (bwselect = cerrd) in place of the baseline MSE-optimal bandwidth. Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. N is the total number of effective observations within the bandwidth. BW: Bandwidth (months). ^{u} Unstable estimate (still larger than half the standard deviation), reported but not interpreted.

Table A33: Effect of CMBs on Marrying in Adulthood (Local Quadratic)

	Without Controls			With Controls		
Subsample	Estimate (SE)	N	BW	Estimate (SE)	N	BW
Full Sample	0.009 (0.009)	235,524	71.7	0.006 (0.009)	239,215	74.1
Individual Level						
Rural	0.018 (0.012)	135,638	73.2	0.018 (0.013)	117,245	64.2
Urban	-0.004 (0.012)	131,860	93.2	0.005 (0.015)	78,094	54.2
Age 18-24	0.022 (0.019)	52,318	36.8	0.023 (0.016)	62,062	44.3
Age 25-34	0.016 (0.017)	64,296	40.5	0.029 (0.024)	33,526	21.3
Age 35+	-0.045 (0.050)	8,768	27.7	-0.021 (0.045)	11,302	34.0
No Primary Education	-0.000 (0.027)	33,536	36.4	-0.008 (0.019)	61,218	67.2
Primary Education	0.022* (0.011)	134,071	55.6	0.013 (0.010)	146,413	62.7
Christian	-0.011 (0.012)	126,410	95.5	0.001 (0.015)	74,974	54.8
Muslim	0.033* (0.014)	87,384	59.4	0.032* (0.014)	88,636	60.7
Country Level						
Low Child Marriage Rate	0.018 (0.015)	57,318	59.0	0.038** (0.014)	63,144	64.5
Medium Child Marriage Rate	0.022 (0.020)	52,953	41.2	0.020 (0.018)	56,466	46.3
High Child Marriage Rate	-0.026 (0.024)	49,555	44.7	-0.023 (0.016)	88,053	86.0
Low Informal Unions	-0.010 (0.013)	112,347	66.9	0.023* (0.011)	141,090	85.3
High Informal Unions	-0.002 (0.015)	88,705	54.5	-0.004 (0.014)	87,000	55.9
Previous Age 16/17	0.033* (0.015)	76,505	60.9	0.023+ (0.013)	85,977	71.8
Previous Age 14/15	0.022 (0.015)	80,580	53.7	0.017 (0.015)	85,071	56.5
No Previous Age	-0.017 (0.022)	40,807	76.7	-0.031 (0.024)	28,130	52.2
Low Rural Share	0.012 (0.011)	155,426	86.4	0.017 (0.013)	102,881	58.3
High Rural Share	0.012 (0.016)	77,890	51.7	0.006 (0.013)	113,010	77.3
Predominantly Christian	0.003 (0.015)	91,853	50.0	-0.003 (0.013)	104,151	59.2
Predominantly Muslim	0.039** (0.015)	79,389	64.8	0.027+ (0.014)	78,049	64.0
Autocracy	-0.106*** (0.032)	35,920	57.6	0.032 (0.028)	34,346	55.2
Anocracy	-0.002 (0.012)	132,930	64.5	0.003 (0.010)	187,866	95.4
Democracy	0.041* (0.019)	29,871	57.2	0.030 (0.023)	20,248	38.1
Low Gov. Effectiveness	0.040* (0.020)	58,676	38.1	0.057** (0.019)	56,960	38.3
High Gov. Effectiveness	-0.001 (0.012)	126,154	70.0	-0.006 (0.011)	136,982	77.8
Strong First Stage (Positive)	0.042* (0.018)	41,743	60.6	0.037* (0.017)	43,249	62.1

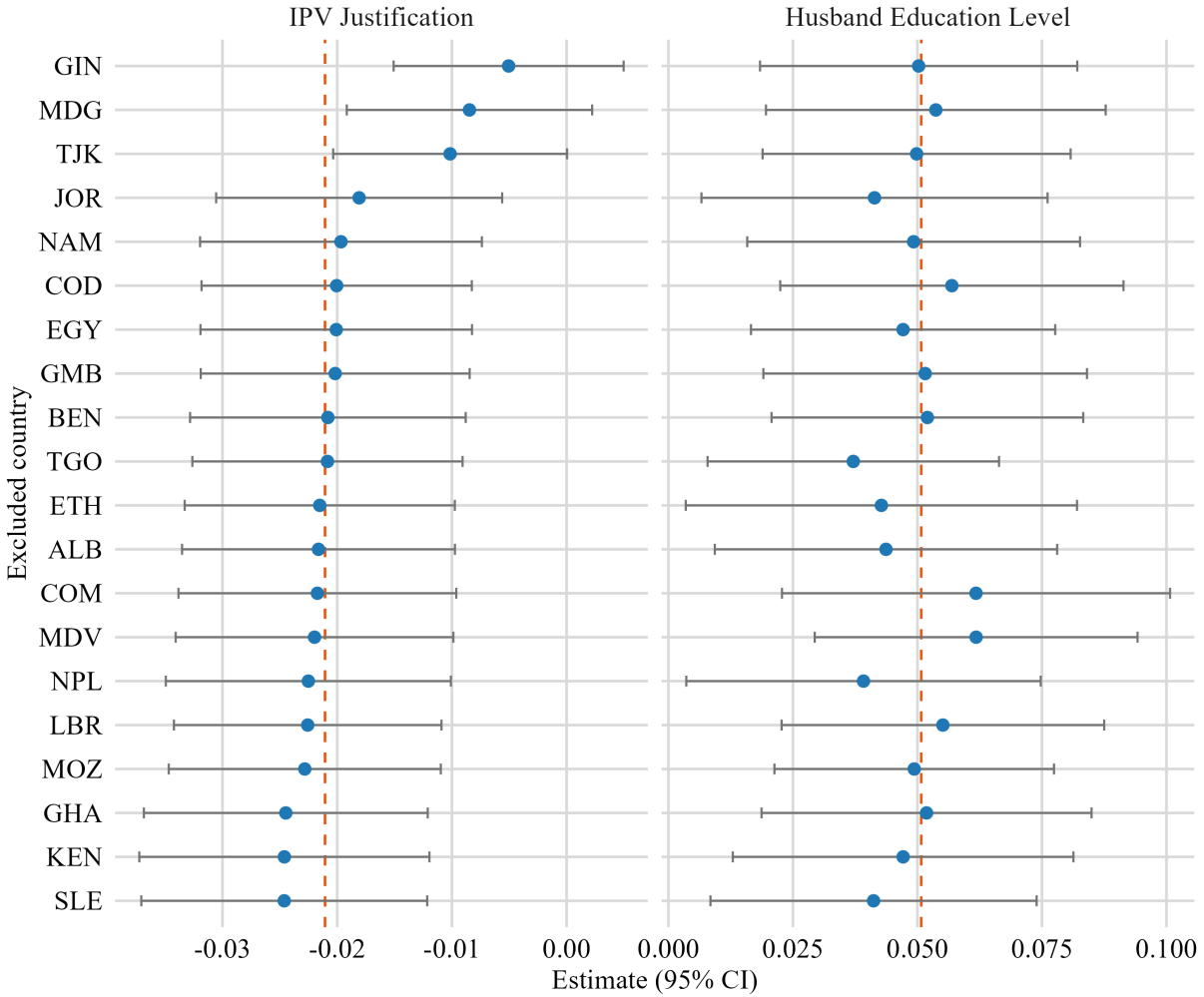
Notes: *** p<0.001, ** p<0.01, * p<0.05, + p<0.1. Estimates are robust bias-corrected RDD coefficients (SE in parentheses) of CMB eligibility on not marrying before age 18, using Calonico et al. (2014), with a local quadratic polynomial (p = 2) in the running variable in place of the baseline local linear specification. Both specifications include country and survey-year fixed effects, the specification with controls additionally includes rural residence and religion and uses the women for whom both are observed. N is the total number of effective observations within the bandwidth. BW: Bandwidth (months).

Figure A5: Placebo Test: Fake Cutoffs



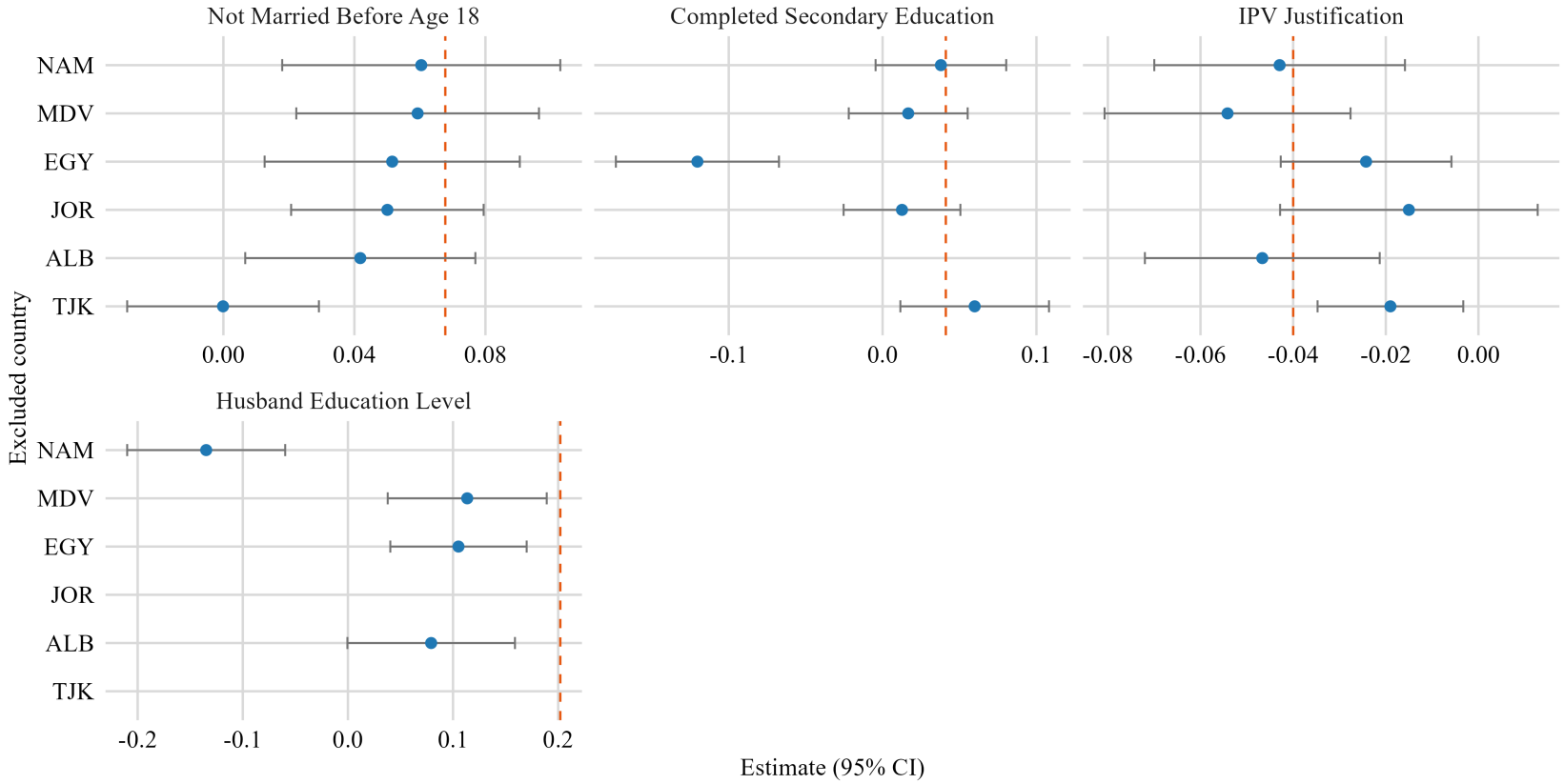
Notes: Each point is the robust bias-corrected first-stage RDD coefficient (Calonico et al., 2014) at a fake cutoff 24 or 12 months before or after the true CMB cutoff, estimated on one side of the true cutoff only so that the window never spans it. The orange point at 0 is the estimate at the true cutoff. Vertical bars are 95% confidence intervals. All regressions control for rural residence, religion, and country and survey-year fixed effects, and cluster standard errors at the primary sampling unit level. Solid lines and filled markers denote significance at the 5% level, dashed lines and hollow markers denote non-significance.

Figure A6: Leave-One-Country-Out: Full Sample



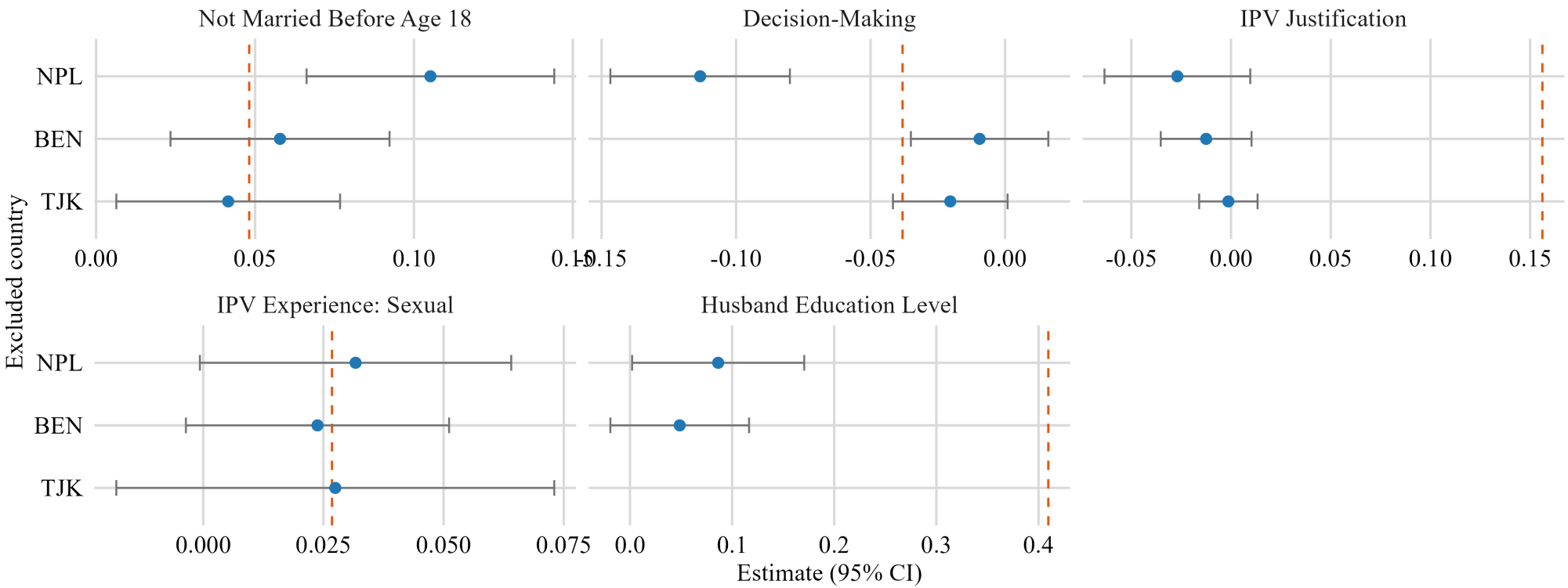
Notes: Each panel re-estimates the effect of CMB eligibility on the labelled outcome, dropping one country at a time from the estimation sample. Only outcomes with a significant effect ($p < 0.1$) in the subsample in the baseline estimation with controls are shown. Points are the point estimate excluding that country, horizontal bars are 95% confidence intervals. The dashed line is the baseline estimate using all countries.

Figure A7: Leave-One-Country-Out: Low Child Marriage Rate



Notes: Each panel re-estimates the effect of CMB eligibility on the labelled outcome, dropping one country at a time from the estimation sample. Only outcomes with a significant effect ($p < 0.1$) in the subsample in the baseline estimation with controls are shown. Points are the point estimate excluding that country, horizontal bars are 95% confidence intervals. The dashed line is the baseline estimate using all countries.

Figure A8: Leave-One-Country-Out: Strong First Stage Countries



Notes: Each panel re-estimates the effect of CMB eligibility on the labelled outcome, dropping one country at a time from the estimation sample. Only outcomes with a significant effect ($p < 0.1$) in the subsample in the baseline estimation with controls are shown. Points are the point estimate excluding that country, horizontal bars are 95% confidence intervals. The dashed line is the baseline estimate using all countries.